



SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Well File No. 25160

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent	Approximate Start Date	☐ Drilling Prognosis	☐ Spill Report
☒ Report of Work Done	Date Work Completed December 29, 2017	☐ Redrilling or Repair	☐ Shooting
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date	☐ Casing or Liner	☐ Acidizing
		☐ Plug Well	☐ Fracture Treatment
		☐ Supplemental History	☐ Change Production Method
		☐ Temporarily Abandon	☐ Reclamation
		☒ Other	NDAC 43-02-03-49 Compliance

Well Name and Number See Attached					
Footages		Qtr-Qtr	Section	Township	Range
F	L	F	L	N	W
Field		Pool		County	

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Continental Resources, Inc. has inspected all the sites listed on the attached list and all are in compliance with NDAC 43-02-03-49. All sites are in compliance based on the fact that our oil storage tanks, flow-through process vessels, recycle pumps, and load lines are all located within secondary containment.

confirmed per RSD

Company Continental Resources, Inc.		Telephone Number (405) 234-9020	
Address PO Box 268870			
City Oklahoma City		State OK	Zip Code 73126
Signature <i>Robert Sandbo</i>	Printed Name Robert Sandbo		
Title Regulatory Compliance Supervisor	Date February 8, 2018		
Email Address robert.sandbo@clr.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 3/22/18	By <i>[Signature]</i>
Title <i>[Signature]</i>	



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas

August 29, 2017

CONTINENTAL RESOURCES
ATTENTION: BOB SANDBO
P.O. BOX 268870
OKLAHOMA CITY, OK 73126

RE: North Dakota Administrative Code (NDAC) Section 43-02-03-49
Perimeter Berm Requirement

NDIC # Please see attached list of 103 Facilities

Dear Bob Sandbo:

Please be advised that the attached list of facilities require a perimeter berm to be constructed within 180 days of this notice because they have:

1. Storage tanks;
2. Daily throughput of more than one hundred barrels of fluid per day; and
3. Includes production equipment or load lines that are not contained within secondary containment dikes

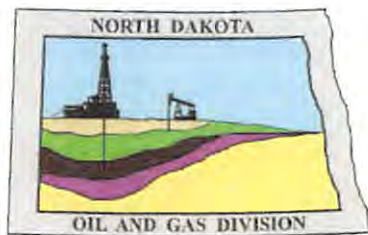
The berm must be at least six inches in height, constructed of sufficiently impermeable material to provide emergency containment, and must be maintained until the facility is either granted a waiver or the site is reclaimed.

Pursuant to NDAC Section 43-02-03-49 - Within one hundred eighty days from the date the operator is notified by the Commission, a perimeter berm, at least six inches in height, must be constructed of sufficiently impermeable material to provide emergency containment and to divert surface drainage away from the site around all storage facilities and production sites that include storage tanks, have a daily throughput of more than one hundred barrels of fluid per day, and include production equipment or load lines that are not contained within secondary containment dikes. The Director may consider an extension of time to implement these requirements if conditions prevent timely construction, or modification of these requirements if other factors are present that provide sufficient protection from environmental impacts.

This perimeter berm requirement may be modified or waived if the operator can demonstrate that other factors are present that provide sufficient protection from environmental impacts. A Sundry Notice (Form 4) outlining any engineering controls or other factors must be submitted to the Commission for approval of this modification or waiver. Should you have any questions regarding this matter, feel free to contact me at 701-770-3554.

Sincerely,


Richard Dunn
Field Inspector



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas

NDIC #	Facility Name	TB/CTB #	NDIC #	Facility Name	TB/CTB #
19126	LANSING 1-25H		24834	DURHAM 3X-2H	
19578	MISSOULA 1-21H		24837	WAHPETON 2-16H2	
19740	SYRACUSE 1-23H		24840	WAHPETON 4-16H1	
19858	JAMESTOWN 1-17H		24842	WAHPETON 5-16H2	
19915	STEELE 1-24H		24843	WAHPETON 6-16H	
19918	CHARLOTTE 1-22H		24844	WAHPETON 7-16H3	
20566	MONTPELIER 1-14H		24908	CHARLOTTE 6-22H2	
20629	PATTERSON 1-13H		25116	BJARNE 2-29H	
20638	KUHN 1-12H		25117	BJARNE 3-29H	
20676	NORFOLK 1-1H		25156	COLUMBUS FEDERAL 1-16H	
21128	CHARLOTTE 2-22H		25157	TALLAHASSEE 3-16H	
21511	BOULDER 1-4H		25158	TALLAHASSEE 2-16H	
21600	ROCHESTER 1-24H		25159	COLUMBUS FEDERAL 2-16H	
22155	LANSING 2-25H		25160	COLUMBUS FEDERAL 3-16H	
22158	KUHN 2-12H		25189	PATTERSON FEDERAL 2-13H	
22273	STEELE 2-24H		25190	PATTERSON FEDERAL 3-13H	
22375	CHICAGO 2-26H		25826	AKRON 6-34H1	
22891	MONROE 1-2H		25827	AKRON 5-34H1	
23048	CHICAGO 3-26H		26190	MONTPELIER 4-14H	
23049	CHICAGO 4-26H		26191	MONTPELIER 3-14H1	
23050	SYRACUSE 3-23H		26420	MONTPELIER 2-14H	
23051	SYRACUSE 4-23H		26476	ROCHESTER FEDERAL 6-24H	
23086	NORFOLK 2-1H		26477	ROCHESTER FEDERAL 7-24H1	
23087	NORFOLK 3-1H		26525	JERRY 2-8H	
23351	MISSOULA 2-21H		26526	JERRY 3-8H	
23352	MISSOULA 3-21H		26530	JERRY 5-8H	
23427	MISSOULA 7-21H		26531	JERRY 4-8H	
23428	MISSOULA 6-21H		26535	JERRY 7-8H	
23429	MISSOULA 5-21H		26536	JERRY 6-8H	
23430	MISSOULA 4-21H		27418	GARFIELD FEDERAL 7-5H1	
23477	DURHAM 2-2H		27419	GARFIELD FEDERAL 6-5H	
23608	CHARLOTTE 5-22H		27420	GARFIELD FEDERAL 5-5H1	
23609	AKRON 3-27AH		27421	GARFIELD 4-5H	
23610	AKRON 2-27AH		27694	BERLAIN 3-30H	
23611	AKRON 4-34H		27695	BERLAIN 2-30H	
23612	CHARLOTTE 4-22H		28202	JAMESTOWN FEDERAL 2-17H	
23664	CHARLOTTE 3-22H		28203	JAMESTOWN FEDERAL 3-17H1	
23747	ROCHESTER 3-24H		28405	HARRISBURG 1-34H	
23748	ROCHESTER 2-24H		28604	JAMESTOWN FEDERAL 6-17H	
23749	ROCHESTER 5-24H1		28605	JAMESTOWN FEDERAL 7-17H	
23750	ROCHESTER 4-24H		28735	DURHAM 7-2H	
24490	DURHAM 4-2H		28736	DURHAM 6-2H1	
24491	DURHAM 5-2H		28737	UHLMAN 1-7H	
24507	NORFOLK 5-1H		28999	NORFOLK 6-1H1	
24508	NORFOLK 4-1H		29000	NORFOLK 7-1H	
24804	WAHPETON 14-16H2	224837-01	31508	AKRON FEDERAL 7-27H	
24805	WAHPETON 13-16H		31838	CHARLOTTE 7X-22H	
24806	WAHPETON 12-16H3		32033	NORFOLK 11-1H	
24807	WAHPETON 11-16H1		32034	NORFOLK 10-1H1	
24808	WAHPETON 10-16H2		32035	NORFOLK 9-1H	
24809	WAHPETON 9-16H		32036	NORFOLK 8-1H1	
24810	WAHPETON 8-16H1				



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas/

September 19, 2017

CONTINENTAL RESOURCES
ATTENTION: BOB SANDBO
P.O. BOX 268870
OKLAHOMA CITY, OK 73126

RE:

MONTPELIER 1-14H
NENE 14-153N-101W
MCKENZIE COUNTY
WELL FILE NO.: 20566

VADER 1-7H
LOT 3 7-153N-100W
MCKENZIE COUNTY
WELL FILE NO.: 21696

COLUMBUS FEDERAL 3-16H
SENE 16-153N-101W
MCKENZIE COUNTY
WELL FILE NO.: 25160

Dear Bob Sandbo:

A Sundry notice (Form 4) is needed for the above wells, detailing the changeover from flowing to well now on rod pump. If you have any questions, feel free to contact our office.

Sincerely,

Tom Delling
Petroleum Engineer - Field Inspector

TKD/RSD/RLR

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

*Received**OCT 05 2016**ND Oil & Gas Division*

Well File No.

25160

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent	Approximate Start Date
☒ Report of Work Done	Date Work Completed July 31, 2016
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☒ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☐ Other	

Well Name and Number Columbus Federal 3-16H					
Footages 2469 F N L 199 F E L		Qtr-Qtr SENE	Section 16	Township 153 N	Range 101 W
Field Baker	Pool Bakken	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	85 Bbls	Oil	86 Bbls
Water	83 Bbls	Water	153 Bbls
Gas	93 MCF	Gas	25 MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Continental Resources, Inc. requests a change in production method on the above mentioned well. The well went from flowing to Rod Pump on 7/31/2016. New Tubing: 2 1/2 Depth: 9871

Company Continental Resources		Telephone Number (405) 234-9688	
Address P.O. Box 268870			
City Oklahoma City		State OK	Zip Code 73126
Signature 	Printed Name Zach Green		
Title Regulatory Specialist	Date September 30, 2016		
Email Address Zach.Green@clr.com			

FOR STATE USE ONLY

☒ Received	☐ Approved
Date 11-2-2016	
By 	
Title TAYLOR ROTH Engineering Technician	



AUTHORIZATION TO PURCHASE AND TRANSPORT OIL FROM LEASE - FORM 8
INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5698 (03-2000)



Well File No.
25160
NDIC CTB No.
<i>125160</i>

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND FOUR COPIES.

Well Name and Number Columbus Federal 3-16H	Qtr-Qtr SENE	Section 16	Township 153 N	Range 101 W	County McKenzie
Operator Continental Resources Inc.	Telephone Number 405-234-9000		Field Baker		
Address P.O. Box 268870	City Oklahoma City			State OK	Zip Code 73126

Name of First Purchaser Continental Resources, Inc.	Telephone Number 405-234-9000	% Purchased 100	Date Effective January 2, 2014
Principal Place of Business P.O. Box 268870	City Oklahoma City	State OK	Zip Code 73126
Field Address	City	State	Zip Code
Name of Transporter Hiland Crude, LLC	Telephone Number 580-616-2053	% Transported	Date Effective January 2, 2014
Address P.O. Box 3886	City Enid	State OK	Zip Code 73702
The above named producer authorizes the above named purchaser to purchase the percentage of oil stated above which is produced from the lease designated above until further notice. The oil will be transported by the above named transporter.			

Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
Other Transporters Transporting From This Lease	% Transported	Date Effective
Other Transporters Transporting From This Lease	% Transported	Date Effective
Comments		

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.			Date January 7, 2014
Signature <i>Becky Barnes</i>	Printed Name Becky Barnes	Title Regulatory Compliance Specialist	
Above Signature Witnessed By			
Witness Signature <i>Christi Scritchfield</i>	Witness Printed Name Christi Scritchfield	Witness Title Regulatory Compliance Specialist	

FOR STATE USE ONLY	
Date Approved	MAY 27 2014
By <i>Annette Trenga</i>	
Title	Oil & Gas Production Analyst

**WELL COMPLETION OR RECOMPLETION REPORT - FORM 6**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 2468 (04-2010)

Well File No.
25160



PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Designate Type of Completion			
☒ Oil Well	☐ EOR Well	☐ Recompletion	☐ Deepened Well
☐ Gas Well	☐ SWD Well	☐ Water Supply Well	☐ Other:
Well Name and Number Columbus Federal 3-16H		Spacing Unit Description Sec 4 & 16-153N-101W See 4, 9, 16, & 21	
Operator Continental Resources, Inc.		Telephone Number 405-234-9000	
Address P.O. Box 268870		Field Baker	
City Oklahoma City		Pool Bakken	
State Ok		Zip Code 73126	
Permit Type ☐ Wildcat ☒ Development ☐ Extension			

LOCATION OF WELL

At Surface 2469 F N L	199 F E L	Qtr-Qtr SENE	Section 16	Township 153 N	Range 101 W	County McKenzie
Spud Date 8/10/2013	Date TD Reached 12/27/2013	Drilling Contractor and Rig Number Cyclone 4		KB Elevation (Ft) 1940	Graded Elevation (Ft) 1920	
Type of Electric and Other Logs Run (See Instructions) CBL/GR, Mud Logs, Geological Report						

CASING & TUBULARS RECORD (Report all strings set in well)

Well Bore	String Type	Size (Inch)	Top Set (MD Ft)	Depth Set (MD Ft)	Hole Size (Inch)	Weight (Lbs/Ft)	Anchor Set (MD Ft)	Packer Set (MD Ft)	Sacks Cement	Top of Cement
Lateral1	Conductor	16		101	20				8 yds	
	Surface	9 5/8		1975	13 1/2	36			1762	
	Intermediate	7		10968	8 3/4	29-32			1109	1680
	Liner	4 1/2	9999	22933	6	11.6				
	Tubing	2 7/8			4	6.5				

PERFORATION & OPEN HOLE INTERVALS

Well Bore	Well Bore TD Drillers Depth (MD Ft)	Completion Type	Open Hole/Perforated Interval (MD,Ft)		Kick-off Point (MD Ft)	Top of Casing Window (MD Ft)	Date Perf'd or Drilled	Date Isolated	Isolation Method	Sacks Cement
Lateral1	23122	Perforations	10968	23122	10070	1680	12/29/2013			

PRODUCTION

Current Producing Open Hole or Perforated Interval(s), This Completion, Top and Bottom, (MD Ft) Bakken 10,968'/23,122'						Name of Zone (If Different from Pool Name)				
Date Well Completed (SEE INSTRUCTIONS) 5/3/2014			Producing Method Flowing		Pumping-Size & Type of Pump			Well Status (Producing or Shut-In) Producing		
Date of Test 5/4/2014	Hours Tested 24	Choke Size 18 /64	Production for Test		Oil (Bbls) 749	Gas (MCF) 557	Water (Bbls) 662	Oil Gravity-API (Corr.) 40.0 °	Disposition of Gas Sold	
Flowing Tubing Pressure (PSI) 2300		Flowing Casing Pressure (PSI) N/A		Calculated 24-Hour Rate	Oil (Bbls) 749	Gas (MCF) 557	Water (Bbls) 662	Gas-Oil Ratio 744		

PLUG BACK INFORMATION

Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								

Well Specific Stimulations

Date Stimulated 3/2/2014	Stimulated Formation Bakken	Top (Ft) 10968	Bottom (Ft) 23122	Stimulation Stages 41	Volume 74589	Volume Units Barrels
Type Treatment Sand Frac	Acid %	Lbs Proppant 3788746	Maximum Treatment Pressure (PSI) 7979		Maximum Treatment Rate (BBLS/Min) 31.0	
Details Pumped 285982# 40/70 mesh, 2418523# 20/40 sand and 1084241# 20/40 ceramic.						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

ADDITIONAL INFORMATION AND/OR LIST OF ATTACHMENTS

--	--	--

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.		Email Address becky.barnes@clr.com	Date 4/23/2014
Signature 		Printed Name Becky Barnes	Title Regulatory Compliance Specialist

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SYNOPSIS

OPERATOR	Continental Resources Inc.
WELL NAME	Columbus Federal 3-16H
SURFACE LOCATION	2469' FNL & 199' FEL: Sec 16, T153N, R101W
BOTTOM HOLE LOCATION	244' FNL & 680.4' FEL: Sec 4, T153N, R101W
FIELD	Williston
COUNTY/STATE	McKenzie Co., North Dakota
API NUMBER	33-053-04856
NorAm FILE NUMBER	NR2690
ELEVATIONS	G.L. = 1920' K.B. = 1940'
SPUD DATE	October 16, 2013
T.D. DATE	December 27, 2013
DRILLING CONTRACTOR	Cyclone 4
HOLE SIZE	10 3/4" to 1975', 8 3/4" to 10983, 6" to 23122'
CASING SIZE & DEPTH	9 5/8" to 1975', 7" to 10967, 5" to
DRILLING MUD COMPANY	GEO Drilling Fluids Inc
DRILLING FLUID TYPE	Invert OBM to 10983', Saltwater Brine to 23122'
DIRECTIONAL COMPANY	Leam Energy Services
WIRELINE LOGGING COMPANY	N/A
LOG RECORD	NorAm Wellsite Services
DRILLING SUPERVISION	Steve Northern, Biggens, Monty Harris
GEOLOGICAL SUPERVISION	Darcy Klessens, NorAm Wellsite Services
MUDLOGGING COMPANY	NorAm Wellsite Services
TOTAL DEPTH	23122'

FORMATION TOPS (ft)

KB =1940'

GL =1920'

FORMATION	PROGNOSIS			SAMPLES		
	MD	TVD	SS	MD	TVD	SS
Top Charles Salts	-	8329	-6389	8299	8298	-6358
Base Last Salt	-	9025	-7085	8998	8997	-7057
Mission Canyon	-	9250	-7310	9222	9221	-7279
Lodgepole	-	9817	-7877	9788	9787	-7847
False Bakken	-	-	-	10599	10483	-8543
Upper Bakken Shale	-	10505	-8565	10621	10494	-8554
Middle Bakken	-	10522	-8582	10672	10506	-8566
7" Casing Point	-	10542	-8602	10967	10524	8584
End of Lateral	-	10542	-8602	23122	10518	-8578

DEVIATION SURVEY RECORD (Teledrift and Wireline Data)

Depth	Incl	Azim	TVD	VS	Coordinates		DLS
(ft)	(?)	(?)	(ft)	(ft)	N/S (ft)	E/W (ft)	(#/100')
1928	0.10	221.00	1927.98	5.38	5.47	2.38	0.00
1990	0.10	213.40	1989.98	5.30	5.38	2.31	0.02
2084	0.80	242.80	2083.98	4.95	5.02	1.69	0.76
2177	1.10	251.10	2176.96	4.42	4.43	0.26	0.35
2271	1.10	253.20	2270.95	3.93	3.88	-1.45	0.04
2364	1.90	247.50	2363.91	3.16	3.03	-3.73	0.87
2457	2.10	248.20	2456.86	2.05	1.81	-6.74	0.22
2550	2.30	248.90	2549.79	0.87	0.50	-10.06	0.22
2645	1.80	236.60	2644.73	-0.53	-1.01	-13.09	0.70
2739	1.20	222.60	2738.70	-2.00	-2.54	-14.98	0.74
2832	1.30	224.50	2831.67	-3.42	-4.01	-16.38	0.12
2925	0.70	277.80	2924.66	-4.04	-4.69	-17.69	1.12
3019	0.90	343.00	3018.65	-3.23	-3.90	-18.47	0.93
3112	0.90	341.60	3111.64	-1.82	-2.51	-18.91	0.02
3205	0.80	346.10	3204.63	-0.49	-1.19	-19.30	0.13
3298	0.50	357.00	3297.63	0.55	-0.15	-19.48	0.35
3391	0.30	355.80	3390.62	1.20	0.49	-19.52	0.22
3484	0.20	22.00	3483.62	1.59	0.89	-19.47	0.16
3578	0.80	49.30	3577.62	2.15	1.47	-18.92	0.67
3671	0.50	67.40	3670.61	2.70	2.05	-18.05	0.39
3765	0.60	87.40	3764.61	2.85	2.23	-17.18	0.23
3858	0.30	72.80	3857.60	2.92	2.32	-16.46	0.34
3952	0.40	64.90	3951.60	3.11	2.53	-15.93	0.12
4045	0.20	51.20	4044.60	3.33	2.77	-15.51	0.23
4139	0.20	86.90	4138.60	3.43	2.88	-15.21	0.13
4232	0.30	148.80	4231.60	3.22	2.68	-14.93	0.29
4326	0.30	161.80	4325.60	2.77	2.24	-14.72	0.07
4419	0.30	142.20	4418.60	2.34	1.82	-14.50	0.11
4512	0.40	181.80	4511.60	1.82	1.30	-14.36	0.27
4606	0.60	201.70	4605.59	1.04	0.51	-14.55	0.28
4699	1.10	199.40	4698.58	-0.23	-0.78	-15.03	0.54
4792	0.90	190.10	4791.57	-1.78	-2.34	-15.45	0.28
4886	0.40	148.60	4885.56	-2.79	-3.35	-15.41	0.70
4979	0.30	38.50	4978.56	-2.88	-3.43	-15.09	0.62
5073	1.20	29.00	5072.55	-1.85	-2.38	-14.46	0.96
5166	1.00	120.30	5165.54	-1.46	-1.94	-13.29	1.70
5260	0.80	55.80	5259.53	-1.55	-1.98	-12.04	1.04
5354	0.90	65.90	5353.52	-0.92	-1.31	-10.82	0.19
5447	1.10	63.50	5446.51	-0.28	-0.62	-9.35	0.22
5540	1.30	59.80	5539.49	0.59	0.31	-7.64	0.23
5634	1.10	56.60	5633.47	1.56	1.34	-5.97	0.22

Depth	Incl	Azim	TVD	VS	Coordinates		DLS
(ft)	(°)	(°)	(ft)	(ft)	N/S (ft)	E/W (ft)	(°/100')
5727	1.90	58.80	5726.43	2.77	2.64	-3.90	0.86
5821	1.90	55.60	5820.38	4.37	4.32	-1.28	0.11
5914	2.00	55.40	5913.33	6.06	6.12	1.32	0.11
6007	1.80	55.10	6006.28	7.73	7.87	3.86	0.22
6101	1.80	49.80	6100.23	9.44	9.67	6.20	0.18
6194	1.00	53.10	6193.20	10.80	11.10	7.96	0.86
6288	0.60	83.70	6287.19	11.31	11.65	9.11	0.61
6382	0.80	98.50	6381.19	11.22	11.60	10.24	0.29
6476	1.10	98.30	6475.17	10.94	11.38	11.79	0.32
6569	0.40	83.40	6568.16	10.81	11.28	12.99	0.78
6662	0.60	3.90	6661.16	11.32	11.81	13.35	0.71
6756	1.10	355.50	6755.15	12.71	13.20	13.31	0.55
6850	1.10	333.00	6849.13	14.43	14.90	12.83	0.46
6943	1.50	341.00	6942.11	16.40	16.85	12.03	0.47
7037	1.70	327.90	7036.07	18.78	19.19	10.89	0.44
7130	2.10	323.50	7129.02	21.38	21.73	9.14	0.46
7223	1.60	295.90	7221.98	23.40	23.67	6.96	1.08
7316	1.30	264.90	7314.95	23.95	24.14	4.74	0.89
7411	1.50	258.80	7409.92	23.70	23.80	2.45	0.26
7504	1.90	264.80	7502.88	23.42	23.43	-0.28	0.47
7597	1.90	262.80	7595.83	23.20	23.09	-3.35	0.07
7690	1.80	245.60	7688.78	22.51	22.30	-6.21	0.60
7784	1.10	237.30	7782.75	21.49	21.20	-8.31	0.78
7877	0.80	208.50	7875.74	20.47	20.15	-9.37	0.60
7970	1.10	199.00	7968.72	19.08	18.73	-9.97	0.36
8063	1.10	200.30	8061.71	17.42	17.05	-10.57	0.03
8156	1.10	209.70	8154.69	15.84	15.44	-11.33	0.19
8250	1.00	212.00	8248.67	14.39	13.96	-12.21	0.12
8343	1.10	227.00	8341.66	13.14	12.66	-13.29	0.31
8436	1.10	230.70	8434.64	12.01	11.49	-14.63	0.08
8529	1.00	213.60	8527.62	10.81	10.25	-15.77	0.35
8623	1.10	219.90	8621.61	9.48	8.87	-16.81	0.16
8716	1.00	226.80	8714.59	8.28	7.63	-17.97	0.17
8810	1.10	238.00	8808.58	7.29	6.59	-19.33	0.24
8903	1.30	248.20	8901.56	6.49	5.73	-21.07	0.31
8997	1.10	236.10	8995.54	5.65	4.83	-22.81	0.34
9090	1.20	233.60	9088.52	4.63	3.75	-24.33	0.12
9183	1.10	236.80	9181.50	3.62	2.69	-25.86	0.13
9277	0.90	213.40	9275.49	2.55	1.58	-27.03	0.48
9370	0.30	111.70	9368.48	1.86	0.88	-27.20	1.08
9464	0.70	59.10	9462.48	2.04	1.08	-26.48	0.61
9557	0.60	58.40	9555.47	2.55	1.63	-25.58	0.11

Depth	Incl	Azim	TVD	VS	Coordinates		DLS
(ft)	(°)	(°)	(ft)	(ft)	N/S (ft)	E/W (ft)	(?/100')
9650	0.70	60.30	9648.47	3.06	2.16	-24.67	0.11
9744	0.60	67.40	9742.46	3.50	2.64	-23.72	0.14
9837	0.60	67.00	9835.46	3.84	3.01	-22.82	0.00
9930	0.30	71.80	9928.45	4.08	3.28	-22.14	0.32
10024	0.30	82.50	10022.45	4.17	3.39	-21.66	0.06
10049	0.20	36.90	10047.45	4.21	3.43	-21.57	0.86
10081	2.70	330.70	10079.44	4.93	4.13	-21.91	8.21
10112	7.80	330.00	10110.30	7.44	6.59	-23.32	16.45
10143	12.50	333.50	10140.80	12.35	11.42	-25.87	15.28
10174	15.80	334.40	10170.86	19.28	18.23	-29.19	10.67
10205	18.50	333.10	10200.48	27.61	26.43	-33.24	8.80
10236	22.00	331.70	10229.56	37.29	35.93	-38.22	11.40
10267	25.10	330.50	10257.97	48.34	46.76	-44.21	10.12
10298	28.80	329.30	10285.60	60.73	58.91	-51.26	12.06
10330	32.40	329.10	10313.14	75.02	72.90	-59.60	11.25
10361	36.80	329.80	10338.65	90.49	88.06	-68.54	14.25
10392	41.80	329.80	10362.63	107.80	105.02	-78.42	16.13
10423	45.80	329.60	10385.00	126.70	123.55	-89.24	12.91
10454	50.00	331.20	10405.78	147.17	143.62	-100.43	14.08
10485	53.50	330.20	10424.97	168.81	164.84	-112.35	11.57
10516	58.10	329.50	10442.39	191.43	187.01	-125.22	14.96
10547	60.90	329.80	10458.13	214.95	210.05	-138.72	9.07
10578	63.40	330.90	10472.61	239.25	233.87	-152.27	8.65
10610	66.40	332.30	10486.18	265.22	259.36	-166.05	10.18
10641	71.10	333.00	10497.41	291.34	285.02	-179.32	15.31
10672	76.20	333.30	10506.14	318.34	311.55	-192.75	16.48
10703	79.30	333.70	10512.71	345.92	338.66	-206.26	10.08
10734	80.90	333.50	10518.04	373.75	366.01	-219.84	5.20
10765	84.40	333.50	10522.01	401.74	393.52	-233.56	11.29
10797	89.80	333.10	10523.63	430.78	422.06	-247.91	16.92
10828	91.00	333.00	10523.41	458.91	449.69	-261.96	3.88
10859	91.40	333.10	10522.76	487.02	477.32	-276.01	1.33
10890	89.40	333.70	10522.54	515.23	505.04	-289.89	6.74
10935	88.70	333.70	10523.29	556.26	545.37	-309.82	1.56
10969	88.60	333.30	10524.09	587.21	575.79	-324.99	1.21
11064	89.30	332.80	10525.83	673.38	660.46	-368.04	0.91
11159	90.00	335.60	10526.41	760.35	745.98	-409.38	3.04
11253	90.20	338.60	10526.25	848.20	832.56	-445.95	3.20
11348	90.70	342.80	10525.50	938.92	922.20	-477.34	4.45
11443	90.70	345.60	10524.34	1031.19	1013.60	-503.20	2.95
11538	90.70	348.60	10523.18	1124.48	1106.18	-524.41	3.16
11633	90.00	350.90	10522.60	1218.51	1199.66	-541.31	2.53

Depth	Incl	Azim	TVD	VS	Coordinates		DLS
(ft)	(°)	(°)	(ft)	(ft)	N/S (ft)	E/W (ft)	(?/100')
11728	90.00	353.40	10522.60	1313.02	1293.76	-554.29	2.63
11823	90.10	356.20	10522.52	1407.87	1388.36	-562.89	2.95
11918	89.50	357.90	10522.85	1502.86	1483.23	-567.78	1.90
12013	87.70	0.00	10525.17	1597.80	1578.17	-569.52	2.91
12108	88.40	359.30	10528.40	1692.70	1673.12	-570.10	1.04
12203	89.20	359.30	10530.39	1787.66	1768.09	-571.26	0.84
12298	91.10	359.90	10530.14	1882.61	1863.08	-571.93	2.10
12393	90.00	1.60	10529.23	1977.48	1958.06	-570.68	2.13
12488	87.70	1.80	10531.14	2072.25	2052.99	-567.87	2.43
12583	87.80	0.00	10534.87	2167.05	2147.91	-566.38	1.90
12678	88.80	359.50	10537.69	2261.96	2242.86	-566.79	1.18
12773	89.90	359.20	10538.76	2356.92	2337.85	-567.87	1.20
12867	88.20	0.70	10540.32	2450.84	2431.83	-567.95	2.41
12960	89.30	1.10	10542.35	2543.69	2524.79	-566.49	1.26
13051	90.00	0.70	10542.91	2634.56	2615.78	-565.06	0.89
13143	88.00	2.00	10544.51	2726.38	2707.73	-562.89	2.59
13235	88.00	1.80	10547.72	2818.10	2799.63	-559.85	0.22
13327	88.60	1.60	10550.45	2909.86	2891.55	-557.12	0.69
13420	88.70	1.60	10552.64	3002.64	2984.48	-554.52	0.11
13511	90.00	1.60	10553.68	3093.45	3075.44	-551.98	1.43
13603	90.70	1.30	10553.11	3185.27	3167.41	-549.65	0.83
13695	90.30	0.60	10552.31	3277.14	3259.39	-548.13	0.88
13787	91.50	1.60	10550.87	3368.98	3351.36	-546.36	1.70
13880	90.00	0.60	10549.65	3461.83	3444.33	-544.58	1.94
13972	89.70	359.50	10549.89	3553.76	3536.33	-544.50	1.24
14064	89.60	358.80	10550.45	3645.74	3628.32	-545.86	0.77
14156	90.70	359.90	10550.21	3737.71	3720.31	-546.90	1.69
14249	91.00	358.50	10548.83	3830.67	3813.29	-548.20	1.54
14341	90.20	358.10	10547.87	3922.66	3905.24	-550.93	0.97
14433	91.10	358.80	10546.82	4014.65	3997.20	-553.42	1.24
14526	91.50	358.30	10544.71	4107.62	4090.14	-555.77	0.69
14619	89.90	357.00	10543.58	4200.61	4183.05	-559.59	2.22
14712	91.50	359.20	10542.44	4293.59	4275.99	-562.67	2.92
14805	90.80	359.00	10540.57	4386.55	4368.96	-564.13	0.78
14897	92.20	359.20	10538.17	4478.50	4460.91	-565.57	1.54
14990	90.50	0.90	10535.98	4571.40	4553.88	-565.49	2.58
15082	91.20	0.60	10534.61	4663.28	4645.86	-564.29	0.83
15174	89.60	1.60	10533.97	4755.13	4737.84	-562.52	2.05
15267	90.00	0.70	10534.29	4847.98	4830.82	-560.66	1.06
15360	90.50	0.90	10533.89	4940.86	4923.81	-559.36	0.58
15453	91.50	0.20	10532.26	5033.75	5016.79	-558.46	1.31
15545	89.80	2.80	10531.22	5125.55	5108.74	-556.06	3.38

Depth	Incl	Azim	TVD	VS	Coordinates		DLS
(ft)	(°)	(°)	(ft)	(ft)	N/S (ft)	E/W (ft)	(°/100')
15640	89.90	2.50	10531.47	5220.23	5203.63	-551.66	0.33
15735	90.70	1.40	10530.97	5314.99	5298.58	-548.43	1.43
15830	90.10	1.60	10530.31	5409.80	5393.54	-545.95	0.67
15925	91.70	3.20	10528.82	5504.49	5488.44	-541.97	2.38
16020	88.90	2.30	10528.32	5599.15	5583.32	-537.41	3.10
16115	90.70	2.80	10528.65	5693.83	5678.22	-533.18	1.97
16210	89.30	3.40	10528.65	5788.44	5773.08	-528.05	1.60
16305	90.30	4.40	10528.98	5882.92	5867.86	-521.59	1.49
16400	87.50	1.10	10530.81	5977.54	5962.71	-517.03	4.55
16495	88.40	1.10	10534.20	6072.33	6057.63	-515.21	0.95
16590	90.00	0.90	10535.53	6167.18	6152.60	-513.55	1.70
16685	91.10	0.20	10534.62	6262.08	6247.59	-512.64	1.37
16780	90.30	0.40	10533.46	6356.99	6342.58	-512.14	0.87
16874	91.10	1.10	10532.31	6450.87	6436.56	-510.91	1.13
16971	92.30	0.40	10529.43	6547.70	6533.51	-509.64	1.43
17066	91.70	0.70	10526.12	6642.54	6628.45	-508.73	0.71
17161	91.40	0.70	10523.55	6737.40	6723.40	-507.57	0.32
17256	91.40	0.40	10521.23	6832.27	6818.37	-506.66	0.32
17351	91.50	359.50	10518.82	6927.18	6913.34	-506.74	0.95
17446	91.70	359.70	10516.17	7022.10	7008.30	-507.40	0.30
17541	92.00	359.50	10513.10	7117.01	7103.25	-508.06	0.38
17636	91.70	358.80	10510.04	7211.94	7198.19	-509.47	0.80
17731	91.50	359.30	10507.38	7306.88	7293.14	-511.05	0.57
17826	90.00	358.60	10506.14	7401.85	7388.11	-512.79	1.74
17921	90.30	358.30	10505.89	7496.85	7483.08	-515.36	0.45
18016	89.50	359.30	10506.06	7591.84	7578.05	-517.35	1.35
18111	89.80	358.80	10506.64	7686.82	7673.04	-518.92	0.61
18206	89.90	358.10	10506.89	7781.81	7768.00	-521.49	0.74
18301	91.40	357.60	10505.81	7876.80	7862.93	-525.06	1.66
18396	90.20	357.60	10504.48	7971.79	7957.83	-529.03	1.26
18491	90.20	356.90	10504.15	8066.78	8052.72	-533.59	0.74
18586	89.20	358.30	10504.65	8161.78	8147.63	-537.57	1.81
18681	89.40	357.60	10505.81	8256.77	8242.56	-540.97	0.77
18776	89.60	359.30	10506.64	8351.76	8337.52	-543.54	1.80
18871	89.30	358.50	10507.55	8446.74	8432.50	-545.36	0.90
18966	89.90	358.30	10508.21	8541.73	8527.46	-548.01	0.67
19061	90.00	357.40	10508.30	8636.73	8622.39	-551.58	0.95
19156	90.30	357.40	10508.05	8731.73	8717.29	-555.89	0.32
19251	90.70	357.00	10507.22	8826.71	8812.18	-560.53	0.60
19346	90.70	358.30	10506.06	8921.70	8907.09	-564.42	1.37
19440	90.60	359.50	10504.99	9015.68	9001.06	-566.23	1.28
19535	89.30	359.50	10505.07	9110.64	9096.06	-567.06	1.37

Depth	Incl	Azim	TVD	VS	Coordinates		DLS
(ft)	(°)	(°)	(ft)	(ft)	N/S (ft)	E/W (ft)	(°/100')
19630	89.9	359.2	10505.737	9205.612	9191.05	-568.133	0.70612
19725	90	359.2	10505.819	9300.588	9286.04	-569.46	0.10526
19820	90.2	359.5	10505.654	9395.558	9381.03	-570.537	0.37953
19915	90.7	359.3	10504.907	9490.523	9476.02	-571.532	0.56686
20009	89.2	358.6	10504.99	9584.505	9570	-573.255	1.76094
20104	88.8	359.3	10506.648	9679.474	9664.97	-574.995	0.84856
20199	90.2	0.7	10507.477	9774.403	9759.96	-574.995	2.08404
20294	90.90	0.60	10506.56	9869.29	9854.95	-573.92	0.74
20389	88.50	0.70	10507.06	9964.17	9949.94	-572.84	2.53
20484	90.00	1.60	10508.31	10059.01	10044.91	-570.93	1.84
20579	91.00	1.40	10507.48	10153.82	10139.87	-568.45	1.07
20674	90.2	2.1	10506.482	10248.6	10234.8	-565.546	1.11893
20769	90.7	1.6	10505.736	10343.38	10329.8	-562.479	0.74431
20864	88.9	1.6	10506.067	10438.18	10424.7	-559.826	1.89474
20959	89.3	1.3	10507.559	10532.98	10519.7	-557.423	0.52629
21054	91.2	2.8	10507.145	10627.73	10614.6	-554.025	2.5481
21150	89.40	3.00	10506.64	10723.36	10710.49	-549.17	1.89
21244	89.00	2.70	10507.95	10817.00	10804.36	-544.49	0.53
21339	89.40	2.50	10509.28	10911.68	10899.25	-540.19	0.47
21434	90.00	2.30	10509.78	11006.39	10994.17	-536.21	0.67
21529	90.70	2.10	10509.20	11101.12	11089.09	-532.56	0.77
21529	90.70	2.10	10509.20	11101.12	11089.09	-532.56	0.77
21624	88.20	2.70	10510.11	11195.82	11184.00	-528.58	2.71
21719	88.30	2.00	10513.01	11290.49	11278.87	-524.69	0.74
21813	88.40	1.80	10515.72	11384.22	11372.78	-521.57	0.24
21905	89.50	2.80	10517.40	11475.93	11464.69	-517.88	1.62
22000	89.70	2.80	10518.07	11570.59	11559.58	-513.24	0.21
22095	90.00	2.50	10518.32	11665.26	11654.47	-508.85	0.45
22190	89.70	1.80	10518.56	11760.00	11749.41	-505.29	0.80
22285	90.40	1.60	10518.48	11854.80	11844.36	-502.47	0.77
22380	90.10	0.90	10518.07	11949.63	11939.34	-500.40	0.80
22475	90.50	0.90	10517.57	12044.50	12034.33	-498.90	0.42
22570	90.30	2.80	10516.91	12139.27	12129.27	-495.84	2.01
22665	89.50	2.50	10517.07	12233.95	12224.17	-491.44	0.90
22760	89.90	2.70	10517.57	12328.63	12319.07	-487.14	0.47
22855	90.50	2.30	10517.24	12423.33	12413.98	-482.99	0.76
22949	90.00	1.40	10516.83	12517.10	12507.92	-479.96	1.10
23044	89.30	0.60	10517.41	12611.96	12602.91	-478.30	1.12
23072	89.40	0.70	10517.73	12639.93	12630.90	-477.98	0.51
23122	89.40	0.70	10518.25	12689.87	12680.90	-477.37	0.00

BIT RECORD

BIT #	TYPE	SIZE (IN)	IN (FT)	OUT (FT)	TOTAL (FT)	HOURS DRLG	WOB (K)	RPM
1	PDC MM65D	8.75	1975	10075	8100	77.5	20-30	50-70
2	Halliburton MMD55M	8.75	10075	10983	913	22.5	30-90	15-25
3	Reed SKHE711M	6	10983	17000	6017	83	15-25	50-60
4	Reed SKHI713M	6	17000	21862	4862	87.5	12-25	45-50
5	Smith MDSi613	6	21862	23122	1260	18.5	20-25	40-45

DRILLING FLUID PARAMETERS

DATE	WT	VIS	PV	YP	PH	NaCL	Ca	% SOL
10/18/2013	9.3	59	12	8	-	37k	2.73	8.83
10/19/2013	9.6	47	9	10	-	36k	4.03	8.91
10/20/2013	9.6	46	10	8	-	35k	3.2	8.94
10/21/2013	10.5	47	12	12	-	44k	3.4	12.79
10/22/2013	10.2	46	13	11	-	44k	3.4	11.53
12/17/2013	9.6	30	1	1	8.5	148,800	30,500	0.66
12/18/2013	9.6	30	1	1	8.5	148,800	30,500	0.66
12/19/2013	9.6	30	1	1	8.5	148,800	30,500	0.66
12/20/2013	9.85	29	1	1	9.0	172,000	32,000	0.39
12/21/2013	9.85	29	1	1	9.0	172,000	32,000	0.39
12/22/2013	9.9	31	1	1	8.5	175,000	33,600	0.49
12/23/2013	9.9	31	1	1	8.5	175,000	33,600	0.49
12/24/2013	9.65	28	1	1	8.0	148,000	32,000	0.54
12/25/2013	10.0	29	1	1	8.5	188,700	33,600	0.54
12/26/2013	10.0	29	1	1	8.5	188,700	33,600	0.54
12/27/2013	10	29	1	1	8.5	188,700	33,600	0.54

DAILY DRILLING CHRONOLOGY

DATE (m/d/y)	DEPTH @ 24.00	PROGRESS (ft/24 hours)	BREAKDOWN 00:00 – 24:00	RIG ACTIVITY
10/16/2013	1895'		00:00-24:00	BOP Testing, Casing pressure Fit test, Safety meetings, Pick up drill pipe
10/17/2013	1975'	80'	00:00-20:00	Pick up Directional tools, pick up drill pipe, safety meetings, rig down laydown truck, drill out cement, float and shoe
			20:00-20:30	Fit test
			20:30-21:00	Drill/Slide 1975'-2130'
			21:00-23:00	Trouble Shoot MWD Tool
			23:00-24:00	Drill 2130'-2225'
10/18/2013	2225'	250'	00:00-01:00	Trouble shoot MWD Tool
			01:00-02:00	Drill 2225'-2411'
			02:00-03:30	TOOH for MWD Tool
			03:30-04:30	BHA Operations
			04:30-06:00	TIH
			06:00-17:00	Drill 2411'-5307'
			17:00-17:30	Rig Service
			17:30-24:00	Drill 5307'-6361'
10/19/2013	6361'	4136'	00:00-03:30	Drill 6361'-6709'
			03:30-04:00	Rig Service
			04:00-17:00	Drill/Slide 6709'-7924'
			17:00-17:30	Rig Service
			17:30-24:00	Drill/Slide 7924'-8356'
10/20/2013	8356'	1995'	00:00-03:30	Drill/Slide 8356'-8576'
			03:30-04:00	Rig Service
			04:00-16:00	Drill/Slide 8576'-9230'
		1	16:00-16:30	Rig Service
			16:30-24:00	Drill/Slide 9230'-9513'
10/21/2013	9513	1157	00:00-00:30	Rig Service
			00:30-13:00	Drill/Slide 9513'-10070'
			13:00-13:30	Circulate, Build Slug
			13:30-14:00	Rig Service
			14:00-20:30	TOOH for Curve Assembly
			20:30-22:00	BHA Operations
			22:00-22:30	Rig Service
			22:30-24:00	TIH
10/22/2013	10070'	557'	00:00-04:00	TIH
			04:00-17:00	Build Curve 10070'-10553'
			17:00-17:30	Rig Service
			17:30-24:00	Build Curve 10553'-10845'

DATE (m/d/y)	DEPTH @ 24.00	PROGRESS (ft/24 hours)	BREAKDOWN 00:00 – 24:00	RIG ACTIVITY
10/23/2013	10983'	913'	00:00-03:00	Build Curve 10845'-10983'
			03:00-03:30	Rig Service
			03:30-04:00	Circulate Bottoms Up
			04:00-05:00	Wiper Trip
			05:00-06:00	Circulate Bottoms Up
			06-11:30	TOOH for Intermediate Casing Operations
12/17/2013	10983'	0'	00:00-11:00	Skid Rig, BOP test, Cut Drilling Line, Directional Work-BHA
			11:00-17:30	TIH
			17:30-18:30	Drill 10983'-10993', FIT Test
			18:30-24:00	Drill/Slide 10993'-11364'
12/18/2013	11364'	381'	00:00-24:00	Drill/Slide 11364'-13454'
12/19/2013	13454'	2090'	00:00-24:00	Drill/Slide 13454'-15180'
12/20/2013	15180'	1726'	00:00-24:00	Drill/Slide 15180'-16588'
12/21/2013	16588'	1408'	00:00-08:30	Drill/Slide 16588'-17000'
			08:30-15:30	Circulate Bottoms Up, TOOH for new BHA
			15:30-16:30	Directional Work - BHA
			16:30-23:00	TIH
			23:00-24:00	Drill/Slide 17000'-17051'
12/22/2013	17051'	463'	00:00-24:00	Drill/Slide 17051'-18577'
12/23/2013	18577'	1523'	00:00-03:30	Drill/Slide 18577'-18781'
			03:30-04:00	Trouble Shoot MWD Tool
			04:00-24:00	Drill/Slide 18781'-20047'
12/24/2013	20047'	1470'	00:00-13:00	Drill/Slide 20047'-20627'
			13:00-14:00	Change out MWD Tool Communication Cable
			14:00-18:00	Drill/Slide 20627'-20743'
			18:00-20:30	Trouble Shoot MWD, Circulate
			20:30-21:00	Re-Log Gamma from 20722'-20743'
			21:00-24:00	Drill/Slide 20743'-20951'
12/25/2013	20951'	904'	00:00-23:30	Drill/Slide 20951'-21,862'
			23:30-24:00	Circulate, TOOH for new Motor
12/26/2013	21862'	911'	00:00-09:00	TOOH
			09:00-10:30	Directional Work-BHA
			10:30-20:30	TIH
			20:30-24:00	Reaming 18475'-19046'; Circulate Gas through Choke
12/27/2013	21862'	0'	00:00-03:00	Reaming 19046'-21800'
			03:00-03:30	Re-Log Gamma 21800'-21862'

DATE (m/d/y)	DEPTH @ 24.00	PROGRESS (ft/24 hours)	BREAKDOWN 00:00 – 24:00	RIG ACTIVITY
			03:30-23:00	Drill/Slide 21862'-23122'
			23:00-24:00	Circulate
12/28/2013	23122'	1260'	00:00-06:00	Circulate, Wiper Trip

LITHOLOGY

30' samples were caught in the vertical and curve by NorAm Wellsite Services, began at 8250'MD on October 19, 2013 at 22:50 hours. 50' samples were caught in the lateral by the Cyclone 4 Rig Crew on December 17th at 19:00 hours.

Formation tops
Sample Interval
(in feet)

SAMPLE DESCRIPTION

8220-8250	LIMESTONE: red orange medium gray brown, buff light to medium gray, firm to bright, microcrystalline, moderately silica, argillaceous, trace sandy material
8250-8280	SALT: clear, translucent trace LIMESTONE: light to medium gray, firm to bright, microcrystalline, moderately silica, argillaceous, trace sandy material

CHARLES: 8299' MD, 8298' TVD (-6358)

8280-8310	SALT: clear, translucent trace LIMESTONE: light to medium gray, firm to bright, microcrystalline, moderately silica, argillaceous, trace sandy material
8310-8400	SALT: clear, translucent; trace LIMESTONE: light to medium gray, firm to bright, microcrystalline, moderately silica, argillaceous, trace sandy material, trace siltstone material; trace ANHYDRITE: cream to buff
8400-8520	SALT: translucent to transparent, off white hard, crystalline; SILTSTONE: orange brown, soft, calcareous, argillaceous
8520-8550	SALT: clear, translucent; trace LIMESTONE: light to medium gray, firm to bright, microcrystalline, moderately silica, argillaceous, trace sandy material, trace siltstone material; trace ANHYDRITE: cream to buff
8550-8580	SALT: clear, translucent; trace LIMESTONE: light to medium gray, firm to bright, microcrystalline, moderately silica, argillaceous, trace sandy material, trace siltstone material; trace ANHYDRITE: cream, off white
8580-8640	LIMESTONE: light gray brown, light to medium gray, firm to bright, microcrystalline, moderately silica, argillaceous, ANHYDRITE: white, soft, amorphous; trace siltstone material; SALT: clear, translucent
8640-8670	LIMESTONE: light gray brown, light to medium gray, firm to bright, microcrystalline, moderately silica, argillaceous, ANHYDRITE: white, soft, amorphous; trace siltstone material; SALT: clear, translucent, hard,

Formation tops
Sample Interval
(in feet)

SAMPLE DESCRIPTION

- 8670-8760 LIMESTONE: light gray brown, light to medium gray, firm to bright, microcrystalline, moderately silty, argillaceous, ANHYDRITE: white, soft, amorphous; trace SILTSTONE: red to orange, friable to firm, calcareous, material; SALT: clear, translucent, hard
- 8760-8820 LIMESTONE: light gray brown, light to medium gray, firm to bright, microcrystalline, moderately silty, argillaceous, ANHYDRITE: white, soft, amorphous; SALT: clear, translucent, hard
- 8820-8880 SALT: clear, translucent, hard; LIMESTONE: light gray brown, light to medium gray, firm to bright, microcrystalline, moderately silty, argillaceous, ANHYDRITE: white to gray, soft, amorphous
- 8880-8910 ANHYDRITE: white to gray, soft, amorphous; LIMESTONE: light gray brown, light to medium gray, firm to bright, microcrystalline, moderately silty, argillaceous
- 8910-8940 LIMESTONE: light gray brown, light to medium gray, firm to bright, microcrystalline, moderately silty, argillaceous, ANHYDRITE: white to gray, soft, amorphous, trace SALT: clear, translucent, hard
- 8940-9000 SALT: clear, translucent, hard; trace LIMESTONE: light to medium gray, firm to hard, microcrystalline, moderately silty, trace argillaceous, mottled in part

BASE LAST SALT: 8998' MD, 8997' TVD (-7057)

- 9000-9030 SALT: clear, translucent, hard; trace LIMESTONE: light to medium gray, tan to light brown, firm to hard, microcrystalline, moderately silty, trace argillaceous, mottled in part
- 9030-9060 LIMESTONE: light to medium gray, light brown, tan, off white in part, firm to hard, microcrystalline, silty, trace argillaceous, mottled in part
- 9060-9210 LIMESTONE: light to medium gray, light brown, tan, off white in part, firm to hard, microcrystalline, silty, trace argillaceous, mottled in part, trace ANHYDRITE: white to gray, soft, amorphous

MISSION CANYON: 9222' MD, 9221' TVD (-7279)

Formation tops
Sample Interval
(in feet)

SAMPLE DESCRIPTION

9210-9300	LIMESTONE: light to medium gray, light brown, tan, off white in part, firm to hard, microcrystalline, silty, trace argillaceous, mottled in part
9300-9360	LIMESTONE: light to medium gray, light brown, tan, off white in part, firm to hard, microcrystalline, silty, trace argillaceous, mottled in part, trace ANHYDRITE: white to gray, soft, amorphous
9360-9410	LIMESTONE: light to medium gray, light brown, tan, off white in part, firm to hard, microcrystalline, silty, trace argillaceous, mottled in part
9410-9500	LIMESTONE: medium to dark brown, light brown to tan in part, trace medium gray, mottled in part, firm, cryptocrystalline, earthy, slightly silty, argillaceous
9500-9560	LIMESTONE: medium gray brown, light to medium gray, firm to bright, microcrystalline, argillaceous, trace sandy material; ANHYDRITE: white, soft,
9560-9650	LIMESTONE: medium gray brown, light to medium gray, firm to bright, microcrystalline, argillaceous, trace sandy material; ANHYDRITE: white, soft, NFSOC
9650-9680	LIMESTONE: medium gray brown, light to medium gray, firm to bright, microcrystalline, argillaceous, trace sandy material;
9680-9740	LIMESTONE: medium to dark brown, light brown to tan in part, trace medium gray, mottled in part, firm, cryptocrystalline, earthy, slightly silty, argillaceous
9740-9770	LIMESTONE: medium to dark brown, light brown to tan in part, trace medium gray, mottled in part, firm, microcrystalline to cryptocrystalline, earthy, slightly silty, argillaceous
9770-9800	LIMESTONE: light to medium gray to brown, light brown to tan in part, trace medium to dark gray, mottled in part, firm, microcrystalline to cryptocrystalline, earthy, slightly silty, argillaceous

LODGEPOLE: 9788' MD, 9787' TVD (-7847)

9800-9830	LIMESTONE: light to medium gray to brown, light brown to tan in part, trace medium to dark gray, mottled in part, firm, microcrystalline to cryptocrystalline, earthy, slightly silty, argillaceous
9830-9950	LIMESTONE: dark to medium gray to brown, light brown to tan in part, mottled in part, firm, microcrystalline to cryptocrystalline, earthy, slightly silty, argillaceous

Formation tops
Sample Interval
(in feet)

SAMPLE DESCRIPTION

9950-10130 LIMESTONE: dark to medium gray, light to medium brown, very trace tan in part, mottled in part, firm to hard, microcrystalline to cryptocrystalline, slightly to moderately silty, argillaceous, earthy texture

10130-10460 LIMESTONE: medium to dark gray, medium brown gray, trace light gray, mottled in part, firm to hard, microcrystalline to cryptocrystalline, moderately silty, argillaceous, earthy texture

10460-10610 LIMESTONE: dark to medium gray, medium to dark brown gray, trace light gray, mottled in part, firm to hard, microcrystalline to cryptocrystalline, moderately silty, argillaceous, earthy texture, sub to blocky

FALSE BAKKEN: 10599' MD, 10483' TVD (-8543)

10610-10640 SHALE: black, firm, sub blocky to sub platy, carbonaceous; trace LIMESTONE: dark gray, medium to dark gray brown, firm to hard, microcrystalline to cryptocrystalline, silty, very argillaceous, sub to blocky

UPPER BAKKEN SHALE: 10621'MD, 10494' TVD (-8554)

10640-10670 SHALE: black, brown, moderately hard, earthy texture carbonaceous, sub blocky,

MIDDLE BAKKEN MEMBER: 10672'MD, 10506' TVD (-8566)

10670-10760 DOLOMITE: light brown, light gray cream tan, gray in part, firm to hard, microcrystalline; SHALE: blue to gray, black in part, soft to firm, sub blocky, green fluorescence, trace stain, diffuse blue white cut

10760-10950 DOLOMITE: brown to dark brown, light gray, tan, gray in part, firm to hard, microcrystalline; SHALE: blue to gray, black in part, soft to firm, sub blocky, green fluorescence, trace stain, diffuse blue white cut

10950-11000 DOLOMITE: light brown to dark, light gray, tan, gray in part, firm to hard, microcrystalline; SHALE: blue to gray, black in part, soft to firm, sub blocky, trace light brown oil stain, green fluorescence, diffuse blue white cut

11000-11150 DOLOMITE: light brown to dark, light gray, tan, gray in part, firm to hard, microcrystalline; SHALE: blue to gray, black in part, soft to firm, sub blocky, trace light brown oil stain, green fluorescence, diffuse blue white cut

Formation tops
Sample Interval
(in feet)

SAMPLE DESCRIPTION

- 11150-11550 DOLOMITE: light to medium gray, medium gray to brown, medium brown, firm, microcrystalline, calcareous, silty, slightly argillaceous, trace sand, intercrystalline porosity, trace carbonaceous material, light yellow mineral fluorescence, yellow to cream slow streaming cut
- 11550-11950 DOLOMITE: light to medium brown, light to medium gray, silty in part, slightly argillaceous, calcareous, laminated in part, microcrystalline, intercrystalline porosity, spotted to even oil show, yellow to green mineral fluorescence, yellow to green flash cut
- 11950-12150 DOLOMITE: light to medium brown, medium gray, cream in part, silty in part, slightly argillaceous, calcareous, laminated in part, microcrystalline, trace intercrystalline porosity, spotty to even oil stain, yellow fluorescence, yellow streaming cut
- 12150-12250 DOLOMITE: light to medium gray, light to medium gray, cream in part, silty in part, slightly argillaceous, calcareous, laminated in part, microcrystalline, trace intercrystalline porosity, spotty to even oil stain, yellow fluorescence, green streaming cut
- 12250-13050 DOLOMITE: light to medium gray, light to medium gray, cream in part, silty in part, slightly argillaceous, calcareous, laminated in part, microcrystalline, trace intercrystalline porosity, spotty to even oil stain, yellow fluorescence, green diffuse cut
- 13050-13500 DOLOMITE: light to medium brown, medium gray, cream in part, silty in part, slightly argillaceous, calcareous, laminated in part, microcrystalline, trace intercrystalline porosity, spotty to even oil stain, yellow fluorescence, yellow streaming cut
- 13500-13700 DOLOMITE: light to medium brown, light to medium gray, silty in part, slightly argillaceous, calcareous, laminated in part, microcrystalline, intercrystalline porosity, spotted to even oil show, yellow to green mineral fluorescence, yellow to green flash cut
- 13700-13800 DOLOMITE: light to medium gray, light to medium gray, cream in part, silty in part, slightly argillaceous, calcareous, laminated in part, microcrystalline, trace intercrystalline porosity, spotty to even oil stain, yellow fluorescence, green streaming cut

Formation tops
Sample Interval
(in feet)

SAMPLE DESCRIPTION

- 13800-14550 DOLOMITE: light to medium gray, light to medium brown, cream to off white in part, silty in part, slightly argillaceous, calcareous, laminated in part, microcrystalline, trace intercrystalline porosity, spotty to even oil stain, yellow to white fluorescence, green to white streaming cut
- 14550-14900 DOLOMITE: light to medium brown, light to medium gray, cream to off white in part, silty in part, slightly argillaceous, calcareous, laminated in part, microcrystalline, trace intercrystalline porosity, spotty to even oil stain, green to white fluorescence, blue to white diffuse cut
- 14900-15100 DOLOMITE: light to medium gray, medium brown to light brown, cream, off white firm, microcrystalline, calcareous, silty, slightly argillaceous, trace sand, intercrystalline porosity, light yellow mineral fluorescence, yellow to cream slow streaming cut
- 15100-15200 DOLOMITE: light to medium gray, light brown to tan cream in part, silty in part, slightly argillaceous, calcareous, laminated in part, microcrystalline, trace intercrystalline porosity, spotty to even oil stain, yellow fluorescence, green diffuse cut
- 15200-15550 DOLOMITE: light to medium gray, light to medium brown, cream to off white in part, silty in part, slightly argillaceous, calcareous, laminated in part, microcrystalline, trace intercrystalline porosity, spotty to even oil stain, yellow to white fluorescence, green to white streaming cut
- 15550-16000 DOLOMITE: light to medium brown, light to medium gray, cream to off white in part, silty in part, slightly argillaceous, calcareous, laminated in part, microcrystalline, trace intercrystalline porosity, spotty to even oil stain, blue white fluorescence, yellow white diffuse cut
- 16000-16450 DOLOMITE: light to medium brown, light to medium gray, cream to off white in part, silty in part, slightly argillaceous, calcareous, laminated in part, microcrystalline, trace intercrystalline porosity, spotty to even oil stain, green white fluorescence, yellow green streaming cut
- 16450-16750 DOLOMITE: light to medium brown, light to medium gray, silty in part, slightly argillaceous, calcareous, laminated in part, microcrystalline, intercrystalline porosity, spotted to even oil show, yellow to green mineral fluorescence, yellow to green flash cut

Formation tops
Sample Interval
(in feet)

SAMPLE DESCRIPTION

- 16750-17550 DOLOMITE: medium to dark gray, light gray in part, light to medium brown, silty in part, moderately argillaceous, calcareous, laminated in part, microcrystalline, intercrystalline porosity, spotty oil stain, yellow fluorescence, green to yellow streaming cut
- 17550-17650 DOLOMITE: light to medium brown, light to medium gray, cream to off white in part, silty in part, moderately argillaceous, calcareous, laminated in part, microcrystalline, intercrystalline porosity, spotty oil stain, green to white fluorescence, white streaming cut
- 17650-17800 DOLOMITE: light to medium brown, light to medium gray, cream to off white in part, silty in part, moderately argillaceous, calcareous, laminated in part, microcrystalline, intercrystalline porosity, spotty oil stain, green to white fluorescence, white streaming cut; sample contaminated with lube
- 17800-17950 DOLOMITE: light to medium brown, light to medium gray, cream to off white in part, silty in part, moderately argillaceous, calcareous, laminated in part, microcrystalline, intercrystalline porosity, spotty oil stain, blue to white fluorescence, white to green diffuse cut
- 17950-18450 DOLOMITE: light to medium brown, light to medium gray, cream to off white in part, silty in part, moderately argillaceous, calcareous, laminated in part, microcrystalline, intercrystalline porosity, spotty oil stain, blue to white fluorescence, white to green diffuse cut; sample contaminated with lube
- 18450-18950 DOLOMITE: light to medium brown, light to medium gray, cream to off white in part, silty in part, slightly argillaceous, calcareous, laminated in part, microcrystalline, trace intercrystalline porosity, spotty to even oil stain, green white fluorescence, yellow green streaming cut
- 18950-19050 DOLOMITE: light to medium brown, light to medium gray, cream to off white in part, silty in part, slightly argillaceous, calcareous, laminated in part, microcrystalline, trace intercrystalline porosity, spotty to even oil stain, yellow fluorescence, green diffuse cut; sample contaminated with lube
- 19050-19350 DOLOMITE: light to medium brown, light to medium gray, cream to off white in part, silty in part, slightly argillaceous, calcareous, laminated in part, microcrystalline, trace intercrystalline porosity, spotty to even oil stain, yellow fluorescence, green diffuse cut

Formation tops
Sample Interval
(in feet)

SAMPLE DESCRIPTION

19350-20000 DOLOMITE: light to medium brown, light to medium gray, cream to off white in part, silty in part, slightly argillaceous, calcareous, laminated in part, microcrystalline, trace intercrystalline porosity, spotty to even oil stain, yellow fluorescence, green diffuse cut; sample contaminated with lube

20000-20200 DOLOMITE: light to medium brown, light to medium gray, cream to off white in part, silty in part, slightly argillaceous, calcareous, laminated in part, microcrystalline, trace intercrystalline porosity, spotty to even oil stain, green white fluorescence, yellow green streaming cut contaminated with lube

20200-21000 DOLOMITE: medium gray, light to medium brown, light gray to cream in part, silty in part, slightly argillaceous, calcareous, laminated in part, microcrystalline, trace intercrystalline porosity, spotty to even oil stain, cream to yellow fluorescence, yellow diffuse cut; sample contaminated with lube

21000-21950 DOLOMITE: light to medium brown, light to medium gray, cream to off white in part, silty in part, slightly argillaceous, calcareous, laminated in part, microcrystalline, trace intercrystalline porosity, spotty to even oil stain, green white fluorescence, yellow green streaming cut sample contaminated with lube

21950-22900 DOLOMITE: medium gray, light to medium brown, light gray to cream in part, silty in part, slightly argillaceous, calcareous, laminated in part, microcrystalline, trace intercrystalline porosity, spotty to even oil stain, cream to yellow fluorescence, yellow diffuse cut; sample contaminated with lube

22900-23122 DOLOMITE: light gray, light to medium brown, light gray to cream in part, silty in part, slightly argillaceous, calcareous, laminated in part, microcrystalline, trace intercrystalline porosity, spotty to even oil stain, cream to yellow fluorescence, yellow diffuse cut; sample contaminated with lube

GEOLOGICAL SUMMARY & CONCLUSIONS

Continental Resources, Inc drilled the northbound Columbus Federal 3-16H in Sections 16, 9, and 4 of Township 153N, Range 101W into the Mississippian/Devonian age Middle Bakken Member. The Well was spud using Cyclone 4 on October 16, 2013 at a surface location of 2469' FNL, and 199' FEL.

The surface was pre-drilled with surface casing set at 1975'MD. From there the 8 3/4" vertical portion was drilled to Kick Off Point (KOP) at 10070'MD. Leam provided Directional Drilling, and Measurement While Drilling (MWD) services. NorAm Wellsite Services provided both Mud Logging and Geosteering Services. One trip was necessary in the vertical section at 2411'MD for MWD tool.

The curve was drilled in 22.5 hours on October 22-23, 2013. The wellbore entered into the Middle Bakken top at 10672'MD, 10506'TVD, and landed at 10983'MD, 10524'TVD, 18' in the Middle Bakken.

The Lateral resumed drilling on December 17, 2013 at 19:00 hours. The first trip for new BHA occurred at 17000'MD on December 21, 2013 at 09:30 hours. Drilling Resumed December 21, 2013 at 21:10 hours. 87.5 hours later, a second trip occurred for a new bit and motor on December 25, 2013 at 23:30 hours. Drilling resumed on December 26, 2013 at 03:30 hours. Gas increased throughout the well, the further the wellbore went. At 19500'MD, background gas stayed above 2000 units, as the wellbore traversed through the middle of the target zone. At 20:55 hours on December 27, 2013, the Columbus Federal reached TD of 23122'MD. Projection to bit is 23122'MD, 10518'TVD, 89.4 INC, and 0.70 AZM. Overall dip was an 89.83 with the wellbore in zone roughly 90% of the time. Bottom hole Location is 244'FNL & 680.4'FEL Section 4, T153N, R101W.

NEWSCO

International Energy Services Inc.

Continental Resources
Company

33326
Job Number

8/11/2013
Date

Cyclone 4
Rig

Columbus Federal 3-16H
Well Name

McKenzie Co., ND
County & State

Surveyed from depth of: Surface to 1927'

GL to KB: 20'

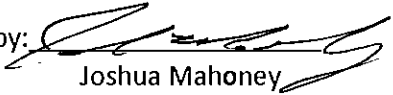
Type of Survey: Nvader/MWD

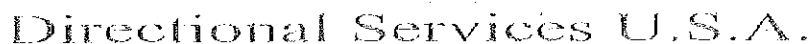
True North

Directional Supervisor/Surveyor: David Hopper

The data and calculations for this survey have been checked by me and conform to the standards and procedures set forth by Newsco International Energy Services Inc. This report represents a true and correct directional survey of this well based on the original data obtained at the well site. Wellbore coordinates are calculated using minimum curvature .

Certified by:


Joshua Mahoney



CLIENT: Continental					NEWSCO JOB # 33326							
DATE: 8/11/2013					WELL NAME: Columbus Federal 3-16H							
STATE: North Dakota					RIG: Cyclone 4							
FIELD: Baker					County: Mckenzie							
TIE-ON DATA					TARGET DATA							
C/L (10,30,100): 100.00 Feet					SENSOR TO BIT: 47.00							
MEASURED DEPTH: 0.00 Feet					KB TO GL : 6.00							
TVD: 0.00 Feet					Cyclone 4: 20.00							
INCLINATION: 0.00 Deg.												
AZIMUTH: 0.00 Deg.												
N(+) S(-): 0.00 Feet												
E(+) W(-): 0.00 Feet					All depths are measured from survey tool to Big Rigs KB							
V/SECTION PLANE: 0.00 Deg.												
V/SECTION : 0.00 Feet												
#	DEPTH	INC	AZM	CRSL	TVD	VS	+N/S-	+E/W-	DLS	100 FEET VBUR	WALK	TF
1	143.00	0.10	60.30	143.00	143.00	0.06	0.06	0.11	0.07	0.07	42.17	MAG
2	225.00	0.40	63.80	82.00	225.00	0.22	0.22	0.43	0.37	0.37	4.27	MAG
3	306.00	0.30	64.60	81.00	306.00	0.44	0.44	0.87	0.12	-0.12	0.99	MAG
4	384.00	0.40	39.60	78.00	384.00	0.74	0.74	1.23	0.23	0.13	-32.05	MAG
5	477.00	0.20	61.70	93.00	476.99	1.06	1.06	1.58	0.24	-0.22	23.76	MAG
6	570.00	0.20	38.90	93.00	569.99	1.27	1.27	1.83	0.09	0.00	-24.52	MAG
7	664.00	0.20	258.60	94.00	663.99	1.36	1.36	1.77	0.40	0.00	-149.26	MAG
8	758.00	0.30	292.00	94.00	757.99	1.42	1.42	1.38	0.18	0.11	35.53	MAG
9	851.00	0.20	291.30	93.00	850.99	1.57	1.57	1.00	0.11	-0.11	-0.75	MAG
10	944.00	0.10	295.50	93.00	943.99	1.67	1.67	0.78	0.11	-0.11	4.52	MAG
11	1036.00	0.40	22.70	92.00	1035.99	2.00	2.00	0.83	0.44	0.33	94.78	MAG
12	1130.00	0.40	6.90	94.00	1129.99	2.63	2.63	0.99	0.12	0.00	-16.81	MAG
13	1222.00	0.20	315.60	92.00	1221.99	3.06	3.06	0.92	0.34	-0.22	-55.76	MAG
14	1315.00	0.30	314.90	93.00	1314.99	3.35	3.35	0.63	0.11	0.11	-0.75	MAG
15	1410.00	0.20	356.70	95.00	1409.99	3.69	3.69	0.45	0.21	-0.11	44.00	MAG
16	1502.00	0.50	35.40	92.00	1501.98	4.18	4.18	0.67	0.40	0.33	42.07	MAG
17	1596.00	0.40	34.30	94.00	1595.98	4.78	4.78	1.09	0.11	-0.11	-1.17	MAG
18	1686.00	0.50	69.50	90.00	1685.98	5.18	5.18	1.64	0.32	0.11	39.11	MAG
19	1780.00	0.20	53.60	94.00	1779.98	5.42	5.42	2.16	0.33	-0.32	-16.91	MAG
20	1873.00	0.10	96.50	93.00	1872.98	5.51	5.51	2.37	0.15	-0.11	46.13	MAG
21	1927.00	0.10	221.00	54.00	1926.98	5.47	5.47	2.38	0.33	0.00	230.56	MAG
22												
23												
24												
25												
26												
27												
28												
#				CRSL	TVD	VS	+N/S-	+E/W-	DLS	100 FEET VBUR	WALK	TF
28												
29												
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Survey Certification Sheet

Continental Resources
Company

20131226D-ND
Job Number

01/13/2014
Date

Section 16, T153N, R101W
LOCATION

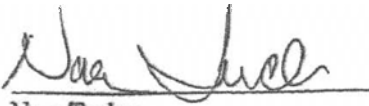
Columbus Federal 3-16H
Well Name

McKenzie County, ND
County & State

Enclosed, please find the survey performed on the referenced well by Leam Drilling Systems, LLC.
Other information required by your office is as follows:

Name & Title Of Surveyor	Drainhole Number	Surveyed Depths	Dates Performed	Type of Survey
Randy W. Rakowitz	OH	1,990'- 23,122'	10/17/2013-12/27/2013	MWD

The data and calculations for this survey have been checked and conform to the standards and procedures set forth by LEAM Drilling Systems, LLC. This report represents a true and correct Survey of this well, to the best of our knowledge, based on the original data obtained at the well site.


Nora Tucker
Well Planner

Physical & Mailing: 2010 East Davis, Conroe, TX 77301

(936) 756-7577 • (936) 756-7595 Fax • 1 (800) 756-7504



LEAM Drilling Systems, Inc.
2027A Airport Rd.
Conroe, TX 77301
(936) 569-1315

Survey Certification Form

State of North Dakota
McKenzie County

I, Randy W. Rakowitz, an employee of LEAM Drilling Systems, Inc., hereby certify that on the dates beginning on 10/17/2013 and ending on 12/27/2013, I conducted or supervised the taking of an MWD Survey from a beginning depth of 1,990' ft. MD to an ending depth of 23,122' ft. MD; that the depth is true, correct, complete, and within the limitations of the tools as set forth by LEAM Drilling Systems, Inc., that I am authorized and qualified to make this report; that this survey was conducted in reference to True North with a declination of 8.52° with respect to the well Columbus Federal 3-16H located in McKenzie County, North Dakota as requested by Continental Resources.

A handwritten signature in black ink, appearing to read 'Randy W. Rakowitz', is written over a solid black horizontal line.

Randy W. Rakowitz
Sr. MWD Operations Coordinator

LEAM Drilling Systems LLC

Survey Report

Company:	Continental Resources	Local Co-ordinate Reference:	Well 3-16H
Project:	McKenzie County, ND	TVD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Site:	Columbus Federal 3	MD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Well:	3-16H	North Reference:	True
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	EDM 5000.1 Multi User DB

Project	McKenzie County, ND		
Map System:	US State Plane 1983	System Datum:	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site	Columbus Federal 3, Sec. 16 - T153N - R101W		
Site Position:		Northing:	408,236.61 usft
From:	Lat/Long	Easting:	1,193,846.91 usft
Position Uncertainty:	0.00 usft	Slot Radius:	0 "
		Latitude:	48° 4' 31.639 N
		Longitude:	103° 40' 11.004 W
		Grid Convergence:	-2.36 °

Well	3-16H		
Well Position	+N/-S	0.00 usft	Northing: 408,236.61 usft
	+E/-W	0.00 usft	Easting: 1,193,846.91 usft
Position Uncertainty	0.00 usft	Wellhead Elevation:	usft
		Latitude:	48° 4' 31.639 N
		Longitude:	103° 40' 11.004 W
		Ground Level:	1,920.00 usft

Wellbore	OH				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	BGGM2013	10/11/13	8.52	72.97	56,411

Design	OH				
Audit Notes:					
Version:	1.0	Phase:	ACTUAL	Tie On Depth:	0.00
Vertical Section:	Depth From (TVD) (usft)	+N/-S (usft)	+E/-W (usft)	Direction (°)	
	0.00	0.00	0.00	357.92	

Survey Program	Date	01/16/14			
From (usft)	To (usft)	Survey (Wellbore)	Tool Name	Description	
143.00	1,927.00	Survey #1 (OH)	MWD-ISCWSA	MWD - Standard	
1,990.00	23,072.00	Survey #2 (OH)	LEAM MWD-ADJ	MWD - Standard	
23,122.00	23,122.00	Survey #3 (OH)	Project	Projection	

Survey										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	
1,927.00	0.10	221.00	1,926.98	5.47	2.38	5.38	0.33	0.00	230.56	
1,990.00	0.10	213.40	1,989.98	5.38	2.32	5.29	0.02	0.00	-12.06	
2,084.00	0.80	242.80	2,083.97	5.01	1.69	4.94	0.76	0.74	31.28	
2,177.00	1.10	251.10	2,176.96	4.42	0.27	4.41	0.35	0.32	8.92	
2,271.00	1.10	253.20	2,270.94	3.87	-1.45	3.92	0.04	0.00	2.23	
2,364.00	1.90	247.50	2,363.91	3.02	-3.73	3.16	0.87	0.86	-6.13	
2,457.00	2.10	248.20	2,456.85	1.80	-6.74	2.04	0.22	0.22	0.75	
2,550.00	2.30	248.90	2,549.79	0.49	-10.06	0.86	0.22	0.22	0.75	
2,645.00	1.80	236.60	2,644.72	-1.01	-13.08	-0.54	0.70	-0.53	-12.95	

LEAM Drilling Systems LLC

Survey Report

Company:	Continental Resources	Local Co-ordinate Reference:	Well 3-16H
Project:	McKenzie County, ND	TVD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Site:	Columbus Federal 3	MD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Well:	3-16H	North Reference:	True
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	EDM 5000.1 Multi User DB

Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
2,739.00	1.20	222.60	2,738.69	-2.55	-14.98	-2.00	0.74	-0.64	-14.89
2,832.00	1.30	224.50	2,831.67	-4.02	-16.38	-3.42	0.12	0.11	2.04
2,925.00	0.70	277.80	2,924.66	-4.69	-17.68	-4.05	1.12	-0.65	57.31
3,019.00	0.90	343.00	3,018.65	-3.91	-18.47	-3.24	0.93	0.21	69.36
3,112.00	0.90	341.60	3,111.64	-2.52	-18.91	-1.83	0.02	0.00	-1.51
3,205.00	0.80	346.10	3,204.63	-1.20	-19.30	-0.49	0.13	-0.11	4.84
3,298.00	0.50	357.00	3,297.62	-0.16	-19.48	0.55	0.35	-0.32	11.72
3,391.00	0.30	355.80	3,390.62	0.49	-19.52	1.20	0.22	-0.22	-1.29
3,484.00	0.20	22.00	3,483.62	0.88	-19.47	1.59	0.16	-0.11	28.17
3,578.00	0.80	49.30	3,577.61	1.46	-18.91	2.15	0.67	0.64	29.04
3,671.00	0.50	67.40	3,670.61	2.04	-18.05	2.69	0.39	-0.32	19.46
3,765.00	0.60	87.40	3,764.60	2.22	-17.18	2.84	0.23	0.11	21.28
3,858.00	0.30	72.80	3,857.60	2.31	-16.46	2.91	0.34	-0.32	-15.70
3,952.00	0.40	64.90	3,951.60	2.53	-15.92	3.10	0.12	0.11	-8.40
4,045.00	0.20	51.20	4,044.60	2.77	-15.50	3.33	0.23	-0.22	-14.73
4,139.00	0.20	86.90	4,138.60	2.88	-15.21	3.43	0.13	0.00	37.98
4,232.00	0.30	148.80	4,231.60	2.68	-14.92	3.22	0.29	0.11	66.56
4,326.00	0.30	161.80	4,325.60	2.23	-14.72	2.77	0.07	0.00	13.83
4,419.00	0.30	142.20	4,418.59	1.81	-14.49	2.33	0.11	0.00	-21.08
4,512.00	0.40	181.80	4,511.59	1.29	-14.36	1.81	0.27	0.11	42.58
4,606.00	0.60	201.70	4,605.59	0.51	-14.55	1.04	0.28	0.21	21.17
4,699.00	1.10	199.40	4,698.58	-0.79	-15.02	-0.24	0.54	0.54	-2.47
4,792.00	0.90	190.10	4,791.56	-2.35	-15.45	-1.79	0.28	-0.22	-10.00
4,886.00	0.40	148.60	4,885.56	-3.35	-15.41	-2.79	0.70	-0.53	-44.15
4,979.00	0.30	38.50	4,978.56	-3.44	-15.09	-2.89	0.62	-0.11	-118.39
5,073.00	1.20	29.00	5,072.55	-2.39	-14.46	-1.86	0.96	0.96	-10.11
5,166.00	1.00	120.30	5,165.54	-1.95	-13.28	-1.46	1.70	-0.22	98.17
5,260.00	0.80	55.80	5,259.53	-1.99	-12.03	-1.55	1.04	-0.21	-68.62
5,354.00	0.90	65.90	5,353.52	-1.32	-10.82	-0.93	0.19	0.11	10.74
5,447.00	1.10	63.50	5,446.50	-0.62	-9.35	-0.28	0.22	0.22	-2.58
5,540.00	1.30	59.80	5,539.48	0.31	-7.64	0.58	0.23	0.22	-3.98
5,634.00	1.10	56.60	5,633.46	1.34	-5.97	1.55	0.22	-0.21	-3.40
5,727.00	1.90	58.80	5,726.43	2.63	-3.90	2.77	0.86	0.86	2.37
5,821.00	1.90	55.60	5,820.38	4.32	-1.28	4.36	0.11	0.00	-3.40
5,914.00	2.00	55.40	5,913.32	6.11	1.33	6.06	0.11	0.11	-0.22
6,007.00	1.80	55.10	6,006.27	7.87	3.86	7.72	0.22	-0.22	-0.32
6,101.00	1.80	49.80	6,100.23	9.66	6.20	9.43	0.18	0.00	-5.64
6,194.00	1.00	53.10	6,193.20	11.09	7.96	10.80	0.86	-0.86	3.55
6,288.00	0.60	83.70	6,287.19	11.64	9.11	11.30	0.61	-0.43	32.55
6,382.00	0.80	98.50	6,381.18	11.60	10.25	11.22	0.29	0.21	15.74
6,476.00	1.10	98.30	6,475.17	11.37	11.79	10.93	0.32	0.32	-0.21
6,569.00	0.40	83.40	6,568.16	11.28	12.99	10.80	0.78	-0.75	-16.02
6,662.00	0.60	3.90	6,661.16	11.80	13.35	11.31	0.71	0.22	-85.48

LEAM Drilling Systems LLC

Survey Report

Company:	Continental Resources	Local Co-ordinate Reference:	Well 3-16H
Project:	McKenzie County, ND	TVD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Site:	Columbus Federal 3	MD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Well:	3-16H	North Reference:	True
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	EDM 5000.1 Multi User DB

Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
6,756.00	1.10	355.50	6,755.15	13.19	13.31	12.70	0.55	0.53	-8.94
6,850.00	1.10	333.00	6,849.13	14.90	12.83	14.42	0.46	0.00	-23.94
6,943.00	1.50	341.00	6,942.11	16.84	12.03	16.39	0.47	0.43	8.60
7,037.00	1.70	327.90	7,036.07	19.19	10.89	18.78	0.44	0.21	-13.94
7,130.00	2.10	323.50	7,129.02	21.72	9.14	21.38	0.46	0.43	-4.73
7,223.00	1.60	295.90	7,221.97	23.66	6.96	23.39	1.08	-0.54	-29.68
7,316.00	1.30	264.90	7,314.94	24.13	4.74	23.95	0.89	-0.32	-33.33
7,411.00	1.50	258.80	7,409.92	23.80	2.45	23.69	0.26	0.21	-6.42
7,504.00	1.90	264.80	7,502.87	23.42	-0.28	23.42	0.47	0.43	6.45
7,597.00	1.90	262.80	7,595.82	23.09	-3.35	23.19	0.07	0.00	-2.15
7,690.00	1.80	245.60	7,688.78	22.29	-6.21	22.50	0.60	-0.11	-18.49
7,784.00	1.10	237.30	7,782.74	21.19	-8.31	21.48	0.78	-0.74	-8.83
7,877.00	0.80	208.50	7,875.73	20.14	-9.37	20.47	0.60	-0.32	-30.97
7,970.00	1.10	199.00	7,968.72	18.73	-9.97	19.08	0.36	0.32	-10.22
8,063.00	1.10	200.30	8,061.70	17.05	-10.57	17.42	0.03	0.00	1.40
8,156.00	1.10	209.70	8,154.69	15.43	-11.32	15.83	0.19	0.00	10.11
8,250.00	1.00	212.00	8,248.67	13.95	-12.21	14.39	0.12	-0.11	2.45
8,343.00	1.10	227.00	8,341.65	12.66	-13.29	13.13	0.31	0.11	16.13
8,436.00	1.10	230.70	8,434.64	11.48	-14.63	12.01	0.08	0.00	3.98
8,529.00	1.00	213.60	8,527.62	10.24	-15.77	10.81	0.35	-0.11	-18.39
8,623.00	1.10	219.90	8,621.61	8.87	-16.80	9.47	0.16	0.11	6.70
8,716.00	1.00	226.80	8,714.59	7.63	-17.97	8.27	0.17	-0.11	7.42
8,810.00	1.10	238.00	8,808.57	6.59	-19.33	7.28	0.24	0.11	11.91
8,903.00	1.30	248.20	8,901.55	5.72	-21.07	6.48	0.31	0.22	10.97
8,997.00	1.10	236.10	8,995.53	4.82	-22.81	5.65	0.34	-0.21	-12.87
9,090.00	1.20	233.60	9,088.52	3.75	-24.33	4.63	0.12	0.11	-2.69
9,183.00	1.10	236.80	9,181.50	2.68	-25.86	3.62	0.13	-0.11	3.44
9,277.00	0.90	213.40	9,275.48	1.57	-27.02	2.55	0.48	-0.21	-24.89
9,370.00	0.30	111.70	9,368.48	0.87	-27.20	1.86	1.08	-0.65	-109.35
9,464.00	0.70	59.10	9,462.47	1.07	-26.48	2.03	0.61	0.43	-55.96
9,557.00	0.60	58.40	9,555.47	1.62	-25.58	2.55	0.11	-0.11	-0.75
9,650.00	0.70	60.30	9,648.46	2.16	-24.67	3.05	0.11	0.11	2.04
9,744.00	0.60	67.40	9,742.46	2.63	-23.72	3.49	0.14	-0.11	7.55
9,837.00	0.60	67.00	9,835.45	3.01	-22.82	3.83	0.00	0.00	-0.43
9,930.00	0.30	71.80	9,928.45	3.27	-22.14	4.08	0.32	-0.32	5.16
10,024.00	0.30	82.50	10,022.45	3.38	-21.66	4.17	0.06	0.00	11.38
10,049.00	0.20	35.90	10,047.45	3.43	-21.57	4.21	0.87	-0.40	-186.40
10,081.00	2.70	330.70	10,079.44	4.13	-21.91	4.92	8.19	7.81	-203.75
10,112.00	7.80	330.00	10,110.29	6.59	-23.32	7.43	16.45	16.45	-2.26
10,143.00	12.50	333.50	10,140.80	11.42	-25.87	12.35	15.28	15.16	11.29
10,174.00	15.80	335.60	10,170.86	18.26	-29.11	19.31	10.77	10.65	6.77
10,205.00	18.50	333.10	10,200.48	26.50	-33.08	27.68	9.03	8.71	-8.06
10,236.00	22.00	331.70	10,229.56	36.00	-38.06	37.35	11.40	11.29	-4.52

LEAM Drilling Systems LLC

Survey Report

Company:	Continental Resources	Local Co-ordinate Reference:	Well 3-16H
Project:	McKenzie County, ND	TVD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Site:	Columbus Federal 3	MD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Well:	3-16H	North Reference:	True
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	EDM 5000.1 Multi User DB

Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
10,267.00	25.10	330.50	10,257.97	46.83	-44.05	48.40	10.12	10.00	-3.87
10,298.00	28.80	329.30	10,285.60	58.98	-51.10	60.80	12.06	11.94	-3.87
10,330.00	32.40	329.10	10,313.14	72.97	-59.44	75.08	11.25	11.25	-0.63
10,361.00	36.80	329.80	10,338.65	88.13	-68.38	90.55	14.25	14.19	2.26
10,392.00	41.80	329.80	10,362.63	105.09	-78.26	107.86	16.13	16.13	0.00
10,423.00	45.80	329.60	10,385.00	123.61	-89.08	126.77	12.91	12.90	-0.65
10,454.00	50.00	331.20	10,405.78	143.61	-100.43	147.16	14.08	13.55	5.16
10,485.00	53.50	330.20	10,424.97	164.84	-112.34	168.81	11.57	11.29	-3.23
10,516.00	58.10	329.50	10,442.39	187.00	-125.22	191.42	14.96	14.84	-2.26
10,547.00	60.90	329.80	10,458.12	210.05	-138.72	214.94	9.07	9.03	0.97
10,578.00	63.40	330.90	10,472.60	233.87	-152.27	239.24	8.65	8.06	3.55
10,610.00	66.40	332.30	10,486.18	259.36	-166.05	265.21	10.18	9.38	4.38
10,641.00	71.10	333.00	10,497.41	285.01	-179.32	291.33	15.31	15.16	2.26
10,672.00	76.20	333.30	10,506.13	311.54	-192.75	318.34	16.48	16.45	0.97
10,703.00	79.30	333.70	10,512.71	338.65	-206.26	345.92	10.08	10.00	1.29
10,734.00	80.90	333.50	10,518.04	366.01	-219.84	373.74	5.20	5.16	-0.65
10,765.00	84.40	333.50	10,522.00	393.52	-233.56	401.73	11.29	11.29	0.00
10,797.00	89.80	333.10	10,523.62	422.06	-247.91	430.78	16.92	16.88	-1.25
10,828.00	91.00	333.00	10,523.41	449.69	-261.96	458.90	3.88	3.87	-0.32
10,859.00	91.40	333.10	10,522.76	477.32	-276.01	487.02	1.33	1.29	0.32
10,890.00	89.40	333.70	10,522.54	505.03	-289.89	515.22	6.74	-6.45	1.94
10,935.00	88.70	333.70	10,523.29	545.37	-309.82	556.25	1.56	-1.56	0.00
10,969.00	88.60	333.30	10,524.09	575.79	-324.99	587.20	1.21	-0.29	-1.18
11,064.00	89.30	332.80	10,525.83	660.46	-368.03	673.38	0.91	0.74	-0.53
11,159.00	90.00	335.60	10,526.41	745.98	-409.38	760.34	3.04	0.74	2.95
11,253.00	90.20	338.60	10,526.24	832.56	-445.95	848.19	3.20	0.21	3.19
11,348.00	90.70	342.80	10,525.50	922.20	-477.34	938.91	4.45	0.53	4.42
11,443.00	90.70	345.60	10,524.34	1,013.59	-503.20	1,031.19	2.95	0.00	2.95
11,538.00	90.70	348.60	10,523.18	1,106.18	-524.41	1,124.48	3.16	0.00	3.16
11,633.00	90.00	350.90	10,522.60	1,199.65	-541.31	1,218.51	2.53	-0.74	2.42
11,728.00	90.00	353.40	10,522.60	1,293.75	-554.28	1,313.02	2.63	0.00	2.63
11,823.00	90.10	356.20	10,522.51	1,388.35	-562.89	1,407.87	2.95	0.11	2.95
11,918.00	89.50	357.90	10,522.85	1,483.22	-567.78	1,502.85	1.90	-0.63	1.79
12,013.00	87.70	0.00	10,525.17	1,578.17	-569.52	1,597.80	2.91	-1.89	2.21
12,108.00	88.40	359.30	10,528.40	1,673.11	-570.10	1,692.70	1.04	0.74	-0.74
12,203.00	89.20	359.30	10,530.39	1,768.08	-571.26	1,787.65	0.84	0.84	0.00
12,298.00	91.10	359.90	10,530.14	1,863.07	-571.93	1,882.60	2.10	2.00	0.63
12,393.00	90.00	1.60	10,529.23	1,958.06	-570.68	1,977.48	2.13	-1.16	1.79
12,488.00	87.70	1.80	10,531.13	2,052.99	-567.87	2,072.25	2.43	-2.42	0.21
12,583.00	87.80	0.00	10,534.86	2,147.90	-566.38	2,167.04	1.90	0.11	-1.89
12,678.00	88.80	359.50	10,537.68	2,242.86	-566.79	2,261.95	1.18	1.05	-0.53
12,773.00	89.90	359.20	10,538.76	2,337.84	-567.87	2,356.91	1.20	1.16	-0.32
12,867.00	88.20	0.70	10,540.32	2,431.82	-567.95	2,450.83	2.41	-1.81	1.60
12,960.00	89.30	1.10	10,542.35	2,524.79	-566.49	2,543.69	1.26	1.18	0.43

LEAM Drilling Systems LLC

Survey Report

Company:	Continental Resources	Local Co-ordinate Reference:	Well 3-16H
Project:	McKenzie County, ND	TVD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Site:	Columbus Federal 3	MD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Well:	3-16H	North Reference:	True
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	EDM 5000.1 Multi User DB

Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
13,051.00	90.00	0.70	10,542.90	2,615.77	-565.06	2,634.56	0.89	0.77	-0.44
13,143.00	88.00	2.00	10,544.51	2,707.73	-562.89	2,726.37	2.59	-2.17	1.41
13,235.00	88.00	1.80	10,547.72	2,799.62	-559.84	2,818.10	0.22	0.00	-0.22
13,327.00	88.60	1.60	10,550.45	2,891.54	-557.12	2,909.86	0.69	0.65	-0.22
13,420.00	88.70	1.60	10,552.64	2,984.48	-554.52	3,002.64	0.11	0.11	0.00
13,511.00	90.00	1.60	10,553.67	3,075.43	-551.98	3,093.44	1.43	1.43	0.00
13,603.00	90.70	1.30	10,553.11	3,167.40	-549.65	3,185.27	0.83	0.76	-0.33
13,695.00	90.30	0.60	10,552.31	3,259.39	-548.13	3,277.13	0.88	-0.43	-0.76
13,787.00	91.50	1.60	10,550.86	3,351.36	-546.36	3,368.98	1.70	1.30	1.09
13,880.00	90.00	0.60	10,549.65	3,444.33	-544.58	3,461.82	1.94	-1.61	-1.08
13,972.00	89.70	359.50	10,549.89	3,536.32	-544.50	3,553.76	1.24	-0.33	-1.20
14,064.00	89.60	358.80	10,550.45	3,628.31	-545.86	3,645.73	0.77	-0.11	-0.76
14,156.00	90.70	359.90	10,550.21	3,720.30	-546.90	3,737.70	1.69	1.20	1.20
14,249.00	91.00	358.50	10,548.83	3,813.28	-548.20	3,830.67	1.54	0.32	-1.51
14,341.00	90.20	358.10	10,547.86	3,905.23	-550.93	3,922.66	0.97	-0.87	-0.43
14,433.00	91.10	358.80	10,546.82	3,997.19	-553.42	4,014.65	1.24	0.98	0.76
14,526.00	91.50	358.30	10,544.71	4,090.14	-555.77	4,107.62	0.69	0.43	-0.54
14,619.00	89.90	357.00	10,543.57	4,183.05	-559.59	4,200.60	2.22	-1.72	-1.40
14,712.00	91.50	359.20	10,542.44	4,275.98	-562.67	4,293.59	2.92	1.72	2.37
14,805.00	90.80	359.00	10,540.57	4,368.95	-564.13	4,386.55	0.78	-0.75	-0.22
14,897.00	92.20	359.20	10,538.16	4,460.91	-565.57	4,478.49	1.54	1.52	0.22
14,990.00	90.50	0.90	10,535.97	4,553.87	-565.49	4,571.40	2.58	-1.83	1.83
15,082.00	91.20	0.60	10,534.61	4,645.85	-564.29	4,663.27	0.83	0.76	-0.33
15,174.00	89.60	1.60	10,533.97	4,737.83	-562.52	4,755.13	2.05	-1.74	1.09
15,267.00	90.00	0.70	10,534.29	4,830.81	-560.66	4,847.98	1.06	0.43	-0.97
15,360.00	90.50	0.90	10,533.88	4,923.80	-559.36	4,940.86	0.58	0.54	0.22
15,453.00	91.50	0.20	10,532.26	5,016.78	-558.46	5,033.74	1.31	1.08	-0.75
15,545.00	89.80	2.80	10,531.22	5,108.73	-556.06	5,125.55	3.38	-1.85	2.83
15,640.00	89.90	2.50	10,531.47	5,203.63	-551.66	5,220.22	0.33	0.11	-0.32
15,735.00	90.70	1.40	10,530.97	5,298.57	-548.43	5,314.99	1.43	0.84	-1.16
15,830.00	90.10	1.60	10,530.31	5,393.54	-545.94	5,409.80	0.67	-0.63	0.21
15,925.00	91.70	3.20	10,528.81	5,488.43	-541.97	5,504.49	2.38	1.68	1.68
16,020.00	88.90	2.30	10,528.32	5,583.31	-537.41	5,599.14	3.10	-2.95	-0.95
16,115.00	90.70	2.80	10,528.65	5,678.21	-533.18	5,693.83	1.97	1.89	0.53
16,210.00	89.30	3.40	10,528.65	5,773.07	-528.05	5,788.43	1.60	-1.47	0.63
16,305.00	90.30	4.40	10,528.98	5,867.85	-521.59	5,882.91	1.49	1.05	1.05
16,400.00	87.50	1.10	10,530.80	5,962.70	-517.03	5,977.54	4.55	-2.95	-3.47
16,495.00	88.40	1.10	10,534.20	6,057.62	-515.21	6,072.33	0.95	0.95	0.00
16,590.00	90.00	0.90	10,535.53	6,152.59	-513.55	6,167.18	1.70	1.68	-0.21
16,685.00	91.10	0.20	10,534.62	6,247.58	-512.64	6,262.07	1.37	1.16	-0.74
16,780.00	90.30	0.40	10,533.45	6,342.57	-512.14	6,356.98	0.87	-0.84	0.21
16,874.00	91.10	1.10	10,532.31	6,436.56	-510.91	6,450.86	1.13	0.85	0.74
16,971.00	92.30	0.40	10,529.43	6,533.50	-509.64	6,547.70	1.43	1.24	-0.72

LEAM Drilling Systems LLC

Survey Report

Company:	Continental Resources	Local Co-ordinate Reference:	Well 3-16H
Project:	McKenzie County, ND	TVD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Site:	Columbus Federal 3	MD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Well:	3-16H	North Reference:	True
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	EDM 5000.1 Multi User DB

Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
17,066.00	91.70	0.70	10,526.11	6,628.44	-508.73	6,642.54	0.71	-0.63	0.32
17,161.00	91.40	0.70	10,523.54	6,723.40	-507.57	6,737.39	0.32	-0.32	0.00
17,256.00	91.40	0.40	10,521.22	6,818.37	-506.66	6,832.26	0.32	0.00	-0.32
17,351.00	91.50	359.50	10,518.82	6,913.34	-506.74	6,927.17	0.95	0.11	-0.95
17,446.00	91.70	359.70	10,516.17	7,008.30	-507.40	7,022.09	0.30	0.21	0.21
17,541.00	92.00	359.50	10,513.10	7,103.24	-508.06	7,117.00	0.38	0.32	-0.21
17,636.00	91.70	358.80	10,510.03	7,198.18	-509.47	7,211.93	0.80	-0.32	-0.74
17,731.00	91.50	359.30	10,507.38	7,293.13	-511.05	7,306.88	0.57	-0.21	0.53
17,826.00	90.00	358.60	10,506.14	7,388.11	-512.79	7,401.85	1.74	-1.58	-0.74
17,921.00	90.30	358.30	10,505.89	7,483.07	-515.36	7,496.84	0.45	0.32	-0.32
18,016.00	89.50	359.30	10,506.05	7,578.05	-517.35	7,591.83	1.35	-0.84	1.05
18,111.00	89.80	358.80	10,506.63	7,673.03	-518.92	7,686.81	0.61	0.32	-0.53
18,206.00	89.90	358.10	10,506.88	7,768.00	-521.49	7,781.81	0.74	0.11	-0.74
18,301.00	91.40	357.60	10,505.80	7,862.92	-525.06	7,876.80	1.66	1.58	-0.53
18,396.00	90.20	357.60	10,504.48	7,957.83	-529.03	7,971.78	1.26	-1.26	0.00
18,491.00	90.20	356.90	10,504.15	8,052.72	-533.59	8,066.78	0.74	0.00	-0.74
18,586.00	89.20	358.30	10,504.64	8,147.63	-537.57	8,161.77	1.81	-1.05	1.47
18,681.00	89.40	357.60	10,505.80	8,242.56	-540.97	8,256.76	0.77	0.21	-0.74
18,776.00	89.60	359.30	10,506.63	8,337.52	-543.54	8,351.75	1.80	0.21	1.79
18,871.00	89.30	358.50	10,507.55	8,432.49	-545.36	8,446.73	0.90	-0.32	-0.84
18,966.00	89.90	358.30	10,508.21	8,527.45	-548.01	8,541.73	0.67	0.63	-0.21
19,061.00	90.00	357.40	10,508.29	8,622.39	-551.58	8,636.72	0.95	0.11	-0.95
19,156.00	90.30	357.40	10,508.04	8,717.29	-555.89	8,731.72	0.32	0.32	0.00
19,251.00	90.70	357.00	10,507.21	8,812.17	-560.53	8,826.71	0.60	0.42	-0.42
19,346.00	90.70	358.30	10,506.05	8,907.08	-564.42	8,921.70	1.37	0.00	1.37
19,440.00	90.60	359.50	10,504.99	9,001.06	-566.23	9,015.68	1.28	-0.11	1.28
19,535.00	89.30	359.50	10,505.07	9,096.05	-567.06	9,110.64	1.37	-1.37	0.00
19,630.00	89.90	359.20	10,505.73	9,191.04	-568.13	9,205.61	0.71	0.63	-0.32
19,725.00	90.00	359.20	10,505.82	9,286.03	-569.46	9,300.58	0.11	0.11	0.00
19,820.00	90.20	359.50	10,505.65	9,381.03	-570.54	9,395.55	0.38	0.21	0.32
19,915.00	90.70	359.30	10,504.90	9,476.02	-571.53	9,490.52	0.57	0.53	-0.21
20,009.00	89.20	358.60	10,504.99	9,570.00	-573.25	9,584.50	1.76	-1.60	-0.74
20,104.00	88.80	359.30	10,506.64	9,664.97	-574.99	9,679.47	0.85	-0.42	0.74
20,199.00	90.20	0.70	10,507.47	9,759.96	-574.99	9,774.40	2.08	1.47	1.47
20,294.00	90.90	0.60	10,506.56	9,854.95	-573.92	9,869.29	0.74	0.74	-0.11
20,389.00	88.50	0.70	10,507.06	9,949.93	-572.84	9,964.17	2.53	-2.53	0.11
20,484.00	90.00	1.60	10,508.30	10,044.90	-570.93	10,059.01	1.84	1.58	0.95
20,579.00	91.00	1.40	10,507.47	10,139.87	-568.45	10,153.82	1.07	1.05	-0.21
20,674.00	90.20	2.10	10,506.48	10,234.81	-565.55	10,248.60	1.12	-0.84	0.74
20,769.00	90.70	1.60	10,505.73	10,329.76	-562.48	10,343.37	0.74	0.53	-0.53
20,864.00	88.90	1.60	10,506.06	10,424.72	-559.83	10,438.17	1.89	-1.89	0.00
20,959.00	89.30	1.30	10,507.56	10,519.68	-557.42	10,532.98	0.53	0.42	-0.32
21,054.00	91.20	2.80	10,507.14	10,614.61	-554.02	10,627.72	2.55	2.00	1.58

LEAM Drilling Systems LLC

Survey Report

Company:	Continental Resources	Local Co-ordinate Reference:	Well 3-16H
Project:	McKenzie County, ND	TVD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Site:	Columbus Federal 3	MD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Well:	3-16H	North Reference:	True
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	EDM 5000.1 Multi User DB

Survey										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	
21,150.00	89.40	3.00	10,506.64	10,710.48	-549.17	10,723.36	1.89	-1.88	0.21	
21,244.00	89.00	2.70	10,507.95	10,804.35	-544.49	10,817.00	0.53	-0.43	-0.32	
21,339.00	89.40	2.50	10,509.28	10,899.25	-540.19	10,911.67	0.47	0.42	-0.21	
21,434.00	90.00	2.30	10,509.78	10,994.16	-536.21	11,006.38	0.67	0.63	-0.21	
21,529.00	90.70	2.10	10,509.19	11,089.09	-532.56	11,101.11	0.77	0.74	-0.21	
21,624.00	88.20	2.70	10,510.11	11,183.99	-528.58	11,195.81	2.71	-2.63	0.63	
21,719.00	88.30	2.00	10,513.01	11,278.87	-524.69	11,290.48	0.74	0.11	-0.74	
21,813.00	88.40	1.80	10,515.71	11,372.78	-521.57	11,384.22	0.24	0.11	-0.21	
21,905.00	89.50	2.80	10,517.40	11,464.69	-517.88	11,475.93	1.62	1.20	1.09	
22,000.00	89.70	2.80	10,518.06	11,559.57	-513.24	11,570.58	0.21	0.21	0.00	
22,095.00	90.00	2.50	10,518.31	11,654.47	-508.85	11,665.26	0.45	0.32	-0.32	
22,190.00	89.70	1.80	10,518.56	11,749.40	-505.29	11,760.00	0.80	-0.32	-0.74	
22,285.00	90.40	1.60	10,518.48	11,844.36	-502.47	11,854.79	0.77	0.74	-0.21	
22,380.00	90.10	0.90	10,518.06	11,939.33	-500.40	11,949.63	0.80	-0.32	-0.74	
22,475.00	90.50	0.90	10,517.57	12,034.32	-498.90	12,044.50	0.42	0.42	0.00	
22,570.00	90.30	2.80	10,516.90	12,129.26	-495.84	12,139.27	2.01	-0.21	2.00	
22,665.00	89.50	2.50	10,517.07	12,224.16	-491.44	12,233.94	0.90	-0.84	-0.32	
22,760.00	89.90	2.70	10,517.57	12,319.06	-487.13	12,328.63	0.47	0.42	0.21	
22,855.00	90.50	2.30	10,517.23	12,413.97	-482.99	12,423.32	0.76	0.63	-0.42	
22,949.00	90.00	1.40	10,516.82	12,507.92	-479.96	12,517.10	1.10	-0.53	-0.96	
23,044.00	89.30	0.60	10,517.40	12,602.90	-478.30	12,611.96	1.12	-0.74	-0.84	
23,072.00	89.40	0.70	10,517.72	12,630.90	-477.98	12,639.92	0.51	0.36	0.36	
23,122.00	89.40	0.70	10,518.25	12,680.89	-477.37	12,689.86	0.00	0.00	0.00	
Projection to Bit										



SUNDRY NOTICES AND REPORTS ON WELLS - FORM 1

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (03-2004)



Well File No.

25160

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date October 14, 2013
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	
Approximate Start Date	

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Flow back exemption

Well Name and Number Columbus Federal 3-16H						
Footages 2469 F N L		Qtr-Qtr 199 F E L	Section SENE	Township 16	Range 153 N	Range 101 W
Field Baker		Pool Bakken		County McKenzie		

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Continental Resources, Inc. requests a waiver from the tubing/pkr requirement included in NDIC 43-02-03-21: Casing, Tubing, and Cementing Requirements during the completion period immediately following the upcoming fracture stimulation. The following assurances apply:

- 1) The well is equipped with 26#/ft P-110 7" casing at surface with an API burst rating of 9960 psig for the 26 #/ft casing.
- 2) The frac design will use a safety factor of 0.85 * API burst rating to determine the max pressure.
- 3) Damage to the casing during the frac would be detected immediately by monitoring equipment.
- 4) The casing is exposed to significantly lower rates and pressures during flow back than during the frac job.
- 5) The frac fluid and formation fluids have very low corrosion and erosion rates
- 6) Production equipment will be installed as soon as possible after the well ceases flowing.
- 7) A 300# gauge will be installed on surface casing during flowback period.

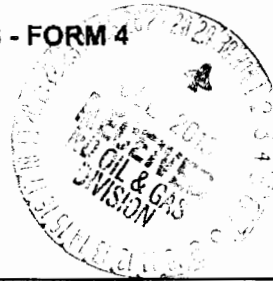
Company Continental Resources, Inc.		Telephone Number 405-234-9000	
Address P.O. Box 269000			
City Oklahoma City	State OK	Zip Code 73126	
Signature 	Printed Name Jim Landrigan		
Title Completion Engineer	Date August 29, 2013		

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date September 4, 2013	
By 	
Title PETROLEUM ENGINEER	



SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
25160

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date August 14, 2013
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Spud with Small Rig

Well Name and Number Columbus Federal 3-16H							
Footages 2469 F N L 199 F E L		Qtr-Qtr SENE	Section 16	Township 153 N	Range 101 W		
Field Bakker	Pool Bakken	County McKenzie					

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s) Advanced Energy Services, LLC			
Address P.O. Box 85	City South Boardman	State Michigan	Zip Code 49680-0085

DETAILS OF WORK

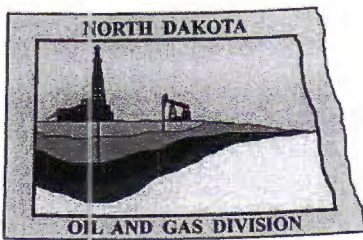
Continental Resources, Inc. requests permission for suspension of drilling for up to 90 days for the referenced well under NDAC 43-02-03-55. Continental Resources, Inc. intends to drill the surface hole with freshwater based drilling mud and set surface casing with a small drilling rig and move off within 3 to 5 days. The casing will be set at a depth pre-approved by the NDIC per the Application for Permit to Drill NDAC 43-02-03-21. No saltwater will be used in the drilling and cementing operations of the surface casing. Once the surface casing is cemented, a plug or mechanical seal will be placed at the top of the casing to prevent any foreign matter from getting into the well. A rig capable of drilling to TD will move onto the location within the 90 days previously outlined to complete the drilling and casing plan as per the APD. The undersigned states that this request for suspension of drilling operations in accordance with the Subsection 4 of Section 43-02-03-55 of the NDAC, is being requested to take advantage of the cost savings and time savings of using an initial rig that is smaller than the rig necessary to drill a well to total depth but is not intended to alter or extend the terms and conditions of, or suspend any obligation under, any oil and gas lease with acreage in or under the spacing or drilling unit for the above-referenced well. Continental Resources, Inc. understands NDAC 43-02-03-31 requirements regarding confidentiality pertaining to this permit. The drilling pit will be fenced immediately after construction if the well pad is located in a pasture (NDAC 43-02-03-19 & 19.1). Continental Resources, Inc. will plug and abandon the well and reclaim the well site if the well is not drilled by the larger rotary rig within 90 days after spudding the well with the smaller drilling rig.

Company Continental Resources, Inc.		Telephone Number (405) 234-9000	
Address P.O. Box 268870			
City Oklahoma City	State OK	Zip Code 73126	
Signature <i>Becky Barnes</i>	Printed Name Becky Barnes		
Title Regulatory Compliance Specialist	Date July 30, 2013		
Email Address becky.barnes@clr.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 8-04-2013	By <i>David Burns</i>
Title David Burns	

David Burns
Engineering Tech.

CR1 must notify NDIC Inspector Richard Runyon 701-770-3554 with spudding TD



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas

25160
TA

BECKY BARNES
CONTINENTAL RESOURCES, INC.
PO BOX 1032
ENID, OK 73702-1032 USA

Date: 3/18/2013

RE: CORES AND SAMPLES

Well Name: COLUMBUS FEDERAL 3-16H Well File No.: 25160
Location: SENE 16-153-101 County: MCKENZIE
Permit Type: Development - HORIZONTAL
Field: BAKER Target Horizon: MIDDLE BAKKEN

Dear BECKY BARNES:

North Dakota Century Code (NDCC) Section 38-08-04 provides for the preservation of cores and samples and their shipment to the State Geologist when requested. The following is required on the above referenced well:

- 1) All cores, core chips and samples must be submitted to the State Geologist as provided for the NDCC Section 38-08-04 and North Dakota Administrative Code 43-02-03-38.1.
- 2) Samples shall include all cuttings from:

Base of the Last Charles Salt

Samples of cuttings shall be taken at 30' maximum intervals through all vertical, build and horizontal sections. Samples must be washed, dried, packed in sample envelopes in correct order with labels showing operator, well name, location and depth, and forwarded in standard boxes to the State Geologist within 30 days of the completion of drilling operations.

- 3) Cores: ALL CORES cut shall be preserved in correct order, properly boxed, and forwarded to the State Geologist within 90 days of completion of drilling operations. Any extension of time must have written approval from the State Geologist.
- 4) All cores, core chips, and samples must be shipped, prepaid, to the State Geologist at the following address:
**ND Geological Survey Core Library
Campus Road and Cornell
Grand Forks, ND 58202**
- 5) NDCC Section 38-08-16 allows for a civil penalty for any violation of Chapter 38 08 not to exceed \$12,500 for each offense, and each day's violation is a separate offense.

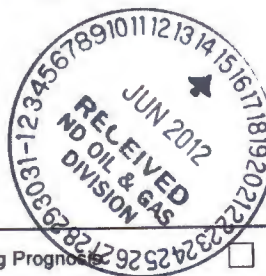
Sincerely


Richard A. Suggs
Geologist



SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.

25160

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.

PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent	Approximate Start Date
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	<u>Open Hole Log Waiver</u>

Well Name and Number Columbus Federal 3-16H					
Footages	Qtr-Qtr	Section	Township	Range	
2469 F N L 199 F E L	SENE	16	153 N	101 W	
Field	Pool	County			
	Bakken	McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Requested variance to not run openhole logs. GR/CBL/CCL will be run from deepest point obtainable to base of surface casing.

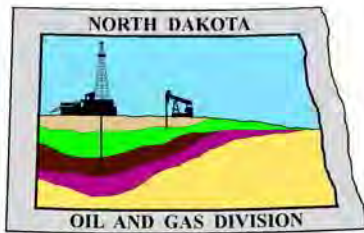
Offset logs used will be the Nance Petroleum, Corps of Engineers 31-10, Sec 10-153N-101W, McKenzie County, ND.

The Gamma Ray Log will be run all the way to surface and all mud logs will be submitted as one digital tiff formatted file and one digital LAS formatted file.

** Approval per logs run on #10710 - NOVAK 1-9-3A*

Company Continental Resources, Inc.		Telephone Number 580-233-8955	
Address P.O. Box 1032			
City Enid	State OK	Zip Code 73702	
Signature <i>Terry L. Olson</i>	Printed Name Terry L. Olson		
Title Regulatory Compliance Specialist	Date June 12, 2012		
Email Address Terry.Olson@clr.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date <i>3/11/2013</i>	
By <i>Richard A. Suggs</i>	
Title Richard A. Suggs	Geologist



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

March 11, 2013

Terry L. Olson
Regulatory Compliance Specialist
CONTINENTAL RESOURCES, INC.
P.O. Box 1032
Enid, OK 73702

**RE: HORIZONTAL WELL
COLUMBUS FEDERAL 3-16H
SENE Section 16-153N-101W
McKenzie County
Well File # 25160**

Dear Terry:

Pursuant to Commission Order No. 1983221551, approval to drill the above captioned well is hereby given. The approval is granted on the condition that all portions of the well bore not isolated by cement, be no closer than the **200' setback** from the north & south boundaries and **500' setback** from the east & west boundaries within the 2560 acre spacing unit consisting of Sections 4, 9, 16, & 21-T153N-R101W. **Tool error is not required pursuant to order.**

PERMIT STIPULATIONS: Liner must be placed under the location. THE MOUSE AND RAT HOLE MUST BE CEMENTED. A spill contingency plan must be in place prior to spudding the wells. Be advised the Mildred 1, NDIC file #12306 well is within 100 feet of the proposed wellbore for the Columbus Federal 3-16H well. Precautions must be taken while designing the frac job for the Columbus Federal 3-16H well as to not adversely affect the Mildred 1, NDIC file #12306 well. No Drilling Pit will be allowed. Due to surficial water adjacent to the well site, a dike is required surrounding the entire location. One horizontal well shall be drilled and completed in the standup 1280-acre spacing unit described as Sections 4 and 9, T153N-R101W, McKenzie County, north Dakota, prior to completing any horizontal well in the 2560-acre spacing unit described as Sections 4, 9, 16, and 21-T153N-R101W McKenzie County, North Dakota. Continental must drill the Columbus Federal 2-16H, 3-16H and Tallahassee 2-16H, 3-16H back to back. CONTINENTAL RESOURCES must contact NDIC Field Inspector Richard Dunn at 701-770-3554 prior to location construction.

Drilling pit

NDAC 43-02-03-19.4 states that "a pit may be utilized to bury drill cuttings and solids generated during well drilling and completion operations, providing the pit can be constructed, used and reclaimed in a manner that will prevent pollution of the land surface and freshwaters. Reserve and circulation of mud system through earthen pits are prohibited. All pits shall be inspected by an authorized representative of the director prior to lining and use. Drill cuttings and solids must be stabilized in a manner approved by the director prior to placement in a cuttings pit."

Form 1 Changes & Hard Lines

Any changes, shortening of casing point or lengthening at Total Depth must have prior approval by the NDIC. The proposed directional plan is at a legal location. The minimum legal coordinate from the well head at casing point is: 301' W. Also, based on the azimuth of the proposed lateral the maximum legal coordinate from the well head is: 12716' N 7 a minimum legal coordinate from the well head of 301' W.

Location Construction Commencement (Three Day Waiting Period)

Operators shall not commence operations on a drill site until the 3rd business day following publication of the approved drilling permit on the NDIC - OGD Daily Activity Report. If circumstances require operations to commence before the 3rd business day following publication on the Daily Activity Report, the waiting period may be waived by the Director. Application for a waiver must be by sworn affidavit providing the information necessary to evaluate the extenuating circumstances, the factors of NDAC 43-02-03-16.2 (1), (a)-(f), and any other information that would allow the Director to conclude that in the event another owner seeks revocation of the drilling permit, the applicant should retain the permit.

Permit Fee & Notification

Payment was received in the amount of \$100 via credit card. It is requested that notification be given immediately upon the spudding of the well. This information should be relayed to the Oil & Gas Division, Bismarck, via telephone. The following information must be included: Well name, legal location, permit number, drilling contractor, company representative, date and time of spudding. Office hours are 8:00 a.m. to 12:00 p.m. and 1:00 p.m. to 5:00 p.m. Central Time. Our telephone number is (701) 328-8020, leave a message if after hours or on the weekend.

Survey Requirements for Horizontal, Horizontal Re-entry, and Directional Wells

NDAC Section 43-02-03-25 (Deviation Tests and Directional Surveys) states in part (that) the survey contractor shall file a certified copy of all surveys with the director free of charge within thirty days of completion. Surveys must be submitted as one electronic copy, or in a form approved by the director. However, the director may require the directional survey to be filed immediately after completion if the survey is needed to conduct the operation of the director's office in a timely manner. Certified surveys must be submitted via email in one adobe document, with a certification cover page to certsurvey@nd.gov.

Survey points shall be of such frequency to accurately determine the entire location of the well bore.

Specifically, the Horizontal and Directional well survey frequency is 100 feet in the vertical, 30 feet in the curve (or when sliding) and 90 feet in the lateral.

Confidential status

Your request for confidential status of all information furnished to the Director, or his representatives, is hereby granted. Such information, except production runs, shall remain confidential for six months commencing on the date the well is spud.

Confidential status notwithstanding, the Director and his representatives shall have access to all well records wherever located. Your company personnel, or any person performing work for your company shall permit the Director and his representatives to come upon any lease, property, well, or drilling rig operated or controlled by them, complying with all safety rules, and to inspect the records and operation of such wells and to have access at all times to any and all records of wells. The Commission's field personnel periodically inspect producing and drilling wells. Any information regarding such wells shall be made available to them at any time upon request. The information so obtained by the field personnel shall be maintained in strict confidence and shall be available only to the Commission and its staff.

Surface casing cement

Tail cement utilized on surface casing must have a minimum compressive strength of 500 psi within 12 hours, and tail cement utilized on production casing must have a minimum compressive strength of 500 psi before drilling the plug or initiating tests.

Logs

NDAC Section 43-02-03-31 requires the running of (1) a suite of open hole logs from which formation tops and porosity zones can be determined, (2) a Gamma Ray Log run from total depth to ground level elevation of the well bore, and (3) a log from which the presence and quality of cement can be determined (Standard CBL or Ultrasonic cement evaluation log) in every well in which production or intermediate casing has been set, this log must be run prior to completing the well. All logs run must be submitted free of charge, as one digital TIFF (tagged image file format) copy and one digital LAS (log ASCII) formatted copy. Digital logs may be submitted on a standard CD, DVD, or attached to an email sent to digitallogs@nd.gov. Thank you for your cooperation.

Sincerely,

David Tabor
Engineering Technician IV



APPLICATION FOR PERMIT TO DRILL HORIZONTAL WELL - FORM 1H

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 54269 (08-2005)

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Type of Work New Location	Type of Well Oil & Gas	Approximate Date Work Will Start 6 / 16 / 2012	Confidential Status Yes
Operator CONTINENTAL RESOURCES, INC.			Telephone Number 580-233-8955
Address P.O. Box 1032		City Enid	State OK Zip Code 73702

☒ Notice has been provided to the owner of any permanently occupied dwelling within 1,320 feet.

☒ This well is not located within five hundred feet of an occupied dwelling.

WELL INFORMATION (If more than one lateral proposed, enter data for additional laterals on page 2)

Well Name COLUMBUS FEDERAL				Well Number 3-16H			
Surface Footages 2469 F N L 199 F E L		Qtr-Qtr SENE	Section 16	Township 153 N	Range 101 W	County McKenzie	
Longstring Casing Point Footages 1952 F N L 500 F E L		Qtr-Qtr SESE	Section 16	Township 153 N	Range 101 W	County McKenzie	
Longstring Casing Point Coordinates From Well Head 517 N From WH 301 W From WH		Azimuth 352 °	Longstring Total Depth 10952 Feet MD 10543 Feet TVD				
Bottom Hole Footages From Nearest Section Line 200 F N L 660 F E L		Qtr-Qtr NENE	Section 4	Township 153 N	Range 101 W	County McKenzie	
Bottom Hole Coordinates From Well Head 12716 N From WH 461 W From WH		KOP Lateral 1 10102 Feet MD	Azimuth Lateral 1 2.27 °		Estimated Total Depth Lateral 1 23199 Feet MD 10543 Feet TVD		
Latitude of Well Head 48 ° 04 ' 31.64 "		Longitude of Well Head -103 ° 40 ' 11.00 "		NAD Reference NAD83	Description of (Subject to NDIC Approval) Spacing Unit: Sec 4, 9, 16, & 21 153 101		
Ground Elevation 1920 Feet Above S.L.	Acres in Spacing/Drilling Unit 2560	Spacing/Drilling Unit Setback Requirement 200 Feet N/S 500 Feet E/W			Industrial Commission Order 1983221551		
North Line of Spacing/Drilling Unit 5279 Feet	South Line of Spacing/Drilling Unit 5300 Feet	East Line of Spacing/Drilling Unit 20988 Feet			West Line of Spacing/Drilling Unit 20969 Feet		
Objective Horizons Middle Bakken						Pierre Shale Top 1841	
Proposed Surface Casing	Size 9 - 5/8 "	Weight 36 Lb./Ft.	Depth 1940 Feet	Cement Volume 732 Sacks	NOTE: Surface hole must be drilled with fresh water and surface casing must be cemented back to surface.		
Proposed Longstring Casing	Size 7 - "	Weight(s) 26-32 Lb./Ft.	Longstring Total Depth 10952 Feet MD 10543 Feet TVD		Cement Volume 1139 Sacks	Cement Top 0 Feet	Top Dakota Sand 4826 Feet
Base Last Charles Salt (If Applicable) 9031 Feet		NOTE: Intermediate or longstring casing string must be cemented above the top Dakota Group Sand.					
Proposed Logs CBL/GR from deepest depth obtainable to ground surface/mud							
Drilling Mud Type (Vertical Hole - Below Surface Casing) Invert				Drilling Mud Type (Lateral) Brine			
Survey Type in Vertical Portion of Well MWD Every 100 Feet		Survey Frequency: Build Section 30 Feet		Survey Frequency: Lateral 90 Feet		Survey Contractor LEAM Drilling Services	

NOTE: A Gamma Ray log must be run to ground surface and a CBL must be run on intermediate or longstring casing string if set.

Surveys are required at least every 30 feet in the build section and every 90 feet in the lateral section of a horizontal well. Measurement inaccuracies are not considered when determining compliance with the spacing/drilling unit boundary setback requirement except in the following scenarios: 1) When the angle between the well bore and the respective boundary is 10 degrees or less; or 2) If Industry standard methods and equipment are not utilized. Consult the applicable field order for exceptions.

If measurement inaccuracies are required to be considered, a 2° MWD measurement inaccuracy will be applied to the horizontal portion of the well bore. This measurement inaccuracy is applied to the well bore from KOP to TD.

REQUIRED ATTACHMENTS: Certified surveyor's plat, horizontal section plat, estimated geological tops, proposed mud/cementing plan, directional plot/plan, \$100 fee.
See Page 2 for Comments section and signature block.

COMMENTS, ADDITIONAL INFORMATION, AND/OR LIST OF ATTACHMENTS

--

Lateral 2

KOP Lateral 2 Feet MD	Azimuth Lateral 2 °	Estimated Total Depth Lateral 2 Feet MD Feet TVD	KOP Coordinates From Well Head From WH From WH		
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

Lateral 3

KOP Lateral 3 Feet MD	Azimuth Lateral 3 °	Estimated Total Depth Lateral 3 Feet MD Feet TVD	KOP Coordinates From Well Head From WH From WH		
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

Lateral 4

KOP Lateral 4 Feet MD	Azimuth Lateral 4 °	Estimated Total Depth Lateral 4 Feet MD Feet TVD	KOP Coordinates From Well Head From WH From WH		
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

Lateral 5

KOP Lateral 5 Feet MD	Azimuth Lateral 5 °	Estimated Total Depth Lateral 5 Feet MD Feet TVD	KOP Coordinates From Well Head From WH From WH		
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

I hereby swear or affirm the information provided is true, complete and correct as determined from all available records.		Date 6 / 12 / 2012
ePermit	Printed Name Terry L. Olson	Title Regulatory Compliance Specialist

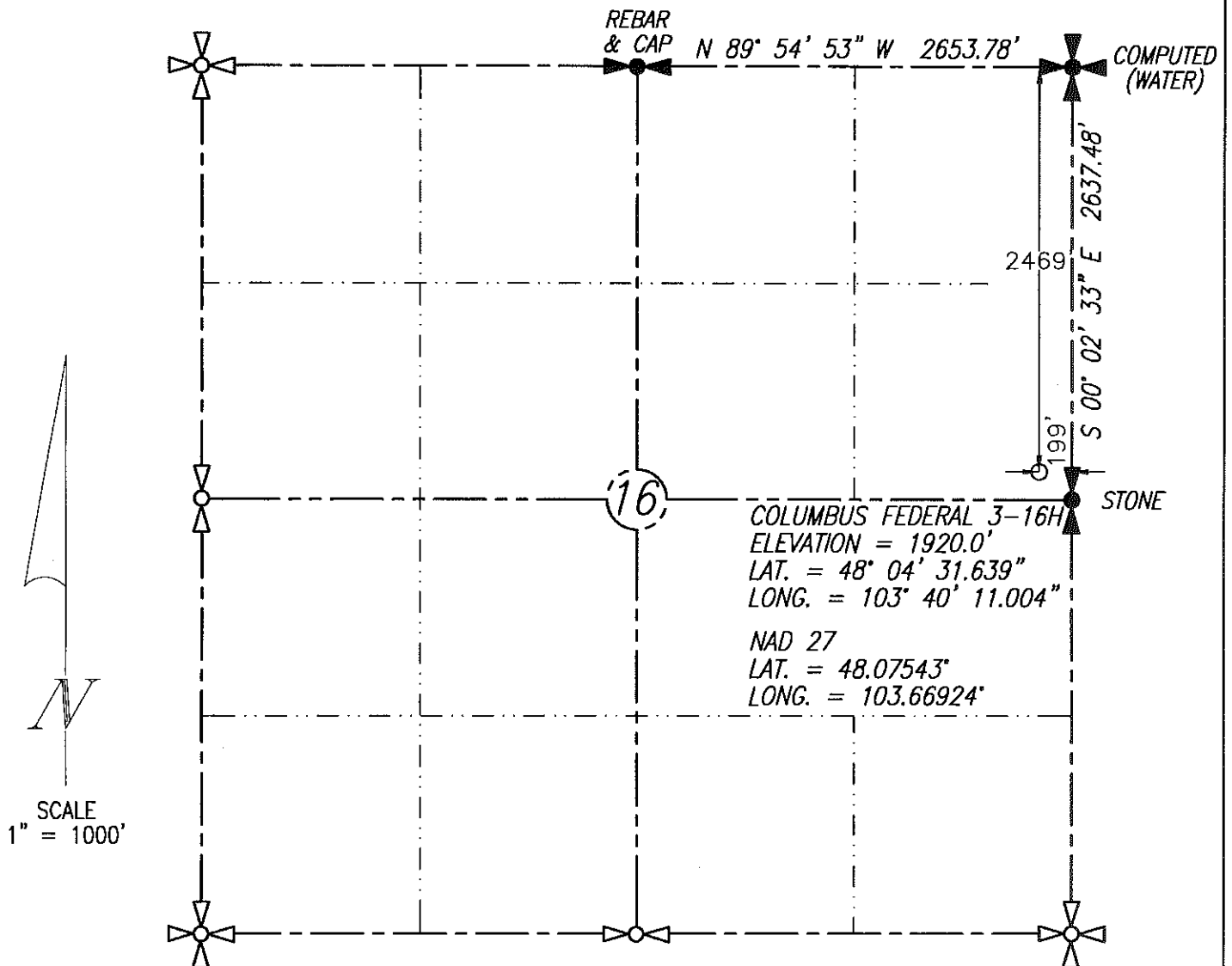
FOR STATE USE ONLY

Permit and File Number 25160	API Number 33 - 053 - 04856
Field BAKER	
Pool BAKKEN	Permit Type DEVELOPMENT

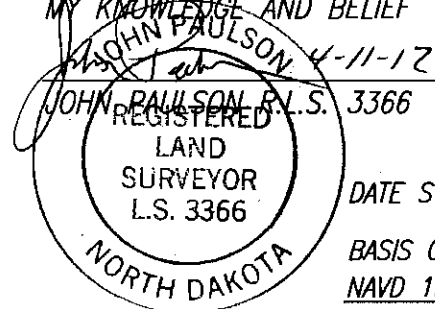
FOR STATE USE ONLY

Date Approved 3 / 11 / 2013
By David Tabor
Title Engineering Technician IV

WELL LOCATION PLAT
CONTINENTAL RESOURCES INC.
COLUMBUS FEDERAL 3-16H
SECTION 16, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA
2469' FNL & 199' FEL



I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS
WORK PERFORMED BY ME OR UNDER MY RESPONSIBLE
CHARGE, AND IS TRUE AND CORRECT TO THE BEST OF
MY KNOWLEDGE AND BELIEF



DATE STAKED: 1-4-2012
BASIS OF VERTICAL DATUM:
NAVD 1988 GEOID 09

PERSON AUTHORIZING SURVEY;
CHAD NEWBY

EXPLANATION AREA: NAD83(CORS96)

BASIS OF BEARING: TRUE NORTH

BROSZ ENGINEERING INC.

BOX 357
BOWMAN, N.D. 58623
PHONE: 701-523-3340
FAX: 701-523-5243
PROJECT NO. 12-10

GEOLOGIC PROGNOSIS

Well Name:

Rig:

Prospect:

Target:

Spacing:

Columbus Fed. 3-16H

Cyclone 20

Williston

Middle Bakken

1280

SHL: 2469' FNL & 199' FEL

Sec. 16 - 153N - 101W

McKenzie , ND

BHL: 200' FNL & 660' FEL

Sec. 4 - 153N - 101W

McKenzie , ND

☐ Pre-Staked

☒ Staked

Rig Grade Elevation:

KB:

RKB:

1920'

21'

1941'

FORMATION	SUBSEA	TVD
Pierre Shale	100	1,841
Greenhorn	-2,484	4,425
Dakota Group (fka Mowry)	-2,885	4,826
Base of Dakota Sand	-3,764	5,705
Dunham Salt Top	-4,792	6,733
Dunham Salt Base	-4,852	6,793
Pine Salt Top	-5,162	7,103
Pine Salt Base	-5,218	7,159
Minnekahta	-5,244	7,185
Opeche Salt Top	NA	
Opeche Salt Base	NA	
Minnelusa Group	-5,530	7,471
Tyler	-5,686	7,627
Kibbey	-6,238	8,179
Top Charles	-6,392	8,333
Base Last Charles Salt	-7,090	9,031
Mission Canyon	-7,314	9,255
Lodgepole	-7,880	9,821
Upper Bakken Shale	-8,570	10,511
Middle Bakken Member	-8,587	10,528
Middle Bakken Target	-8,602	10,543
End of Lateral	-8,602	10,543

==>

20' into MB

up 30' and down 30'

Not Flat

DRILLING PROGRAM

06/08/12

Lease and Well No. Columbus Fed. 3-16H

MUD PROGRAM

Depth	Type	Weight	Remarks
0 ' - 1940 '	Fresh water	8.4-8.8	Add Soap Sticks for Mud Rings
1940 ' - 6500 '	Invert	9.3-9.5	35-50 sec, 10-30 cc's
6500 ' - 13052 '	Invert	9.6-10.0	40-55 sec, 10-15 cc's O/W 70/30 to 80/20
13052 ' - 23107 '	Brine	8.7-10.0	Cuttings pit

TUBULAR PROGRAM

String Type	Hole Size	Depth	Feet	Casing Diameter	Weight, Grade, Connection	ERW/ Seamless	Critical Inspection
Surf	13 1/2 "	1940 '	1940 '	9 5/8 "	9-5/8", 36 #, J-55, STC	ERW	BCI & Drift
Float shoe, shoe joint & float collar. Centralize bottom 3 jts and every 4th jt to surface.							
Int	8 3/4 "	80 '	80 '	7 "	7", 32#, P-110 IC, LTC	ERW	BCI & Drift
		4000 '	3920 '	7 "	7", 26#, P-110 IC, LTC	ERW	BCI & Drift
		8130 '	4130 '	7 "	7", 29#, P-110 IC, LTC	ERW	BCI & Drift
		9230 '	1100 '	7 "	7", 32#, P-110 IC, LTC	ERW	BCI & Drift
		13052 '	3822 '	7 "	7", 29#, P-110 IC, LTC	ERW	BCI & Drift
Float shoe, shoe joint & float collar. Centralize bottom 3 joints. Centralize thru curve and across all salts.							
Liner	6 "	23107 '	13005 '	4 1/2 "	4-1/2", 11.6 #, P-110, BTC		
Tubing		10100 '	10100 '	2 7/8 "	2-7/8", 6.5 #, L-80, EUE		

Notes: Pipe to end up in hole from top to bottom as shown.

CEMENT PROGRAM

String Type	SHOE/DV Depth	Stage Lead/Tail	Cement Bottom	Cement Top	No Sacks	Cement System	Cement Yield	Cement Weight
		Lead	1360 '	0 '	431	35/65 Poz/Class "C", 3% CaCl, 12% gel	2.39	12
Surf	1940	Tail	1940 '	1360 '	301	Class "C", 2% CaCl	1.46	14.3
(Basis: Gauge hole + 55% excess, tail 30% of length, lead to surface.)								
Int	13052	Lead	7830 '	0 '	473	35/65 Poz/Class "C", 3% KCl, 5 #/sk Silica	3.21	11.3
		Tail	13052 '	7830 '	666	Class "G", 3% KCl, 35% Silica	1.59	15.6
(Basis: Gauge hole + 35% excess, Tail to 500 ft above top of Charles Salt, Lead to Surface)								

BOP PROGRAM

Hole Size	Configuration	Pressure Rating	BOP Test Press	Casing Test Press
13 1/2 "	Circulate Conductor			
8 3/4 "	RRA	5M	5000	
6 "	RRA	5M	5000	
Utilize commercial testers every 30 days.				

LOGGING PROGRAM

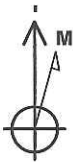
FROM	TO	TOOLS
BLS	TD	2 man mud logger trailer, 30 ft samples
Surf csg	TD	MWD Surveys, 90 ft & 30 ft thru curve
KOP	TD	MWD Gamma Ray
Surface	DDO	CBL, CCL, Gr

Remarks: 4 1/2 inch Liner to be run. Liner Top @ KOP (10102 ft TVD) and 29 swell packers to be run with liner.
Liner top to be tested successfully to 4500 psi prior to rigging down.
Casing Caliper to be run to determine if a frac string is needed.

Prepared By: Pat McCollom Date: 05/02/12

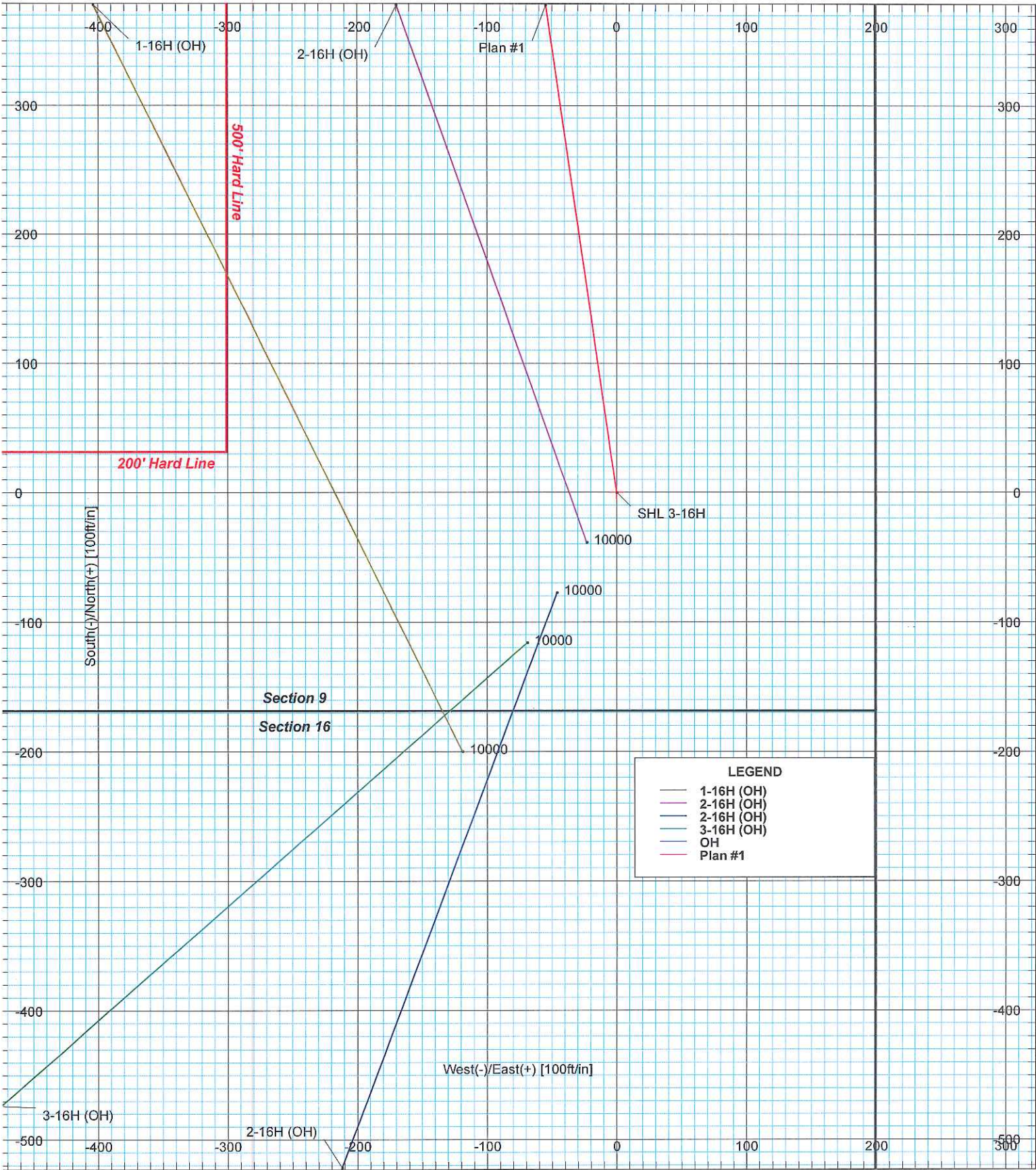
Continental Resources

Field: McKenzie County, ND
Site: Columbus Federal 3
Well: 3-16H
Wellpath: OH
Plan: Plan #1



Azimuths to True North
Magnetic North: 8.65°

Magnetic Field
Strength: 56676nT
Dip Angle: 73.09°
Date: 04/18/2012
Model: IGRF2010



LEAM DRILLING SYSTEMS, INC.
2010 East Davis Conroe, Texas 77301
Phone: 936-756-7577 Fax: 936-756-7595

Plan: Plan #1 (3-16H/OH)
Created By: Justin Andoe Date: 04/23/2012
Checked: Date:
Reviewed: Date:

Continental Resources, Inc. (CRI) respectfully submits the following information concerning the prevention of a wellbore collision while drilling the Columbus Federal 3-16H, SENE of Sec. 16, T153N, R101W, Dunn County, North Dakota.

The Mildred 1, NDIC File No. 12306, is a plugged and abandoned, Red River formation, directional well with a SHL of 1325' FSL, 660' FEL of Sec. 4, T153N, R101W. The directional wellbore lies approximately 660' FEL of Sec. 4, T153N, R101W, terminating at a BHL 769' N. of SHL, 2094' FSL of Sec. 4, T153N, R101W. Bottom hole TVD of the existing directional well is 13451'. The deepest casing string is 8-5/8" set at a depth of 3035' TVD.

The proposed Columbus Federal 3-16H is a horizontal well targeting the Middle Bakken formation. Surface hole location for the Columbus Federal 3-16H is 2469' FSL and 199' FEL of Sec. 16, T153N, R101W. The majority of the Columbus Federal 3-16H lateral will lie 760' FEL in order to avoid the existing Mildred 1 wellbore by approximately 100'. The lateral will taper back to 660' FEL, at 10155' N. of SHL, after adequately clearing the segment of the Mildred 1 wellbore within this Baker-Bakken pool.

The Baker-Bakken pool, as defined by NDIC Order No. 21551, is the interval from 50 feet above the top of the Bakken Formation to above the top of the Birdbear Formation. CRI has assessed the geology of the area and has determined this Bakken interval to be approximately 10461' TVD to 10,771' TVD. On the surface planar, the Mildred 1 wellbore exist in this interval for approximately 38', from 1874' FSL to 1912' FSL of Sec. 4, T153N, R94W.

The completion of the Columbus Federal 3-16H will protect the integrity of the Mildred 1 wellbore. Stimulation plans will be made to include a buffer zone with swell packers of a minimum 500' from either side of the Mildred 1 wellbore existing within the Baker-Bakken pool. This zone will not be perforated or completed.

CRI believes adequate precautions have been taken to prevent the possibility of a wellbore collision and accepts all responsibility should such a collision occur.


Sarah Madden, PD Engineer
Continental Resources, Inc.

STATE OF OKLAHOMA)
)ss:
COUNTY OF OKLAHOMA)

On the 6th day of March 2013, before me, a Notary Public in and for said County and State, personally appeared **Sarah Madden**, known to me to be a PD Engineer of Continental Resources, Inc., the Corporation that executed the within instrument, and acknowledged to me that such Corporation executed the same.


Notary Public

Oklahoma County, Oklahoma
My Commission Expires: 7/5/2015
Commission No.: 11006023





To: Todd Holweger, NDIC

From: Shawn Svob

Date: 4/5/2012

Re: Continental Resources standard CCL, CBL, 4-1/2" liner running and testing procedures

Continental Resources' standard practice for running the cement bond log and casing caliper log is to run both logs immediately after coming out of the hole after TD, prior to running the 4-1/2" liner, to the deepest depth obtainable; however, if there are well control concerns that require us to run the liner sooner, only the CBL will be run and the CCL will be run after setting the liner.

Based on the CCL results, we determine the actual API minimum burst allowance for the 7" casing. If the downgraded API burst pressure is below our minimum required frac pressures, we will run a 4-1/2" frac string; if severe wear or holes are found in the casing, we will run a 5" cemented, to surface, tie back string.

The CBL log is run in order to determine the top of cement, as required by the NDIC. Our current 4-1/2" liner program for a 1280 unit is 30, evenly spaced, stages with 29 swellable packers. The liner shoe is set approximately 180 feet off bottom. The shoe stage below the last packer has 2 joints, a double valved float, one joint, and a ported guide shoe – appx 130 ft. The liner is run using a running tool on the end of 4" DP. The 7" packer/hanger is set about 40 ft above KOP between two casing collars but conditions occasionally occur that require setting higher, either through unexpected failure or in order to isolate casing wear close to KOP. Recently we have tried 40 stage liners and the trend to explore the optimum stage count will continue.. Once the liner is at depth, a ball is dropped through the DP, the ball is pressured up against the setting tool to approximately 2500 psi, and the 7" packer/hanger is set.

A push pull test is done to confirm the hanger has set. Then, a 4500 psi pressure test is completed on the back side of the 4" DP to confirm the packer has set. The setting tool is then backed off and the 4" DP/running tool is laid down.

Immediately after the rotary rig has been moved off the well location, the 7" csg and liner packer/ hanger are tested to the frac pressure. The testers will rig up and test the tubing head to 5000 psi. Next a test plug will be run and set, using wire line, in the top of the 7" packer/hanger. Testers will pressure up to our frac pressure, typically 8500 psi, to confirm the 7" is ready for completion.

Shawn Svob
Drilling Operations Coordinator

Continental Resources

Field: McKenzie County, ND
Site: Columbus Federal 3
Well: 3-16H
Wellpath: OH
Plan: Plan #3



Azimuths to True North
Magnetic North: 8.65°

Magnetic Field
Strength: 56676nT
Dip Angle: 73.09°
Date: 04/18/2012
Model: IGRF2010



SITE DETAILS

Columbus 3
Sec. 16 - T153N - R101W
SHL 2469' FNL & 199' FEL
PBHL 200' FNL & 1980' FEL

Latitude: 48°04'31.639N
Longitude: 103°40'11.004W

Ground Level: 1920.00
KB: 21.00

CASING DETAILS

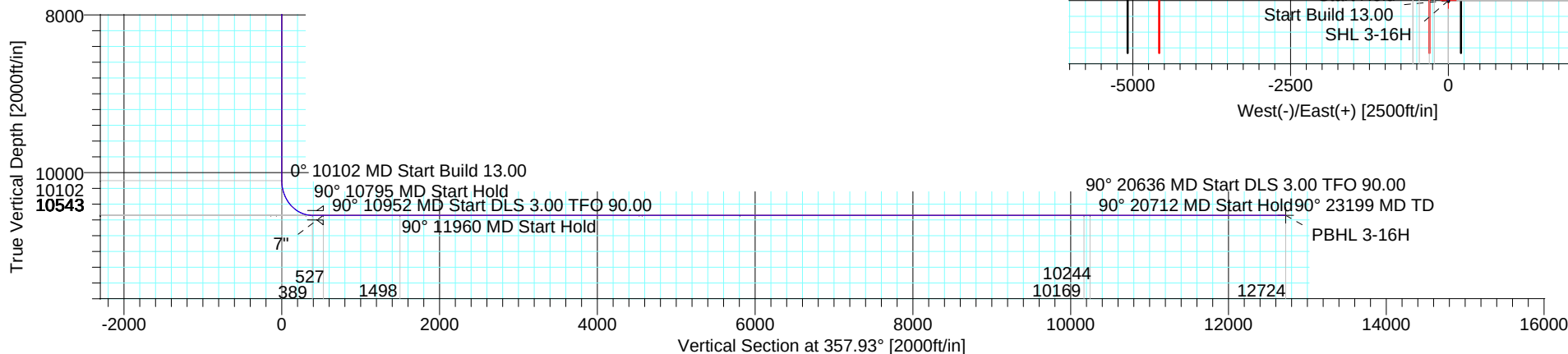
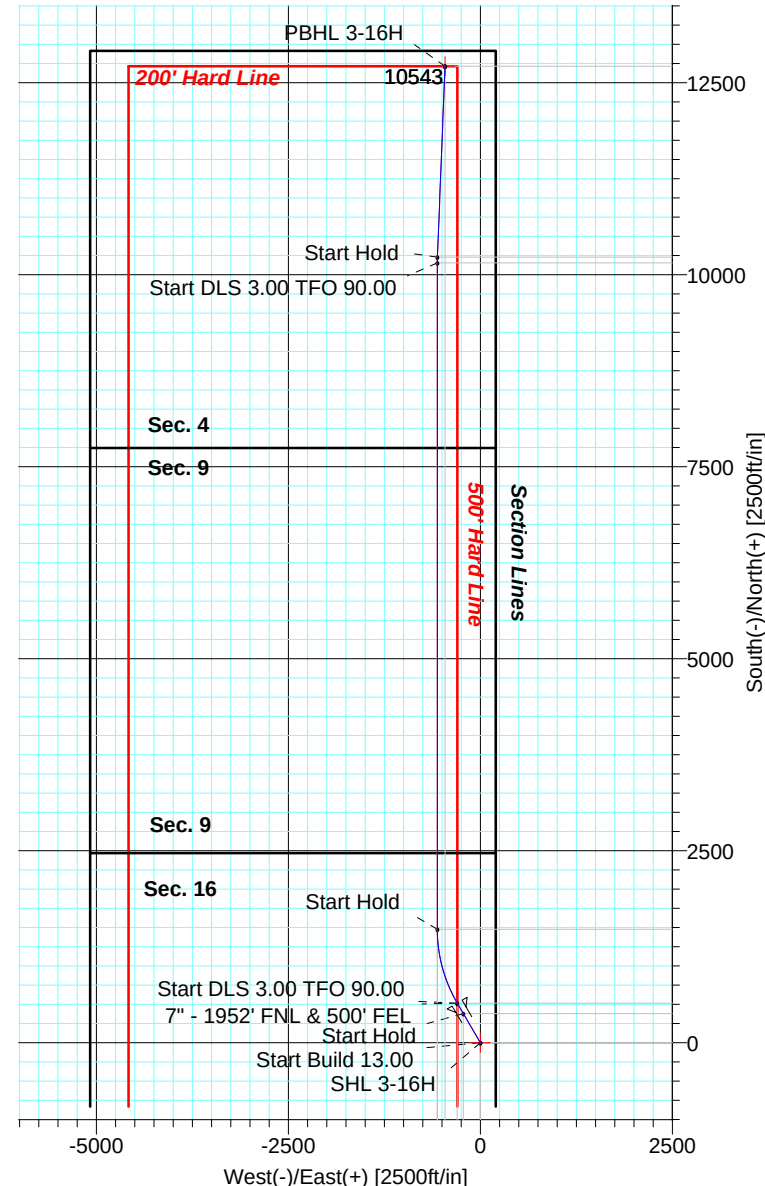
No.	TVD	MD	Name	Size
1	10543.00	10951.72	7"	7.00

TARGET DETAILS

Name	TVD	+N/-S	+E/-W	Northing	Easting	Latitude	Longitude	Shape
SHL 3-16H	0.00	0.00	0.00	408237.43	1193849.30	48°04'31.639N	103°40'11.004W	Point
PBHL 3-16H10543.00		12716.00	-461.00	420961.63	1193912.02	48°06'37.133N	103°40'17.797W	Point

SECTION DETAILS

Sec	MD	Inc	Azi	TVD	+N/-S	+E/-W	DLeg	TFace	VSec	Target
1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
2	10102.26	0.00	0.00	10102.26	0.00	0.00	0.00	0.00	0.00	
3	10794.57	90.00	329.76	10543.00	380.76	-221.97	13.00	329.76	388.53	
4	10951.72	90.00	329.76	10543.00	516.53	-301.11	0.00	0.00	527.07	
5	11959.72	90.00	360.00	10543.00	1478.38	-561.00	3.00	90.00	1497.68	
6	20636.34	90.00	360.00	10543.00	10155.00	-561.00	0.00	0.00	10168.64	
7	20711.99	90.00	2.27	10543.00	10230.63	-559.50	3.00	90.00	10244.17	
8	23199.31	90.00	2.27	10543.00	12716.00	-461.00	0.00	0.00	12724.35	PBHL 3-16H



LEAM DRILLING SYSTEMS LLC
2010 East Davis, Conroe, Texas 77301
Phone: 936-756-7577 Fax: 936-756-7595

Plan: Plan #3 (3-16H/OH)

Created By: David Vogler Date: 03/05/2013
Checked: _____ Date: _____
Reviewed: _____ Date: _____
Approved: _____ Date: _____

LEAM Drilling Systems LLC

Planning Report

Company: Continental Resources Field: McKenzie County, ND Site: Columbus Federal 3 Well: 3-16H Wellpath: OH	Date: 03/05/2013 Co-ordinate(NE) Reference: Vertical (TVD) Reference: Section (VS) Reference: Survey Calculation Method:	Time: 16:13:18 Well: 3-16H, True North GL 1920+KB 21 1941.0 Well (0.00N,0.00E,357.93Azi) Minimum Curvature	Page: 1 Db: Adapti
--	---	---	-------------------------------------

Field: McKenzie County, ND			
Map System: US State Plane Coordinate System 1983		Map Zone: North Dakota, Northern Zone	
Geo Datum: GRS 1980		Coordinate System: Well Centre	
Sys Datum: Mean Sea Level		Geomagnetic Model: IGRF2010	

Site: Columbus Federal 3 Sec. 16 - T153N - R101W 2469' FNL & 199' FEL			
Site Position:	Northing: 408237.43 ft	Latitude: 48 4 31.639 N	
From: Geographic	Easting: 1193849.30 ft	Longitude: 103 40 11.004 W	
Position Uncertainty: 0.00 ft		North Reference: True	
Ground Level: 1920.00 ft		Grid Convergence: -2.36 deg	

Well: 3-16H				Slot Name:			
Well Position:	+N/-S 0.00 ft	Northing: 408237.43 ft	Latitude: 48 4 31.639 N	+E/-W 0.00 ft	Easting : 1193849.30 ft	Longitude: 103 40 11.004 W	
Position Uncertainty: 0.00 ft							

Wellpath: OH			
Current Datum: GL 1920+KB 21	Height 1941.00 ft	Drilled From: Surface	
Magnetic Data: 04/18/2012		Tie-on Depth: 0.00 ft	
Field Strength: 56676 nT		Above System Datum: Mean Sea Level	
Vertical Section:		Declination: 8.65 deg	
Depth From (TVD)	+N/-S	Mag Dip Angle: 73.09 deg	
ft	ft	+E/-W	Direction
		ft	deg
0.00	0.00	0.00	357.93

Plan: Plan #3		Date Composed: 03/05/2013	
Principal: Yes		Version: 1	
		Tied-to: User Defined	

Plan Section Information										
MD ft	Incl deg	Azim deg	TVD ft	+N/-S ft	+E/-W ft	DLS deg/100ft	Build deg/100ft	Turn deg/100ft	TFO deg	Target
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
10102.26	0.00	0.00	10102.26	0.00	0.00	0.00	0.00	0.00	0.00	
10794.57	90.00	329.76	10543.00	380.76	-221.97	13.00	13.00	0.00	329.76	
10951.72	90.00	329.76	10543.00	516.53	-301.11	0.00	0.00	0.00	0.00	
11959.72	90.00	360.00	10543.00	1478.38	-561.00	3.00	0.00	3.00	90.00	
20636.34	90.00	360.00	10543.00	10155.00	-561.00	0.00	0.00	0.00	0.00	
20711.99	90.00	2.27	10543.00	10230.63	-559.50	3.00	0.00	3.00	90.00	
23199.31	90.00	2.27	10543.00	12716.00	-461.00	0.00	0.00	0.00	0.00	PBHL 3-16H

Survey										
MD ft	Incl deg	Azim deg	TVD ft	N/S ft	E/W ft	VS ft	DLS deg/100ft	Build deg/100ft	Turn deg/100ft	Tool/Comment
10102.26	0.00	0.00	10102.26	0.00	0.00	0.00	0.00	0.00	0.00	
10125.00	2.96	329.76	10124.99	0.51	-0.30	0.52	13.00	13.00	0.00	
10150.00	6.21	329.76	10149.91	2.23	-1.30	2.28	13.00	13.00	0.00	
10175.00	9.46	329.76	10174.67	5.17	-3.02	5.28	13.00	13.00	0.00	
10200.00	12.71	329.76	10199.20	9.32	-5.44	9.51	13.00	13.00	0.00	
10225.00	15.96	329.76	10223.42	14.67	-8.55	14.97	13.00	13.00	0.00	
10250.00	19.21	329.76	10247.25	21.19	-12.35	21.63	13.00	13.00	0.00	
10275.00	22.46	329.76	10270.61	28.87	-16.83	29.46	13.00	13.00	0.00	
10300.00	25.71	329.76	10293.43	37.68	-21.97	38.45	13.00	13.00	0.00	
10325.00	28.96	329.76	10315.64	47.60	-27.75	48.57	13.00	13.00	0.00	
10350.00	32.21	329.76	10337.16	58.59	-34.15	59.78	13.00	13.00	0.00	
10375.00	35.46	329.76	10357.92	70.61	-41.16	72.05	13.00	13.00	0.00	
10400.00	38.71	329.76	10377.86	83.63	-48.75	85.34	13.00	13.00	0.00	
10425.00	41.96	329.76	10396.92	97.61	-56.90	99.60	13.00	13.00	0.00	

LEAM Drilling Systems LLC

Planning Report

Company: Continental Resources	Date: 03/05/2013	Time: 16:13:18	Page: 2
Field: McKenzie County, ND	Co-ordinate(NE) Reference: Well: 3-16H, True North		
Site: Columbus Federal 3	Vertical (TVD) Reference: GL 1920+KB 21 1941.0		
Well: 3-16H	Section (VS) Reference: Well (0.00N,0.00E,357.93Azi)		
Wellpath: OH	Survey Calculation Method: Minimum Curvature	Db: Adapti	

Survey

MD ft	Incl deg	Azim deg	TVD ft	N/S ft	E/W ft	VS ft	DLS deg/100ft	Build deg/100ft	Turn deg/100ft	Tool/Comment
10450.00	45.21	329.76	10415.03	112.49	-65.58	114.79	13.00	13.00	0.00	
10475.00	48.46	329.76	10432.13	128.24	-74.76	130.86	13.00	13.00	0.00	
10500.00	51.71	329.76	10448.17	144.81	-84.41	147.76	13.00	13.00	0.00	
10525.00	54.96	329.76	10463.10	162.13	-94.51	165.44	13.00	13.00	0.00	
10550.00	58.21	329.76	10476.86	180.15	-105.02	183.83	13.00	13.00	0.00	
10575.00	61.46	329.76	10489.43	198.82	-115.90	202.88	13.00	13.00	0.00	
10600.00	64.71	329.76	10500.74	218.08	-127.13	222.53	13.00	13.00	0.00	
10625.00	67.96	329.76	10510.78	237.86	-138.66	242.71	13.00	13.00	0.00	
10625.59	68.03	329.76	10511.00	238.33	-138.94	243.19	13.00	13.00	0.00	Upper Bakken Shale
10650.00	71.21	329.76	10519.50	258.10	-150.46	263.36	13.00	13.00	0.00	
10675.00	74.46	329.76	10526.88	278.73	-162.48	284.42	13.00	13.00	0.00	
10679.26	75.01	329.76	10528.00	282.28	-164.56	288.04	13.00	13.00	0.00	Middle Bakken Member
10700.00	77.71	329.76	10532.89	299.69	-174.70	305.80	13.00	13.00	0.00	
10725.00	80.96	329.76	10537.52	320.91	-187.07	327.46	13.00	13.00	0.00	
10750.00	84.21	329.76	10540.75	342.33	-199.56	349.31	13.00	13.00	0.00	
10775.00	87.46	329.76	10542.56	363.86	-212.11	371.29	13.00	13.00	0.00	
10794.57	90.00	329.76	10543.00	380.76	-221.97	388.53	13.00	13.00	0.00	Middle Bakken Target
10800.00	90.00	329.76	10543.00	385.46	-224.70	393.32	0.00	0.00	0.00	
10900.00	90.00	329.76	10543.00	471.85	-275.06	481.48	0.00	0.00	0.00	
10951.72	90.00	329.76	10543.00	516.53	-301.11	527.07	0.00	0.00	0.00	7"
11000.00	90.00	331.21	10543.00	558.54	-324.90	569.91	3.00	0.00	3.00	
11100.00	90.00	334.21	10543.00	647.40	-370.74	660.37	3.00	0.00	3.00	
11200.00	90.00	337.21	10543.00	738.54	-411.88	752.93	3.00	0.00	3.00	
11300.00	90.00	340.21	10543.00	831.70	-448.18	847.35	3.00	0.00	3.00	
11400.00	90.00	343.21	10543.00	926.64	-479.56	943.36	3.00	0.00	3.00	
11500.00	90.00	346.21	10543.00	1023.09	-505.93	1040.69	3.00	0.00	3.00	
11600.00	90.00	349.21	10543.00	1120.78	-527.22	1139.10	3.00	0.00	3.00	
11700.00	90.00	352.21	10543.00	1219.46	-543.36	1238.29	3.00	0.00	3.00	
11800.00	90.00	355.21	10543.00	1318.85	-554.32	1338.01	3.00	0.00	3.00	
11900.00	90.00	358.21	10543.00	1418.67	-560.06	1437.97	3.00	0.00	3.00	
11959.72	90.00	360.00	10543.00	1478.38	-561.00	1497.68	3.00	0.00	3.00	
12000.00	90.00	360.00	10543.00	1518.66	-561.00	1537.93	0.00	0.00	0.00	
12100.00	90.00	360.00	10543.00	1618.66	-561.00	1637.87	0.00	0.00	0.00	
12200.00	90.00	360.00	10543.00	1718.66	-561.00	1737.80	0.00	0.00	0.00	
12300.00	90.00	360.00	10543.00	1818.66	-561.00	1837.74	0.00	0.00	0.00	
12400.00	90.00	360.00	10543.00	1918.66	-561.00	1937.67	0.00	0.00	0.00	
12500.00	90.00	360.00	10543.00	2018.66	-561.00	2037.61	0.00	0.00	0.00	
12600.00	90.00	360.00	10543.00	2118.66	-561.00	2137.54	0.00	0.00	0.00	
12700.00	90.00	360.00	10543.00	2218.66	-561.00	2237.48	0.00	0.00	0.00	
12800.00	90.00	360.00	10543.00	2318.66	-561.00	2337.41	0.00	0.00	0.00	
12900.00	90.00	360.00	10543.00	2418.66	-561.00	2437.35	0.00	0.00	0.00	
13000.00	90.00	360.00	10543.00	2518.66	-561.00	2537.28	0.00	0.00	0.00	
13100.00	90.00	360.00	10543.00	2618.66	-561.00	2637.21	0.00	0.00	0.00	
13200.00	90.00	360.00	10543.00	2718.66	-561.00	2737.15	0.00	0.00	0.00	
13300.00	90.00	360.00	10543.00	2818.66	-561.00	2837.08	0.00	0.00	0.00	
13400.00	90.00	360.00	10543.00	2918.66	-561.00	2937.02	0.00	0.00	0.00	
13500.00	90.00	360.00	10543.00	3018.66	-561.00	3036.95	0.00	0.00	0.00	
13600.00	90.00	360.00	10543.00	3118.66	-561.00	3136.89	0.00	0.00	0.00	
13700.00	90.00	360.00	10543.00	3218.66	-561.00	3236.82	0.00	0.00	0.00	
13800.00	90.00	360.00	10543.00	3318.66	-561.00	3336.76	0.00	0.00	0.00	
13900.00	90.00	360.00	10543.00	3418.66	-561.00	3436.69	0.00	0.00	0.00	
14000.00	90.00	360.00	10543.00	3518.66	-561.00	3536.63	0.00	0.00	0.00	

LEAM Drilling Systems LLC

Planning Report

Company: Continental Resources Field: McKenzie County, ND Site: Columbus Federal 3 Well: 3-16H Wellpath: OH	Date: 03/05/2013 Co-ordinate(NE) Reference: Well: 3-16H, True North Vertical (TVD) Reference: GL 1920+KB 21 1941.0 Section (VS) Reference: Well (0.00N,0.00E,357.93Azi) Survey Calculation Method: Minimum Curvature Db: Adapti
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Survey

MD ft	Incl deg	Azim deg	TVD ft	N/S ft	E/W ft	VS ft	DLS deg/100ft	Build deg/100ft	Turn deg/100ft	Tool/Comment
14100.00	90.00	360.00	10543.00	3618.66	-561.00	3636.56	0.00	0.00	0.00	
14200.00	90.00	360.00	10543.00	3718.66	-561.00	3736.50	0.00	0.00	0.00	
14300.00	90.00	360.00	10543.00	3818.66	-561.00	3836.43	0.00	0.00	0.00	
14400.00	90.00	360.00	10543.00	3918.66	-561.00	3936.37	0.00	0.00	0.00	
14500.00	90.00	360.00	10543.00	4018.66	-561.00	4036.30	0.00	0.00	0.00	
14600.00	90.00	360.00	10543.00	4118.66	-561.00	4136.24	0.00	0.00	0.00	
14700.00	90.00	360.00	10543.00	4218.66	-561.00	4236.17	0.00	0.00	0.00	
14800.00	90.00	360.00	10543.00	4318.66	-561.00	4336.11	0.00	0.00	0.00	
14900.00	90.00	360.00	10543.00	4418.66	-561.00	4436.04	0.00	0.00	0.00	
15000.00	90.00	360.00	10543.00	4518.66	-561.00	4535.97	0.00	0.00	0.00	
15100.00	90.00	360.00	10543.00	4618.66	-561.00	4635.91	0.00	0.00	0.00	
15200.00	90.00	360.00	10543.00	4718.66	-561.00	4735.84	0.00	0.00	0.00	
15300.00	90.00	360.00	10543.00	4818.66	-561.00	4835.78	0.00	0.00	0.00	
15400.00	90.00	360.00	10543.00	4918.66	-561.00	4935.71	0.00	0.00	0.00	
15500.00	90.00	360.00	10543.00	5018.66	-561.00	5035.65	0.00	0.00	0.00	
15600.00	90.00	360.00	10543.00	5118.66	-561.00	5135.58	0.00	0.00	0.00	
15700.00	90.00	360.00	10543.00	5218.66	-561.00	5235.52	0.00	0.00	0.00	
15800.00	90.00	360.00	10543.00	5318.66	-561.00	5335.45	0.00	0.00	0.00	
15900.00	90.00	360.00	10543.00	5418.66	-561.00	5435.39	0.00	0.00	0.00	
16000.00	90.00	360.00	10543.00	5518.66	-561.00	5535.32	0.00	0.00	0.00	
16100.00	90.00	360.00	10543.00	5618.66	-561.00	5635.26	0.00	0.00	0.00	
16200.00	90.00	360.00	10543.00	5718.66	-561.00	5735.19	0.00	0.00	0.00	
16300.00	90.00	360.00	10543.00	5818.66	-561.00	5835.13	0.00	0.00	0.00	
16400.00	90.00	360.00	10543.00	5918.66	-561.00	5935.06	0.00	0.00	0.00	
16500.00	90.00	360.00	10543.00	6018.66	-561.00	6035.00	0.00	0.00	0.00	
16600.00	90.00	360.00	10543.00	6118.66	-561.00	6134.93	0.00	0.00	0.00	
16700.00	90.00	360.00	10543.00	6218.66	-561.00	6234.87	0.00	0.00	0.00	
16800.00	90.00	360.00	10543.00	6318.66	-561.00	6334.80	0.00	0.00	0.00	
16900.00	90.00	360.00	10543.00	6418.66	-561.00	6434.74	0.00	0.00	0.00	
17000.00	90.00	360.00	10543.00	6518.66	-561.00	6534.67	0.00	0.00	0.00	
17100.00	90.00	360.00	10543.00	6618.66	-561.00	6634.60	0.00	0.00	0.00	
17200.00	90.00	360.00	10543.00	6718.66	-561.00	6734.54	0.00	0.00	0.00	
17300.00	90.00	360.00	10543.00	6818.66	-561.00	6834.47	0.00	0.00	0.00	
17400.00	90.00	360.00	10543.00	6918.66	-561.00	6934.41	0.00	0.00	0.00	
17500.00	90.00	360.00	10543.00	7018.66	-561.00	7034.34	0.00	0.00	0.00	
17600.00	90.00	360.00	10543.00	7118.66	-561.00	7134.28	0.00	0.00	0.00	
17700.00	90.00	360.00	10543.00	7218.66	-561.00	7234.21	0.00	0.00	0.00	
17800.00	90.00	360.00	10543.00	7318.66	-561.00	7334.15	0.00	0.00	0.00	
17900.00	90.00	360.00	10543.00	7418.66	-561.00	7434.08	0.00	0.00	0.00	
18000.00	90.00	360.00	10543.00	7518.66	-561.00	7534.02	0.00	0.00	0.00	
18100.00	90.00	360.00	10543.00	7618.66	-561.00	7633.95	0.00	0.00	0.00	
18200.00	90.00	360.00	10543.00	7718.66	-561.00	7733.89	0.00	0.00	0.00	
18300.00	90.00	360.00	10543.00	7818.66	-561.00	7833.82	0.00	0.00	0.00	
18400.00	90.00	360.00	10543.00	7918.66	-561.00	7933.76	0.00	0.00	0.00	
18500.00	90.00	360.00	10543.00	8018.66	-561.00	8033.69	0.00	0.00	0.00	
18600.00	90.00	360.00	10543.00	8118.66	-561.00	8133.63	0.00	0.00	0.00	
18700.00	90.00	360.00	10543.00	8218.66	-561.00	8233.56	0.00	0.00	0.00	
18800.00	90.00	360.00	10543.00	8318.66	-561.00	8333.50	0.00	0.00	0.00	
18900.00	90.00	360.00	10543.00	8418.66	-561.00	8433.43	0.00	0.00	0.00	
19000.00	90.00	360.00	10543.00	8518.66	-561.00	8533.36	0.00	0.00	0.00	
19100.00	90.00	360.00	10543.00	8618.66	-561.00	8633.30	0.00	0.00	0.00	
19200.00	90.00	360.00	10543.00	8718.66	-561.00	8733.23	0.00	0.00	0.00	
19300.00	90.00	360.00	10543.00	8818.66	-561.00	8833.17	0.00	0.00	0.00	

LEAM Drilling Systems LLC

Planning Report

Company: Continental Resources	Date: 03/05/2013	Time: 16:13:18	Page: 4
Field: McKenzie County, ND	Co-ordinate(NE) Reference: Well: 3-16H, True North		
Site: Columbus Federal 3	Vertical (TVD) Reference: GL 1920+KB 21 1941.0		
Well: 3-16H	Section (VS) Reference: Well (0.00N,0.00E,357.93Azi)		
Wellpath: OH	Survey Calculation Method: Minimum Curvature	Db: Adapti	

Survey

MD ft	Incl deg	Azim deg	TVD ft	N/S ft	E/W ft	VS ft	DLS deg/100ft	Build deg/100ft	Turn deg/100ft	Tool/Comment
19400.00	90.00	360.00	10543.00	8918.66	-561.00	8933.10	0.00	0.00	0.00	
19500.00	90.00	360.00	10543.00	9018.66	-561.00	9033.04	0.00	0.00	0.00	
19600.00	90.00	360.00	10543.00	9118.66	-561.00	9132.97	0.00	0.00	0.00	
19700.00	90.00	360.00	10543.00	9218.66	-561.00	9232.91	0.00	0.00	0.00	
19800.00	90.00	360.00	10543.00	9318.66	-561.00	9332.84	0.00	0.00	0.00	
19900.00	90.00	360.00	10543.00	9418.66	-561.00	9432.78	0.00	0.00	0.00	
20000.00	90.00	360.00	10543.00	9518.66	-561.00	9532.71	0.00	0.00	0.00	
20100.00	90.00	360.00	10543.00	9618.66	-561.00	9632.65	0.00	0.00	0.00	
20200.00	90.00	360.00	10543.00	9718.66	-561.00	9732.58	0.00	0.00	0.00	
20300.00	90.00	360.00	10543.00	9818.66	-561.00	9832.52	0.00	0.00	0.00	
20400.00	90.00	360.00	10543.00	9918.66	-561.00	9932.45	0.00	0.00	0.00	
20500.00	90.00	360.00	10543.00	10018.66	-561.00	10032.39	0.00	0.00	0.00	
20600.00	90.00	360.00	10543.00	10118.66	-561.00	10132.32	0.00	0.00	0.00	
20636.34	90.00	360.00	10543.00	10155.00	-561.00	10168.64	0.00	0.00	0.00	
20700.00	90.00	1.91	10543.00	10218.65	-559.94	10232.21	3.00	0.00	3.00	
20711.99	90.00	2.27	10543.00	10230.63	-559.50	10244.17	3.00	0.00	3.00	
20800.00	90.00	2.27	10543.00	10318.57	-556.02	10331.92	0.00	0.00	0.00	
20900.00	90.00	2.27	10543.00	10418.49	-552.06	10431.63	0.00	0.00	0.00	
21000.00	90.00	2.27	10543.00	10518.41	-548.10	10531.35	0.00	0.00	0.00	
21100.00	90.00	2.27	10543.00	10618.34	-544.14	10631.06	0.00	0.00	0.00	
21200.00	90.00	2.27	10543.00	10718.26	-540.18	10730.77	0.00	0.00	0.00	
21300.00	90.00	2.27	10543.00	10818.18	-536.22	10830.49	0.00	0.00	0.00	
21400.00	90.00	2.27	10543.00	10918.10	-532.26	10930.20	0.00	0.00	0.00	
21500.00	90.00	2.27	10543.00	11018.02	-528.30	11029.91	0.00	0.00	0.00	
21600.00	90.00	2.27	10543.00	11117.94	-524.34	11129.63	0.00	0.00	0.00	
21700.00	90.00	2.27	10543.00	11217.87	-520.38	11229.34	0.00	0.00	0.00	
21800.00	90.00	2.27	10543.00	11317.79	-516.41	11329.05	0.00	0.00	0.00	
21900.00	90.00	2.27	10543.00	11417.71	-512.45	11428.77	0.00	0.00	0.00	
22000.00	90.00	2.27	10543.00	11517.63	-508.49	11528.48	0.00	0.00	0.00	
22100.00	90.00	2.27	10543.00	11617.55	-504.53	11628.19	0.00	0.00	0.00	
22200.00	90.00	2.27	10543.00	11717.47	-500.57	11727.91	0.00	0.00	0.00	
22300.00	90.00	2.27	10543.00	11817.39	-496.61	11827.62	0.00	0.00	0.00	
22400.00	90.00	2.27	10543.00	11917.32	-492.65	11927.33	0.00	0.00	0.00	
22500.00	90.00	2.27	10543.00	12017.24	-488.69	12027.05	0.00	0.00	0.00	
22600.00	90.00	2.27	10543.00	12117.16	-484.73	12126.76	0.00	0.00	0.00	
22700.00	90.00	2.27	10543.00	12217.08	-480.77	12226.47	0.00	0.00	0.00	
22800.00	90.00	2.27	10543.00	12317.00	-476.81	12326.19	0.00	0.00	0.00	
22900.00	90.00	2.27	10543.00	12416.92	-472.85	12425.90	0.00	0.00	0.00	
23000.00	90.00	2.27	10543.00	12516.85	-468.89	12525.61	0.00	0.00	0.00	
23100.00	90.00	2.27	10543.00	12616.77	-464.93	12625.33	0.00	0.00	0.00	
23199.31	90.00	2.27	10543.00	12716.00	-461.00	12724.35	0.00	0.00	0.00	PBHL 3-16H

Targets

Name	Description		TVD	+N/-S	+E/-W	Map	Map	<---- Latitude ---->			<--- Longitude ---->				
	Dip.	Dir.	ft	ft	ft	Northing	Easting	Deg	Min	Sec	Deg	Min	Sec		
SHL 3-16H			0.00	0.00	0.00	408237.43	1193849.30	48	4	31.639	N	103	40	11.004	W
PBHL 3-16H			10543.00	12716.00	-461.00	420961.63	1193912.02	48	6	37.133	N	103	40	17.797	W

LEAM Drilling Systems LLC

Planning Report

Company: Continental Resources Field: McKenzie County, ND Site: Columbus Federal 3 Well: 3-16H Wellpath: OH	Date: 03/05/2013 Co-ordinate(NE) Reference: Vertical (TVD) Reference: Section (VS) Reference: Survey Calculation Method:	Time: 16:13:18 Well: 3-16H, True North GL 1920+KB 21 1941.0 Well (0.00N,0.00E,357.93Azi) Minimum Curvature	Page: 5 Db: Adapti
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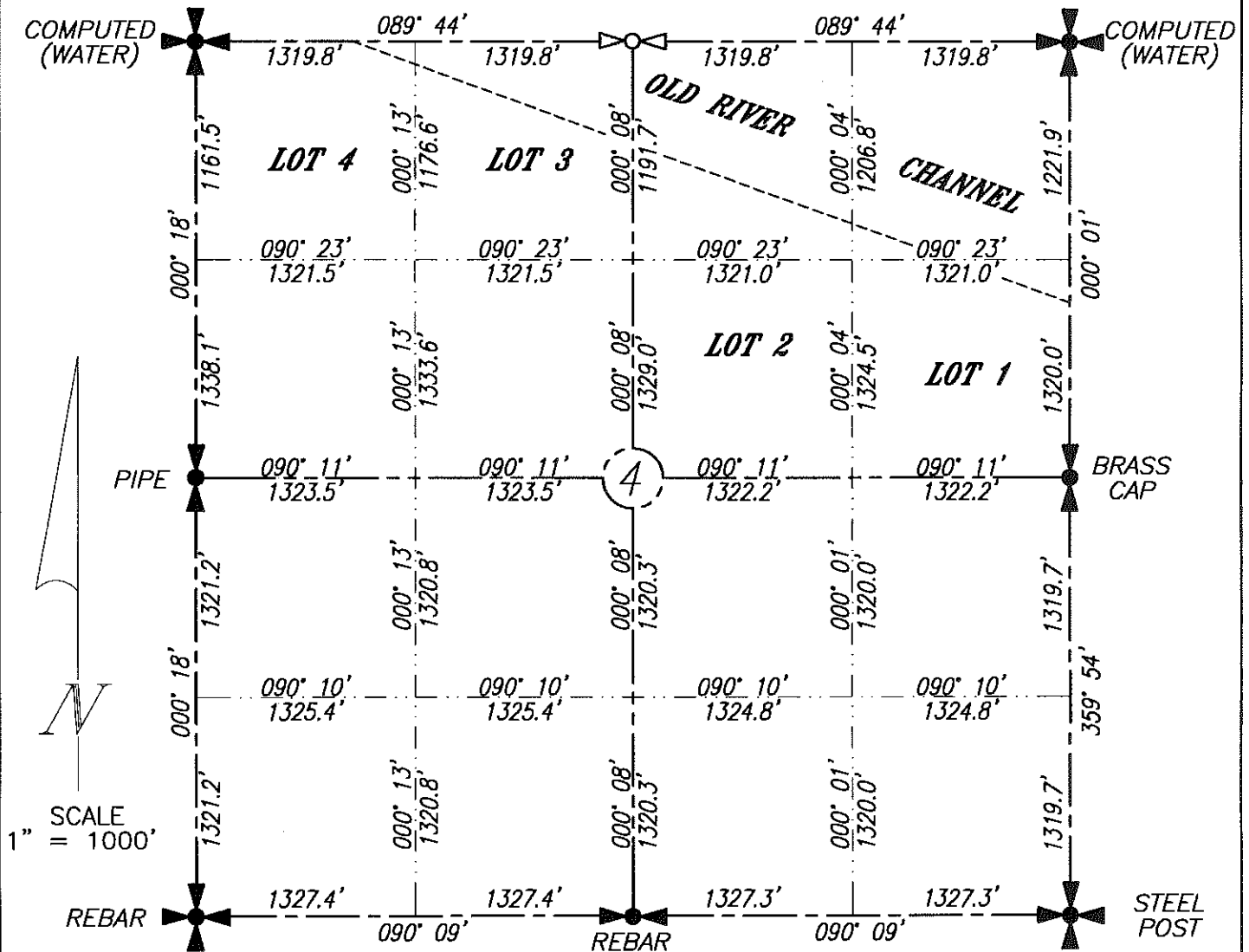
Casing Points

MD ft	TVD ft	Diameter in	Hole Size in	Name
10951.72	10543.00	7.00	8.50	7"

Formations

MD ft	TVD ft	Formations	Lithology	Dip Angle deg	Dip Direction deg
1841.00	1841.00	Pierre Shale		0.00	0.00
4425.00	4425.00	Greenhorn		0.00	0.00
4826.00	4826.00	Dakota Group		0.00	0.00
5705.00	5705.00	Dunham Salt Top		0.00	0.00
6793.00	6793.00	Dunham Salt Base		0.00	0.00
7103.00	7103.00	Pine Salt Top		0.00	0.00
7159.00	7159.00	Pine Salt Base		0.00	0.00
7185.00	7185.00	Minnekahta		0.00	0.00
7471.00	7471.00	Minnelusa Group		0.00	0.00
7627.00	7627.00	Tyler		0.00	0.00
8179.00	8179.00	Kibbey		0.00	0.00
8333.00	8333.00	Top Charles		0.00	0.00
9031.00	9031.00	Base Last Charles Salt		0.00	0.00
9255.00	9255.00	Mission Canyon		0.00	0.00
9821.00	9821.00	Lodgepole		0.00	0.00
10625.59	10511.00	Upper Bakken Shale		0.00	0.00
10679.26	10528.00	Middle Bakken Member		0.00	0.00
10794.57	10543.00	Middle Bakken Target		0.00	0.00

HORIZONTAL SECTION PLAT
CONTINENTAL RESOURCES INC.
COLUMBUS FEDERAL 3-16H
SECTION 4, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA



ALL CORNERS SHOWN ON THIS PLAT WERE FOUND IN THE FIELD
DISTANCES TO ALL OTHERS ARE CALCULATED.
ALL BEARINGS SHOWN ARE ASSUMED.

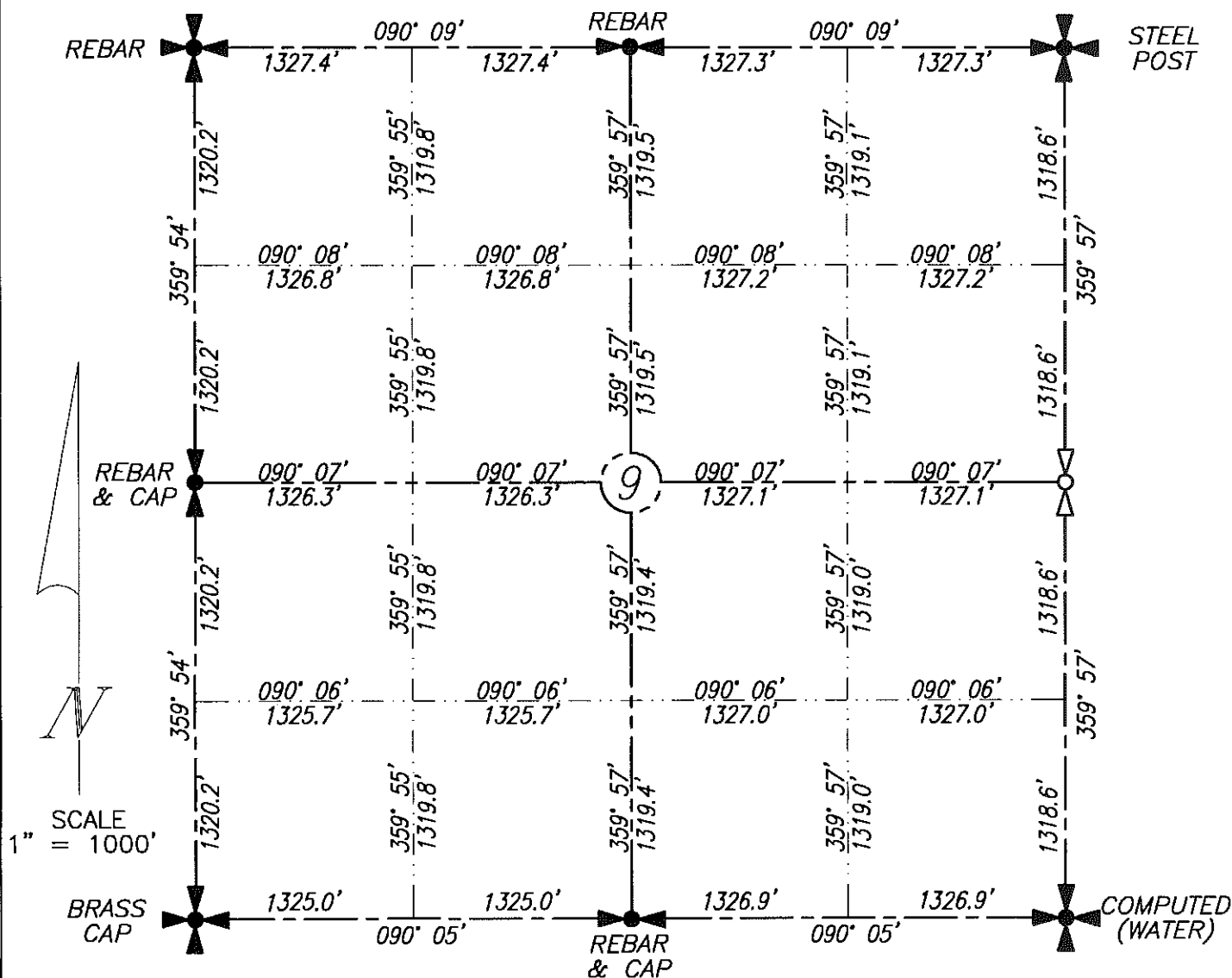
I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS
WORK PERFORMED BY ME OR UNDER MY RESPONSIBLE
CHARGE, AND IS TRUE AND CORRECT TO THE BEST OF
MY KNOWLEDGE AND BELIEF

JOHN PAULSON R.L.S. 3366

BROSZ ENGINEERING INC.

BOX 357
BOWMAN, N.D. 58623
PHONE: 701-523-3340
FAX: 701-523-5243
PROJECT NO. 12-10

HORIZONTAL SECTION PLAT
CONTINENTAL RESOURCES INC.
COLUMBUS FEDERAL 3-16H
SECTION 9, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA



ALL CORNERS SHOWN ON THIS PLAT WERE FOUND IN THE FIELD
DISTANCES TO ALL OTHERS ARE CALCULATED.
ALL BEARINGS SHOWN ARE ASSUMED.

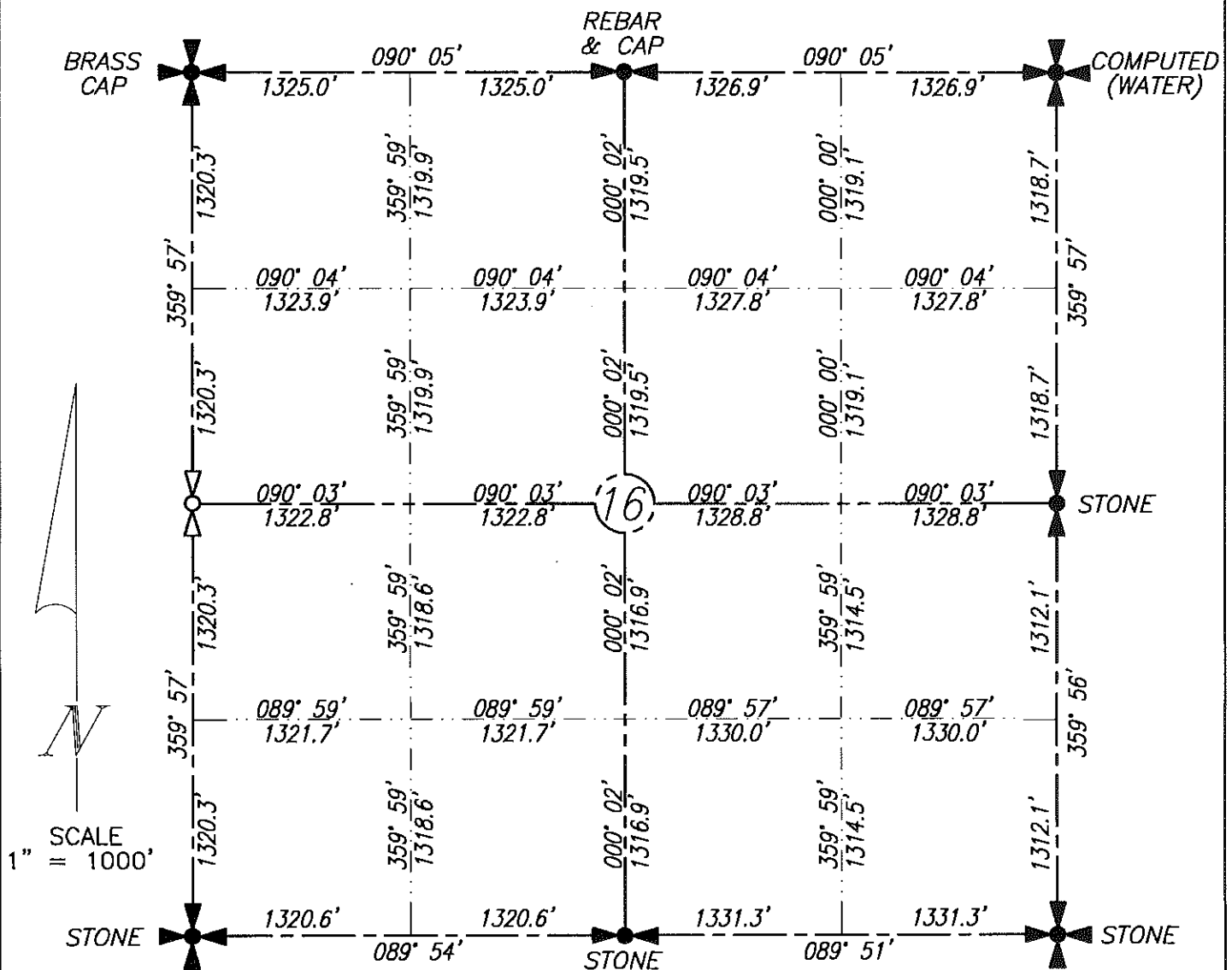
I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS
WORK PERFORMED BY ME OR UNDER MY RESPONSIBLE
CHARGE, AND IS TRUE AND CORRECT TO THE BEST OF
MY KNOWLEDGE AND BELIEF

JOHN PAULSON, DAKOTA 3366

BROSZ ENGINEERING INC.

BOX 357
BOWMAN, N.D. 58623
PHONE: 701-523-3340
FAX: 701-523-5243
PROJECT NO. 12-10

HORIZONTAL SECTION PLAT
CONTINENTAL RESOURCES INC.
COLUMBUS FEDERAL 3-16H
SECTION 16, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA



ALL CORNERS SHOWN ON THIS PLAT WERE FOUND IN THE FIELD
DISTANCES TO ALL OTHERS ARE CALCULATED.
ALL BEARINGS SHOWN ARE ASSUMED.

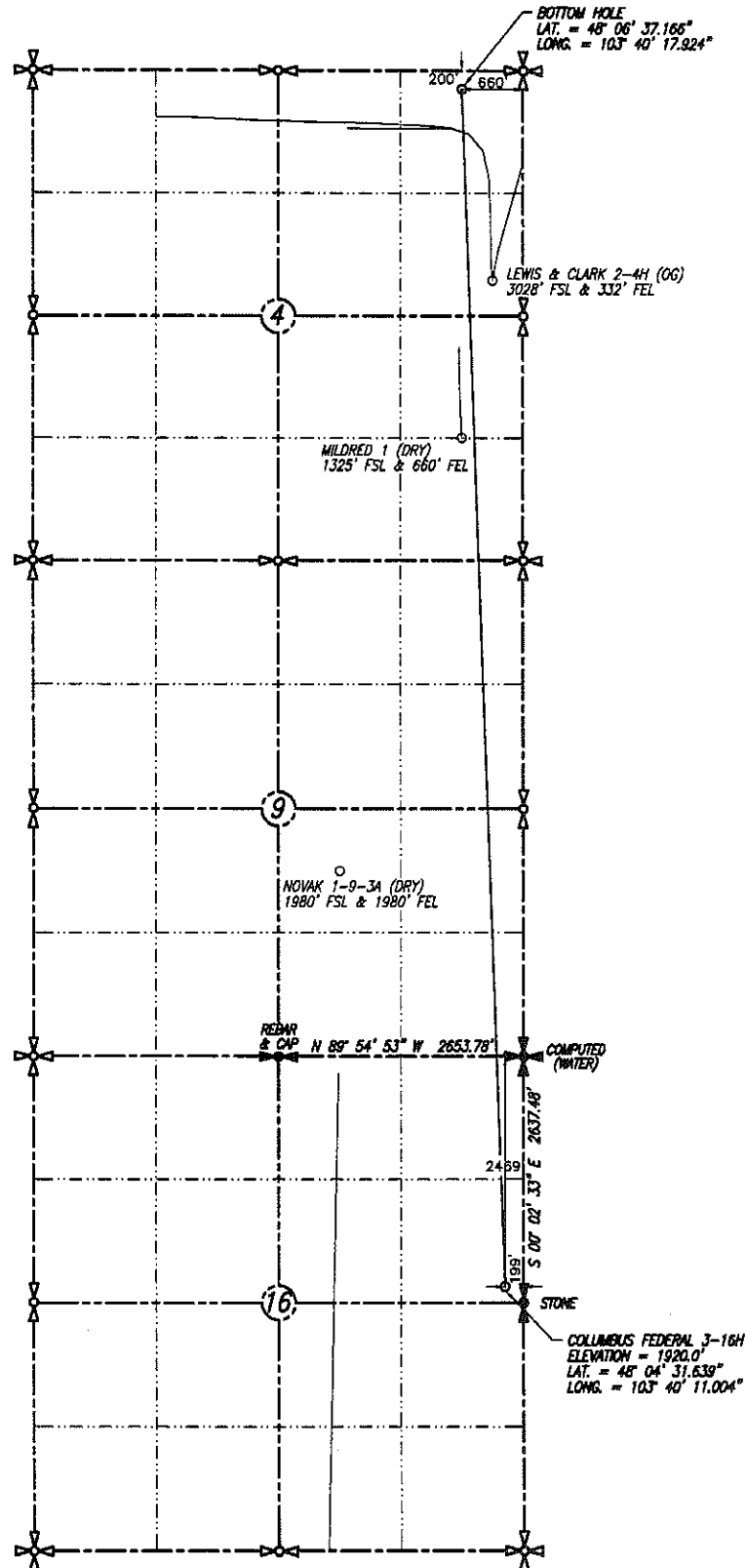
I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS
WORK PERFORMED BY ME OR UNDER MY RESPONSIBLE
CHARGE, AND IS TRUE AND CORRECT TO THE BEST OF
MY KNOWLEDGE AND BELIEF

JOHN PAULSON R.D.S. 3366

BROSZ ENGINEERING INC.

BOX 357
BOWMAN, N.D. 58623
PHONE: 701-523-3340
FAX: 701-523-5243
PROJECT NO. 12-10

BOTTOM HOLE LOCATION PLAT
 CONTINENTAL RESOURCES INC.
 COLUMBUS FEDERAL 3-16H
 SECTION 16, T153N, R101W
 MCKENZIE COUNTY, NORTH DAKOTA
 2469' FNL & 199' FEL



N
 SCALE
 1" = 2000'

I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS
 WORK PERFORMED BY ME OR UNDER MY RESPONSIBLE
 CHARGE, AND IS TRUE AND CORRECT TO THE BEST OF
 MY KNOWLEDGE AND BELIEF

JOHN FAUGSTON
 LAND
 SURVEYOR
 L.S. 3366
 NORTH DAKOTA

DATE STAKED: 1-4-2012
 BASIS OF VERTICAL DATUM:
 NAVD 1988 GEOID 09

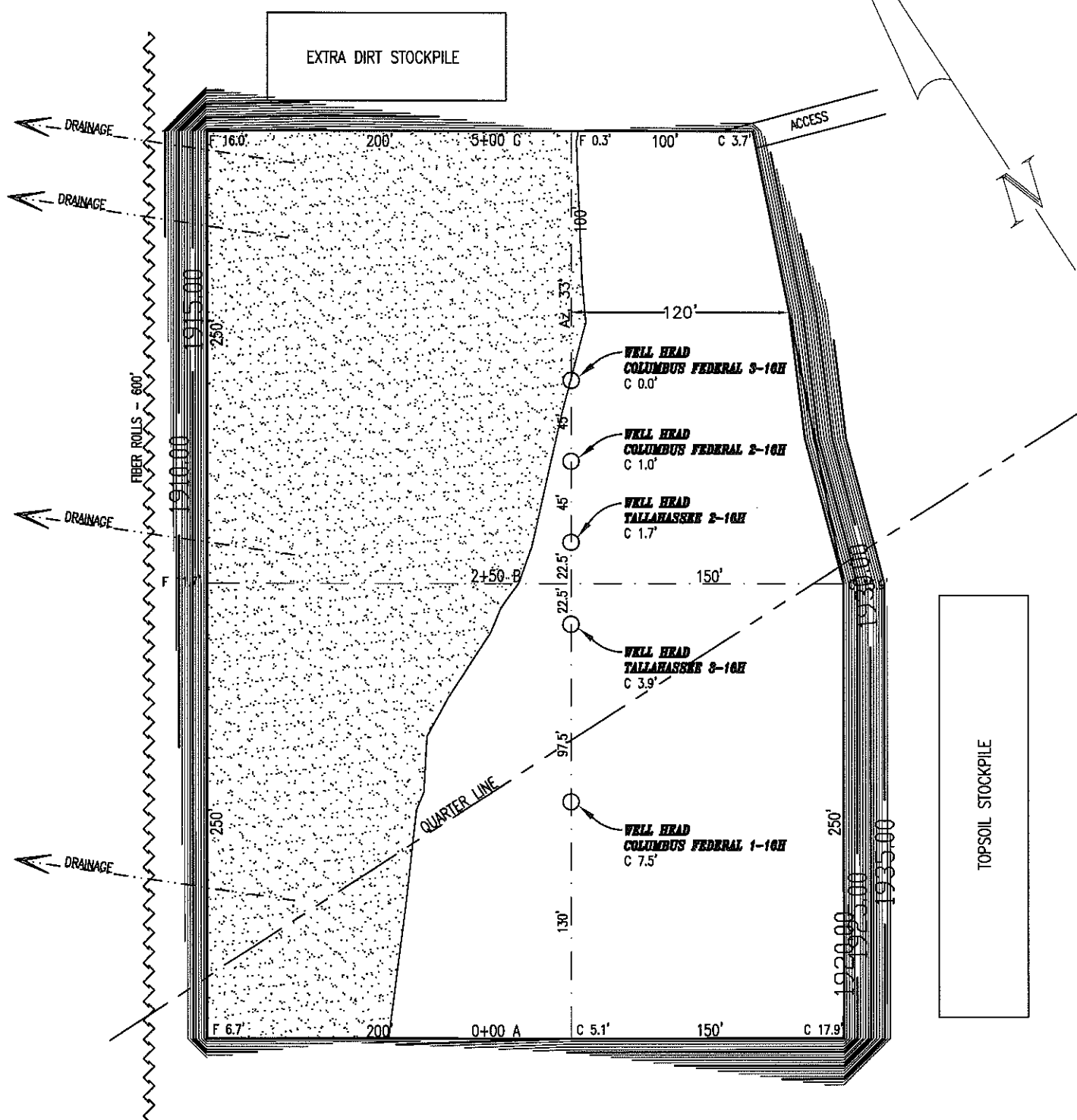
PERSON AUTHORIZING SURVEY;
 CHAD NEWBY

EXPLANATION AREA: NAD83(CORS96)

BASIS OF BEARING: TRUE NORTH

BROSZ ENGINEERING INC.

BOX 357
 BOWMAN, N.D. 58623
 PHONE: 701-523-3340
 FAX: 701-523-5243
 PROJECT NO. 12-10

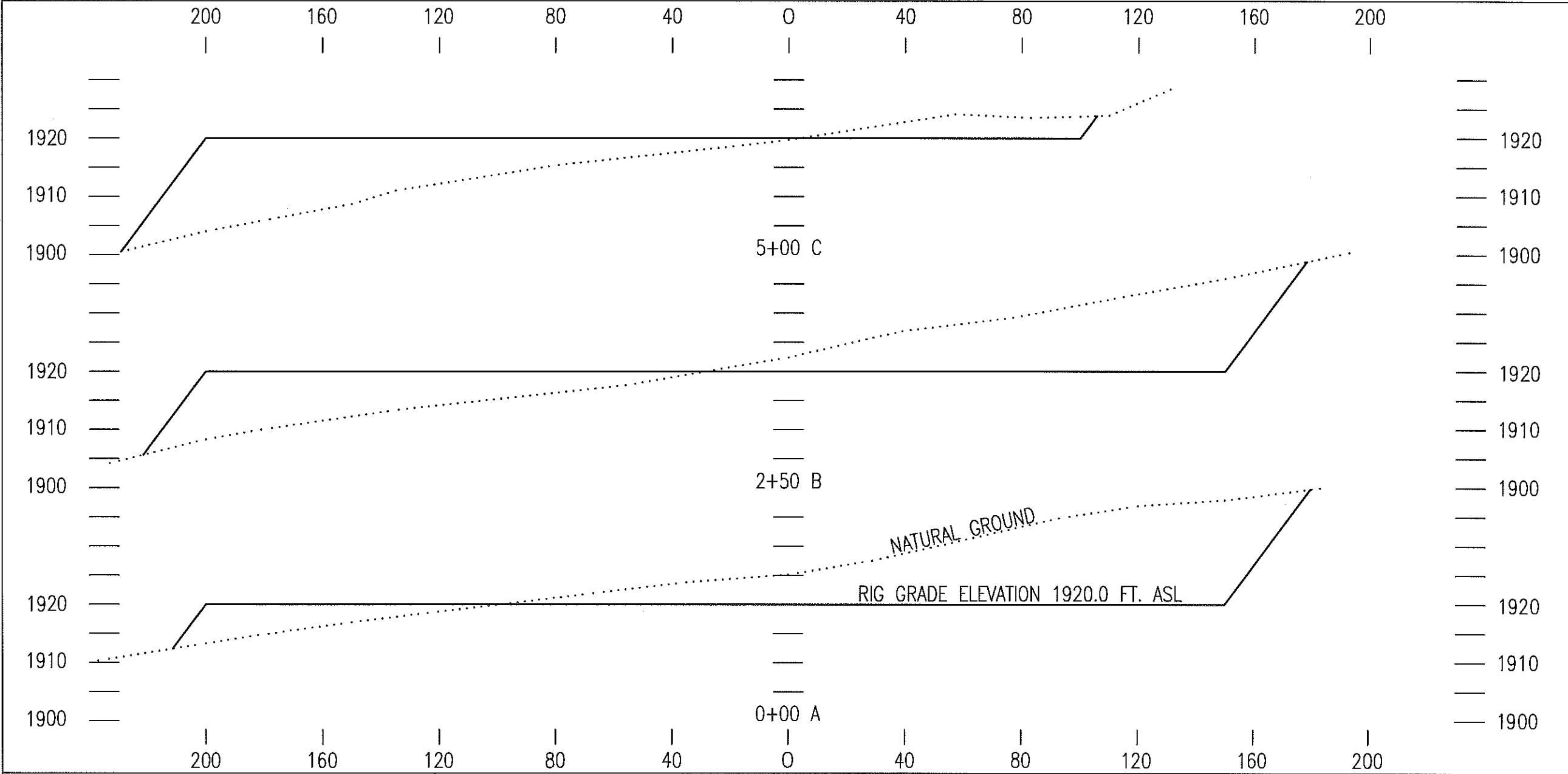


COLUMBUS FEDERAL 3-16H
SECTION 16, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA

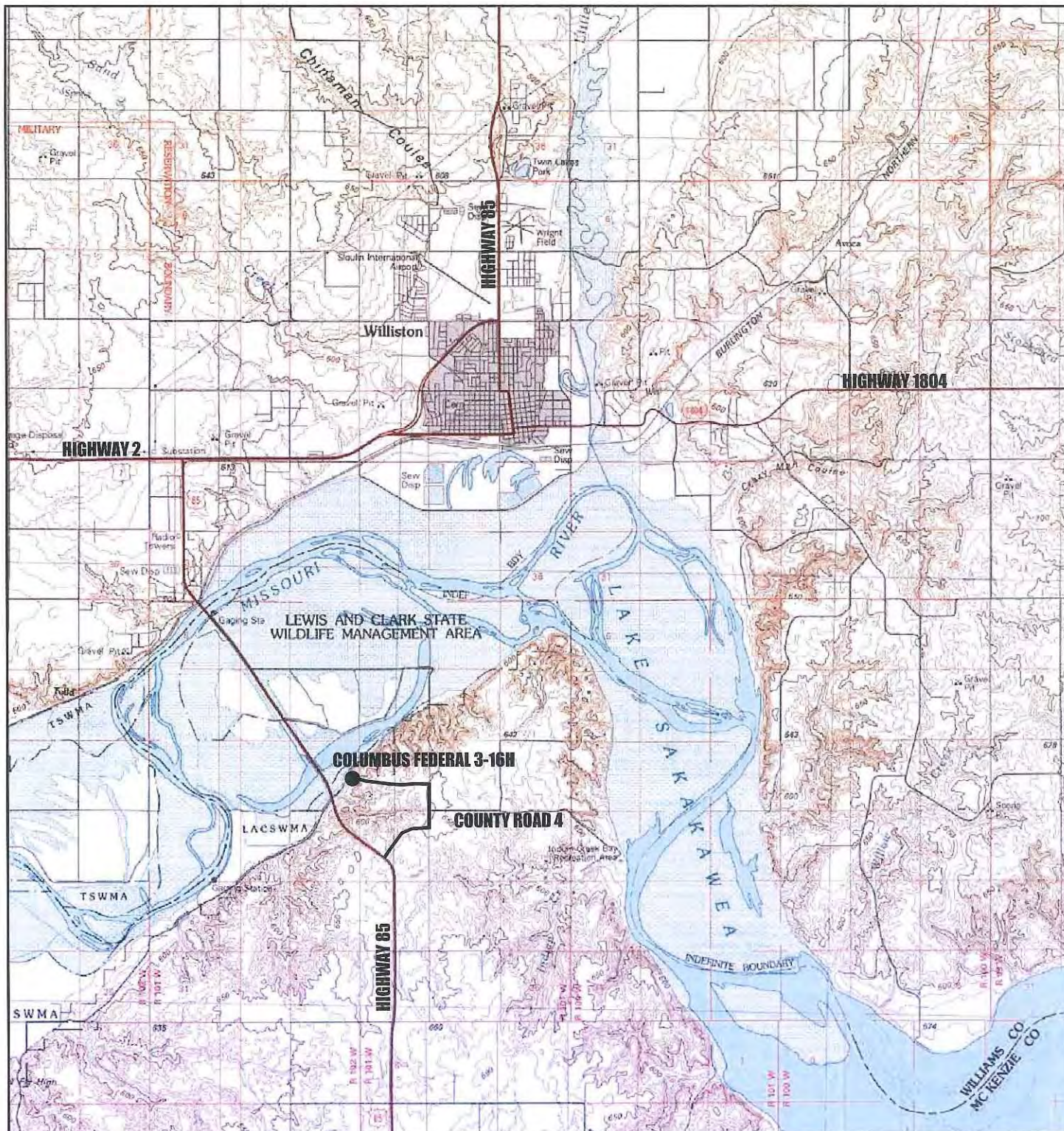
Use excess materials in access road fill

Ground Elevation at Well Head: 1920.0 ft. ASL
Finished Rig Grade Elevation: 1920.0 ft. ASL

BACKSLOPES
ASSUMED
AT 1 1/2 : 1 %

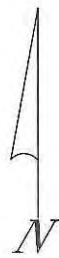
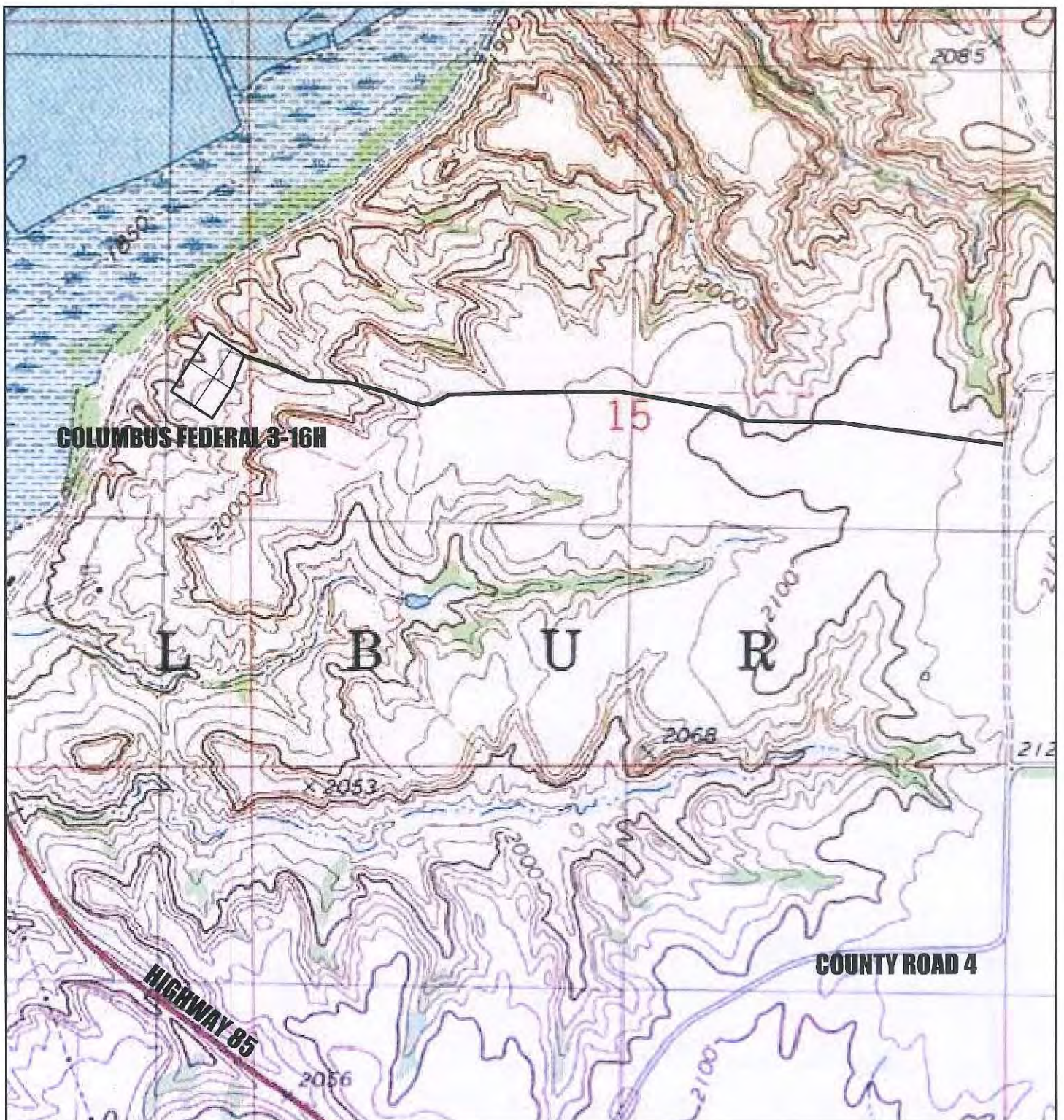


CONTINENTAL RESOURCES INC. PO BOX 1032 ENID, OKLAHOMA 73702	DRILL PAD PROFILE EXHIBIT 7 COLUMBUS FEDERAL 3-16H SECTION 16, T153N, R101W MCKENZIE COUNTY, NORTH DAKOTA	DRAWN BY: JH	PROJ. NO. 12-10	ESTIMATED QUANTITIES		
		SCALE: Horz. 1" = 40' Vert: 1" =20'				
		<i>BROSZ ENGINEERING INC.</i>		TOP-SOIL:	3,125	CUBIC YARDS
		BOX 357 BOWMAN, N.D. 58623 PHONE: 701-523-3340		SUB-SOIL:	28,719	CUBIC YARDS
		TOTAL CUT:	31,844	CUBIC YARDS		
		TOTAL FILL:	27,716	CUBIC YARDS		



CONTINENTAL RESOURCES INC.
EXHIBIT 1
VICINITY MAP

COLUMBUS FEDERAL 3-16H
 SECTION 16, T153N, R101W
 MCKENZIE COUNTY, NORTH DAKOTA



CONTINENTAL RESOURCES INC.

EXHIBIT 2

QUAD ACCESS

COLUMBUS FEDERAL 3-16H
SECTION 16, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA

Tabor, David

From: Tabor, David
Sent: Wednesday, March 06, 2013 2:25 PM
To: Tabor, David
Subject: FW: Columbus Federal 3-16H and pad.

From: Robert Sandbo [<mailto:Robert.Sandbo@clr.com>]
Sent: Friday, March 01, 2013 4:48 PM
To: Tabor, David
Subject: RE: Columbus Federal 3-16H and pad.

Thanks David. I got the email. Here is our schedule for the pad as of right now:

Columbus Federal 1-16H to spud on 3/24/2013 (may spud with the small rig around one week earlier if possible). We had planned on drilling the Columbus 1-16H first and then come back in a year or so and drill the other 4 but if we get the permits for all and the order on the 1280 allows us to drill them all back to back, we will most likely go ahead and drill them all at once. We will take a look at the 1280 order that covers the Columbus 1-16H and see if we can batch drill them.

You should have everything on the 3-16H and the affidavit sometime Monday.

Thanks for the help,

Bob Sandbo
Regulatory Compliance Supervisor

Continental Resources, Inc.
20 N. Broadway
OKC, OK 73102
P: 405-234-9020
F: 405-234-9562
C: 405-708-0691
robert.sandbo@clr.com
www.clr.com



March 7, 2013

Industrial Commission of North Dakota
Oil & Gas Division
600 East Boulevard, Dept 405
Bismarck, North Dakota 58505

Continental Resources, Inc. (CRI) respectfully submits the following information concerning the drilling of the Columbus Federal 1-16H, 2-16H & 3-16H1, and the Tallahassee 2-21H & 3-21H1

Township 153N, Range 101W

Section 16, NE/4 SE/4 & SE/4 NE/4 McKenzie County, North Dakota.

The Columbus Federal and Tallahassee well(s) are located in an environmentally sensitive area in close proximity to the Missouri River. Therefore, Continental Resources Inc. would like to propose the following automatic shut down equipment and level sensing monitoring equipment be deployed on the site to aid in the prevention of any accidental release or safety issue. A schematic diagram and pictures of an existing tank battery with emergency shutdown equipment has been included with this affidavit.

- 1) Tank Side (oil & water tanks)
 - a. ABB Levelmaster dual float
 - b. High level switch as backup
 - c. Battery box with solar backup
- 2) Treater & Separator
 - a. Temperature device in each vessel
 - b. Pressure transducer in each vessel
 - c. Level switch in each vessel
 - d. Battery box with solar backup
- 3) Wellhead
 - a. TotalFlow Controller
 - b. Emergency Shutdown Valve Package
 - c. Battery box with solar backup
 - d. Pressure transducer on casing & tubing
- 4) Flare
 - a. Scrubber pot high level switch
 - b. Monitoring flare for low temp output – alarm if flare goes out

- 5) System Automation through the proposed equipment will provide an independent control system on all equipment on site which will be able to shut the well(s) in should any of the other equipment be incapacitated or functioning improperly.

Don Kennedy
Don Kennedy, Sr. Production Engineer
Continental Resources, Inc.

STATE OF OKLAHOMA)
COUNTY OF OKLAHOMA)ss:

On the 7th day of March, 2013, before me, a Notary Public in and for said County and State, personally appeared **Don Kennedy**, known to me to be the Sr. Production Engineer of Continental Resources, Inc., the Corporation that executed the within instrument, and acknowledged to me that such Corporation executed the same.

Becky Barnes
Notary Public

Oklahoma County, Oklahoma

My Commission Expires: 7/5/2015

Commission No.: 11006023

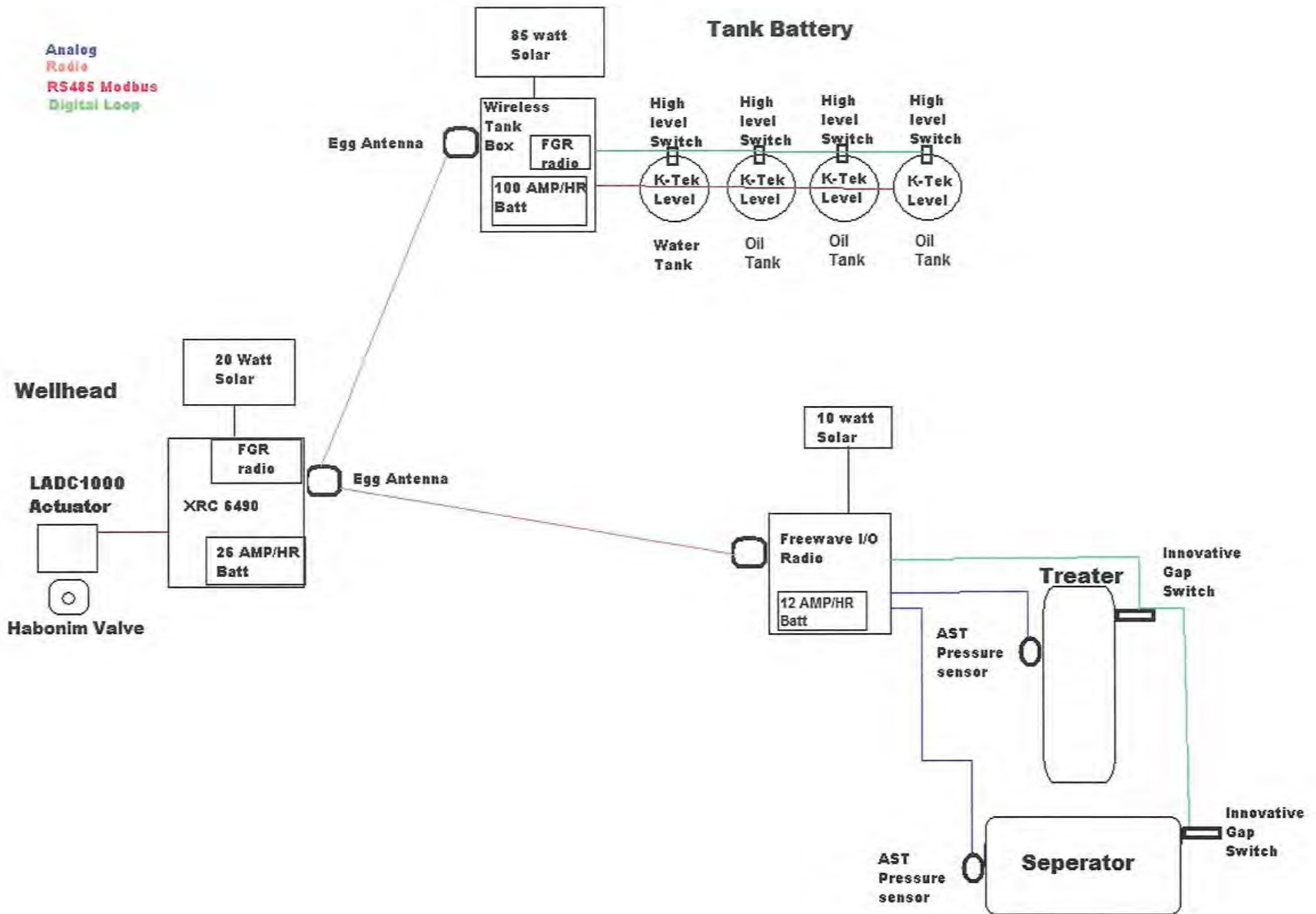


Sincerely,

CONTINENTAL RESOURCES, INC.

Becky Barnes
Regulatory Compliance Specialist

Analog
Radio
RS485 Modbus
Digital Loop





U003 Gap Switch and AST 4600 Transducer monitor pressures and liquid content of Flare and Sales Lines, transmitted to XRC via FreeWave Radio.

ABB TotalFlow XRC 6490

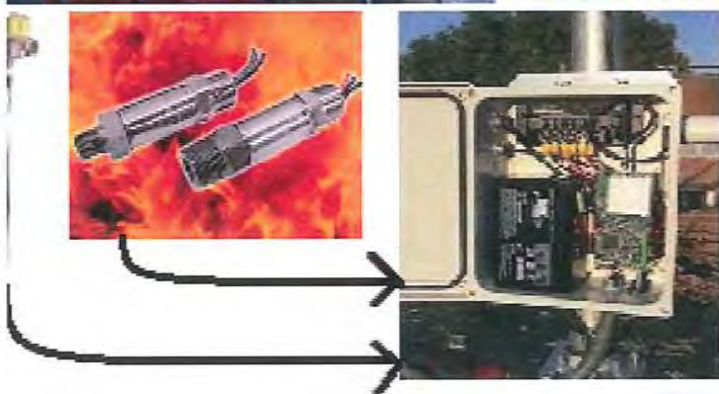


All well information is passed to your SCADA system via FreeWave Network (Future)

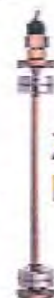
Winn-Marion's Well Head Kit with FGRI0 Radio and Power Supply



Tank Level information is passed from Tank Battery to XRC via FreeWave Radio



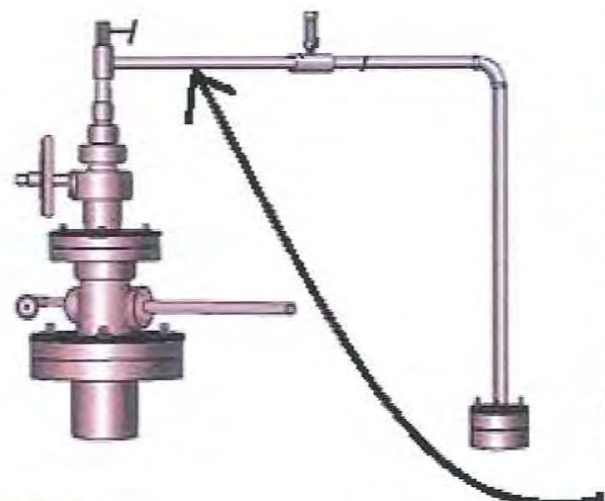
K-Tek MT5100 Guided Wave Radar



Z-Bend High Level Switch



ESD Valve Package
Standard Port Ball Valve rated to 6000 psi topped with a 12 VDC Actuator w/ Battery Backup



Winn-Marion, Inc.

CLR Spill Trailer Inventory

(To be Checked After Each Use)

Supplies	Quantity	Actual	Supplies	Quantity	Actual
Personal Protection			Miscellaneous		
Trauma/1st Aid Kit	1		EnviroClean (5-gal units of concentrate)	2	
Eye Wash	1		Duct Tape (Case)	3	
Hand Cleaner	2		Flashlights	6	
Nitrile Gloves (L & XL Case)	2		Flood Lights	2	
FRC Rain Coat - Extra Large	3		Extension Cord 50' 12-gauge	5	
FRC Rain Coat - Large	3		55-gal. Drums w/lids	2	
Rubber Safety Toed Boots - Size 10	2		Large Trash Cans	2	
Rubber Safety Toed Boots - Size 11	2		HD Drum Liners - boxes	2	
Rubber Safety Toed Boots - Size 12	2		Hoses - Kit (Blue & Green)	5	
FRC Tyvex Suits - Case XL	1		Plastic Buckets	5	
Neoprene Chest Waders - L	1		Propane Cylinders - 20-lb.	2	
Neoprene Chest Waders - L	1		Propane Weed Burner W/Hose	1	
Containment			Pump - Trash	2	
Absorbent (sphag)	10		Pump - 115V Water Transfer	2	
Absorbent Boom 3" x 10'	2		Gas Powered Generator (3-5K Watt)	1	
Absorbent Boom 5" x 10'	10		Misc. Ratchet Straps	6	
Absorbent Boom 8" x 10'	8		Rope 1/2" x 100'	2	
Containment Boom - Fast Water	3		Rope 1/4" x 50'	4	
Absorbent Pads (Hydrocarbon)	10		Rope 3/8" x 100'	2	
Absorbent Pads (Universal)	5		Shop Towels - box	2	
Absorbent Pillows 18" x 18" box	3		Caulking Gun	2	
Absorbent Pom Pom Cube	7		Silicon Tubes	10	
Absorbent Sweep - 16" x 100' - Bag	5		Metal Stakes/Spikes	8	
Miscellaneous			Metal T-Posts	6	
Antifreeze	2		Bungee Cords	3	
Push Broom	2		Wire - 25' roll - smooth	1	
Shovels	2		Fire Extinguisher	1	
Rake	5		Equipment Hooks	6	
Squeegees	2		Shelving	4	
Scoop	2		Drawers	1	
Spark Resistant Scoop	1		Misc. Building Supplies	1	

Continental Resources, Inc. (CRI) respectfully submits the following information concerning the drilling of the Columbus Federal 1-16H, 2-16H and 3-16H and the Tallahassee 2-16H and 3-16H, Sec. 16, T153N, R101W, Williams County, North Dakota.

CRI would like to propose the following safeguards and precautions to be taken while drilling the Columbus Federal-Tallahassee wells to prevent any contamination to freshwater sources during the drilling and completion of the wells:

- 1) During construction of the location, the entire location will be constructed per NDIC permit stipulations, ensuring any spills or runoff which occur on location do not penetrate the fresh ground water and are contained on the surface of the location. These modifications include, but are not limited to, the inclusion of a liner being placed under the location, and cementing of the rathole and mousehole.
- 2) Drainage will be re-routed to avoid the location, and erosion controls will be employed, as appropriate, around the site to reduce erosion and the resulting sediment contamination to freshwater runoff due to weather events.
- 3) The earthen berm, constructed to keep any freshwater runoff off the location, will also eliminate any spills from leaving the location
- 4) No reserve pit or dry cuttings pit will be utilized on location.
- 5) The conductor will be drilled to a depth of 80', and 20" pipe will be run to depth and cemented to surface.
- 6) During drilling operations, a freshwater protection string of 13-3/8" 48# H40 casing will be set to a depth of 500' and cemented to surface to protect the shallow freshwater zones. Standard 9-5/8" 36# J-55 surface casing will be set 100' into the Pierre Shale, to a depth of 1940', and cemented to surface.
- 7) A frac string will be used to protect the intermediate casing during hydraulic fracturing of the well.
- 8) CRI is submitting a comprehensive rig specific Spill Prevention Containment and Countermeasure Plan to prepare for any event which may occur during drilling and completion operations. A spill trailer will be located on location for spill response, if necessary.

CRI believes adequate planning and precautions are being taken to prevent any contamination to ground water and surface waters.


Sarah Madden, Project Development Engineer
Continental Resources, Inc.

STATE OF OKLAHOMA)
)ss:
COUNTY OF OKLAHOMA)

On the 7th day of March 2013, before me, a Notary Public in and for said County and State, personally appeared **Sarah Madden**, known to me to be a Project Development Engineer of Continental Resources, Inc., the Corporation that executed the within instrument, and acknowledged to me that such Corporation executed the same.


Notary Public

Oklahoma County, Oklahoma

My Commission Expires: 7/5/2015

Commission No.: 11006023



**SPILL PREVENTION CONTAINMENT
AND COUNTERMEASURE PLAN**

FOR

**CYCLONE DRILLING, INC.
P.O. BOX 908
GILLETTE, WYOMING 82717-908**

PREPARED BY;

**TOP LINE ENGINEERING, LLC
12635 HWY 200, P.O. BOX 884
SIDNEY, MONTANA 59270
PH; (701)570-2844**

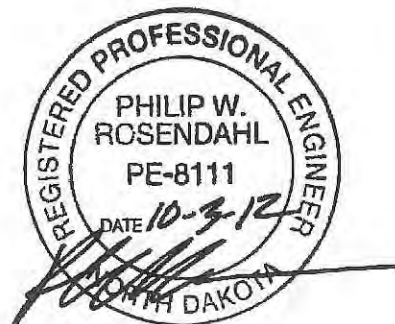


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APPENDIX

TYPICAL DRILL RIG LOCATION PLAN

DISCHARGE REPORTING FORM

DISCHARGE SOURCE INSPECTION RECORD

SPILL PREVENTION CONTAINMENT AND COUNTERMEASURE PLAN (SPCC)

PER 40 CFR 112.7 DATED: SEPTEMBER 14, 2012

This plan will follow the guidelines and format of 40 CFR part 112.7 and will replace the self-certified plan currently in affect.

112.7(a)(1) Discussion of facilities' conformance with the requirements listed in this part. The storage tanks on site store petroleum based drilling mud, diesel fuel, the invert tanks being the base for the drilling mud, salt water tanks, fresh water tanks, crude oil storage tanks and empty tanks for emergency storage. In addition to the storage tanks, other operations that can produce contaminants are the drilling operation itself and the cuttings drying operation. It is the purpose of this plan to define the prevention containment and countermeasure plan for discharges from any of these sources on the site. All the actions taken in this regard are per approved policies of this part of 40 CFR 112. Universally and in setting up the drilling operation on the site prevention measures are taken by grading the site to contain discharges, building in effective valves and other related operational and control mechanisms to prevent discharges. This equipment is installed and tested prior to beginning drilling operations. The tanks and equipment are installed within secondary containment berms and containment trenches and sumps are installed around the facilities prior to beginning drilling operations with capacity to contain with freeboard a rupture of the largest vessel associated with the containment. In the event of a discharge the sites are equipped with vacuums that are capable of picking up the spilled material and loading it into containers where it can be transferred to an approved disposal location. In the event of a discharge larger than can be handled by Cyclone and other manpower on location or that leaks offsite and potentially into navigable waters of the US, the operator, under his directions will arrange for trained contract personnel to immediately contain and clean-up the discharge.

112.7(a)(2) This plan will comply with all applicable items of this part. With no deviations anticipated.

112.7(a)(3) The attached sketch shows a typical layout of the drill rig sites depicting the location, size and contents of the tanks. There are no underground tanks nor are there any mobile or portable containers associated with these drilling locations. The site is prepared in advance of the drill rig coming onto the site. The sites are designed and constructed to contain spills and tank ruptures on the site. The drill rig and associated tanks and equipment are laid out in an efficient and effective manner to streamline the drilling operation. Frack tanks are located in such a manner that they have easy access to the mud tanks with valves located on both ends of transfer lines. The drilling mud tanks are plumbed to the drill rig with valves appropriately located to

stop flow from the tanks to the rig or from the rig to the tanks. The cuttings from the drilling operation are screened from the recycled drilling mud and directed to a container where they are dried using fly ash or some other equally effective absorbent material and are then transferred to the lined cuttings disposal pit onsite. The fuel tank is tied into the fuel lines providing fuel flow to the generators. Blowout preventers are placed over the drill hole casing and tested prior to beginning the drilling.

112.7(a)(3)(i) The type of oil in each fixed container and its storage capacity. No mobile or portable containers anticipated.

SOURCE	MAJOR TYPE OF FAILURE	TOTAL CAP. (BBLs.) (Type Fluid)	RATE (BBLs/MIN)	DIRECTION OF FLOW	SECONDARY CONTAINMENT
Diesel Fuel Tank	Leak or rupture	14,000 Maximum (Diesel)	Depends on size	Dams and/or Trenches	Site Perimeter Containment
Pumps & Engines	Leakage	½ - gal/hr. (lube oil)	½ - 1 gal/hr.	Drip Pans or Trenches	Site Perimeter Containment
Crude Oil Tanks	Rupture or Leak	400 BBLs	Depends on Type of Failure	Contained within Berm	Tank Battery Berm
Mud Tanks, Vibrator Rotary Hoses	Leak or rupture	450 BBLs Maximum per tank (Petroleum based drilling fluid)	Varies	Drip Pans for Hoses, Trench & Sump or Berms for Tanks	Site Perimeter Containment
Wellhead Blowout Preventer	Blowout or kicking well	Depends on severity (Well bore fluids)	Will vary	Away from well head to Trench and Sump	Site Perimeter Containment

112.7(a)(3)(ii) Discharge Prevention Measures including procedures for routine handling of products (loading, unloading and facility transfers, etc.); Before the drill rig moves on, the site is prepared for discharges associated with tank ruptures, as the site is graded to contain all discharges and storm runoff. When the rig is moved onto the site and set up additional discharge prevention and containment measures are taken. Equipment inspection records

are reviewed and the equipment is inspected following written procedures, the written procedures are attached in the appendix of the plan. A dated record of the inspection, signed by the appropriate supervisor or inspector is recorded and filed with time sheets or safety meeting minutes, when the drill rig changes locations and notes of deficiencies or acceptance are included. Operation and maintenance of equipment is designed to prevent discharges. Personnel are trained and made aware of the discharge prevention procedures and applicable pollution control laws, rules and regulations. The tool pusher in charge of the drill rig will have in his possession a copy of the Cyclone Drilling SPCC Plan and it is his responsibility to instruct the employees and make them aware of the discharge prevention measures contained in the plan. Scheduled briefings and refreshers are to be conducted with all the employees prior to and during rig up operations, such briefings are to be held during regularly scheduled safety meetings.

Employees are assigned to make routine inspections of valves, hose connections and other fluid connections for leaks. These inspections are made and documented on a monthly basis at a minimum. These leaks are provided drip pans and reported to the appropriate personnel for repair. If repairs cannot be readily made then regular emptying of the drip pans is required until the repairs can be made. Inspections of trenches and sumps are made on a regular basis to insure that they are free flowing and functional. The containment berms are also inspected periodically to insure their stability and function.

112.7(a)(3)(iii) Discharge or drainage controls such as secondary containment around containers and other structures, equipment, and procedures for the control of the discharge; Containment berms are placed around the tanks containing crude oil, the invert drilling fluid, salt water tanks and the empty tanks to be used for emergency storage, as secondary containment. The containment berms are of adequate height to contain a rupture and total failure of the largest tank contained within it amounting to 400 BBLS requiring a containment capacity of 3,000 cu. ft. Similarly, berms are built around the crude oil storage tanks on the site and requiring the same containment volume. Berms or trenches and sumps are constructed around the drill rig itself, the drilling mud tanks and diesel fuel tank, the sumps and trenches are sized to contain the volume of the largest tank in the case of a rupture and complete failure. Additionally, drip pans are placed under leaking valves, hose joints and other sources of small leaks until the item can be replaced or repaired to not leak. Regular inspections of these facilities are made to insure that they will function per plan.

112.7(a)(3)(iv) Countermeasures for discharge discovery, response, and cleanup (both the facility's capability and those that might be required of a contractor); Regular inspections may reveal a discharge such as a leaking valve or hose joint. These instances can readily be cleaned up by employees of Cyclone Drilling using on site vacuums and containers. Larger spills up to and

beyond the reporting limits can also be handled by Cyclone Drilling employees. Spills that exceed the limitations of the onsite cleanup equipment or in the event of a spill migrating offsite, will require the tool pushers' notification of the operator/owner of the site and they will contact the clean-up contractor they have contracted with to contain and clean-up such events.

112.7(a)(3)(v) Methods of disposal of recovered materials in accordance with applicable legal requirements: When spills occur the fluids will be either soaked up with absorptive material and placed in a container for transporting to an approved disposal site or vacuumed up and placed in a storage container for transportation to an approved disposal site. It is understood by all Cyclone Drilling employees that there is not an approved disposal site on or around the drilling rig to dispose of these recovered materials or fluids, other than drying drilling cuttings and disposing of them in the cuttings pit.

112.7(a)(3)(vi) Contact list and phone numbers for the facility response coordinator, National Response Center, cleanup contractors with whom operator has an agreement for response, and all appropriate Federal, State, and local agencies who must be contacted in the event of a discharge as described in § 112.1(b). Facility Response Coordinator for Cyclone Drilling HSE Dept. (Ph. 307-682-4161) is responsible for contacting the operators environmental group so they can contact the appropriate Federal, State and Local personnel in the event of a discharge as described in § 112.1(b).

112.7(a)(4) Information and procedures to enable a person reporting a discharge as described in § 112.1(b) to relate information on the exact address or location and phone number of the facility; the date and time of the discharge, the type of material discharged; estimates of the total quantity discharged; estimates of the quantity discharged as described in § 112.1(b); the source of the discharge: a description of all affected media; the cause of the discharge; any damages or injuries caused by the discharge; actions being used to stop, remove, and mitigate the effects of the discharge; whether an evacuation may be needed; and, the names of individuals and/or organizations who have also been contacted. In the event of a discharge as described in §112.1(b) it is the responsibility of the tool pusher to relate the above information to the operators environmental group so they can contact, any emergency organizations and provide the reporting necessary to the Federal, State and Local agencies that require reporting of such a discharge. The tool pusher on the rig is the person in charge and either he or the drilling superintendent have the responsibility of reporting the answers to the above questions. These questions are spelled out on a "Discharge Reporting Form" that each has at his disposal in the appendix of "Cyclone Drilling Oil Spill Contingency Plan" In reporting a discharge a list of all individuals and or organizations that received the report will be listed.

112.7(a)(5) Plan procedures to be used when a discharge occurs:

- 1) Immediately extinguish any heater or fire that may ignite the spill.
- 2) No smoking during spill control operations.
- 3) Close all associated valves.
- 4) Direct discharge to ditches or drains that will carry the discharge to a safe holding sump or reserve pit.
- 5) Distribute hill, fibertex, gel, barite, or any other absorptive material available as required to contain the discharge not entering the ditch, sump or reserve pit.
- 6) Inspect area to ensure that all of the discharge is contained in ditches, sumps or reserve pits. Add ditches or diversion structures as required to contain the discharge onsite.
- 7) Start jet or sump pumps and transfer discharged material from sumps to reserve pit or holding tanks.
- 8) After discharge is stopped, collect all used hulls, fibertex and similar absorptive material for disposal per instruction from the drilling superintendent or the tool pusher. No oil is to remain in the ditches or sumps that may create a fire hazard.
- 9) Hold collected discharged material for hauling and disposal in an approved location.
- 10) If discharge migrates from the drill rig location Cyclone Drilling personnel will use these same procedures stop and recover the discharge. Additionally the tool pusher will notify the owners representative to notify the contract clean-up company for clean-up and mitigation of the offsite discharge migration.
- 11) Fill out Site Discharge Reporting Form and distribute to appropriate agencies and personnel. (Form included in Appendix of Plan)

112.7(a)(5)(b) Flow diagram of potential discharges from significant sources as a result of facilities failure: Flow diagrams and quantities included in appendix of the plan.

112.7(a)(5)(c) Site discharge containment system: The site discharge containment system for the Cyclone Drilling Rigs is a combination of containment berms, confinement trenches, sumps and reserve pits. Drip pans will be used for small discharges until repairs are made or equipment replaced. (See typical site plan included in appendix.)

112.7(a)(5)(d) Not applicable as all measures spelled out in this part are practical as describe in each section.

112.7(a)(5)(e) Inspections, tests and records: Written procedures are provided on the "Discharge Source Inspection Record", these inspections are recorded at least on a monthly basis or whenever a leak or discharge is detected and reported, using the "Discharge Source Inspection Record" in the appendix. These records are signed by the tool pusher onsite and maintained by Cyclone Drilling for a period of 3 years.

112.7(a)(5)(f) Personnel Training and Discharge Prevention Procedures:

The personnel working on the drilling crews are trained in the maintenance and operation of all the equipment to prevent discharges, the discharge procedures and general facility operations. They are also made aware of the contents of the SPCC Plan. The tool pusher on each crew is the person on the location who is accountable for discharge prevention and who reports to the Cyclone Drilling Management. The tool pusher is responsible for conducting discharge prevention briefings and assuring an adequate understanding of the Cyclone Drilling SPCC Plan for the facility and any recent developments of new precautionary measures due to failures or malfunctions are implemented.

112.7(a)(5)(g) Site Security: All persons entering the site are required to check in with the tool pusher of the on duty crew. Unauthorized persons are not allowed on the site. Unauthorized visitors are easily detected, questioned regarding their presents and appropriately escorted to take care of their business.

112.7(a)(5)(h) Facility Tank Truck Loading/Unloading: The tank truck loading and unloading areas on the Cyclone Drilling Sites are located adjacent to the frack tank or crude oil tank batteries the area adjacent to the tank battery, where tank trucks are loaded and unloaded will be graded to drain into the containment trenches and associated sump, surrounding the drill rig, mud tanks, fuel tank, etc. Wheel chocks are used to prevent the truck from moving while connected to the tanks during loading/unloading operations. Before departure the vehicle is inspected for leaks in the lowermost drains and outlets and if they are any detected the associated valves are tightened or adjusted to prevent discharge while in transit. All above ground containers are checked and evaluated for risk of discharge or failure and as necessary appropriate action shall be taken.

APPENDIX



June 12, 2012

Industrial Commission of North Dakota
Oil & Gas Division
600 East Boulevard, Dept 405
Bismarck, North Dakota 58505

Re: Columbus Federal 3-16H

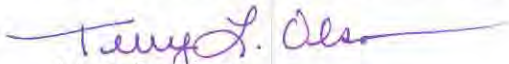
Continental Resources, Inc., would like to request all filings and information regarding the above captioned well be considered "Tight Hole".

Please charge the Continental Resources, Inc., credit card that is on file with your agency for the application fee of this well.

Thank you for your prompt attention to this matter. If you have any questions, you may contact me at 580-548-5139 or email the following Terry.Olson@clr.com.

Sincerely,

CONTINENTAL RESOURCES, INC.


Terry L. Olson
Regulatory Compliance Specialist